

Electric Circuits 2

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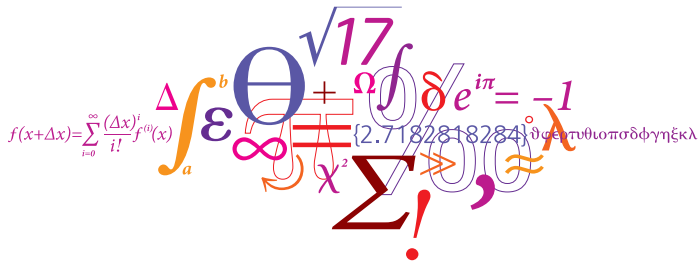


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Literature

IMPORTANT Primary course literature

The primary literature for the course is [1].

Further reading

For some selected parts of the course [2] is a good book.

A more colorful and applied approach for some of the topics in this course is [3].

If you feel like practicing your german [4] is a very good book for the contents of this course.

For assistance with LTSpice, [5] is a good reference.

More references, especially in the section [Applications](#) are provided throughout the teaching material.

Section 1

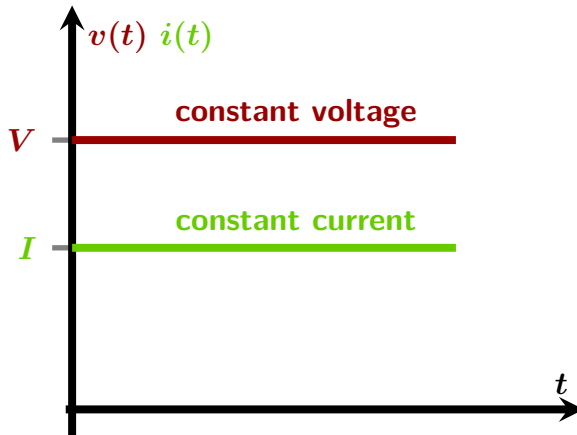
Signals

- **Signals**
 - Signals in Time Domain
 - Signals in Frequency Domain
 - Sources
 - Power
 - Three Phase Circuits

Time independent

- voltages are static
- current are static
- $\Rightarrow$ circuit behaves the same way at any point in time

Examples



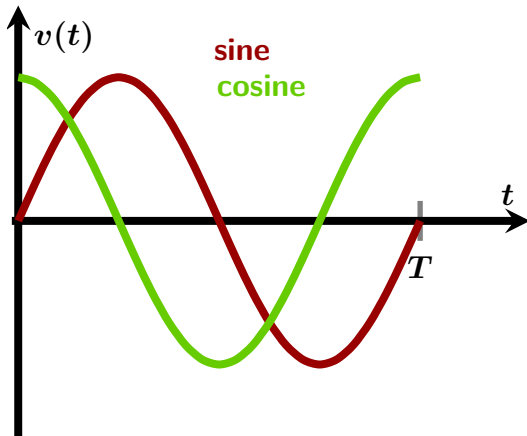
Characterization of signals

Time dependent

This course, for example $v(t)$, $i(t)$

- periodic, i.e. $v(t) = v(t + T)$ with the period T
 - sinusoidal:
 - mathematically well described by trigonometric functions
 - basis for the analysis of other periodic functions
 - practical impact, e.g. grid, wireless, wireless transmissions and test signals
 - non-sinusoidal
 - can mathematically be constructed by superimposing various periodic functions ((see Fourier Series))
 - well-suited for describing other electronic signals, like digital signals
- non-periodic
 - important at start-up and shutdown of electrical circuits
 - response of circuits to one-time events, like triggers and interrupts
 - noise

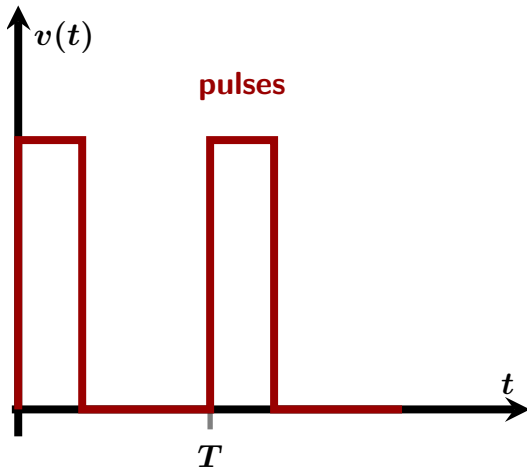
Periodic signals



Application Examples

- Analog electronics
- Audio
- Grid (AC mains)

Periodic signals

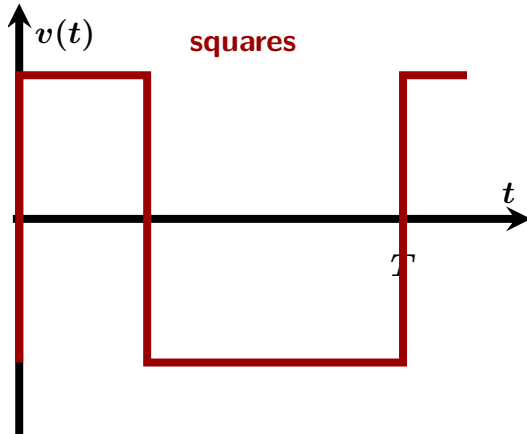


Application Example

- Digital electronics

Examples of signals

Periodic signals

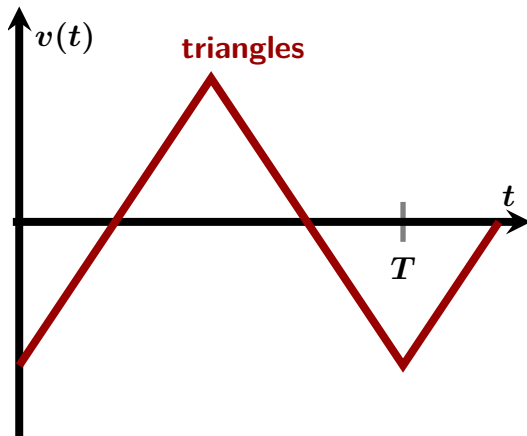


Application Example

- Power electronics

Examples of signals

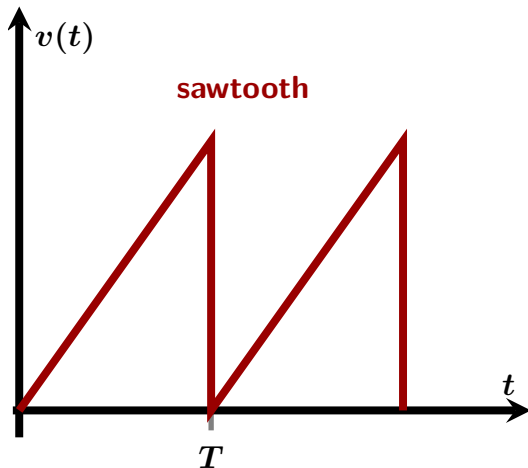
Periodic signals



Application Example

- Analog electronics

Periodic signals

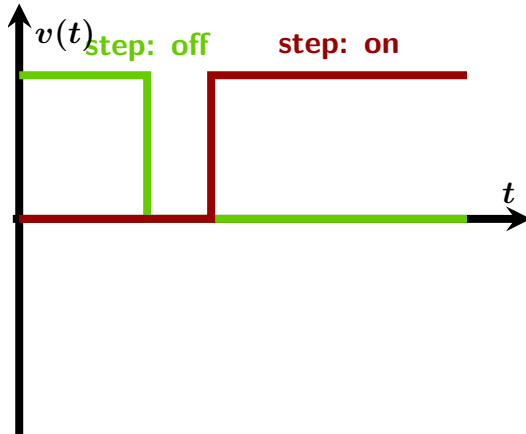


Application Example

- Power electronics
- Counters

Examples of signals

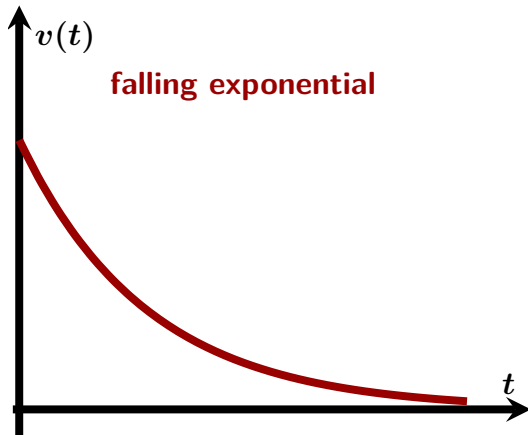
Non periodic signals



Application Example

- Turn on of circuits
- Turn off of circuits

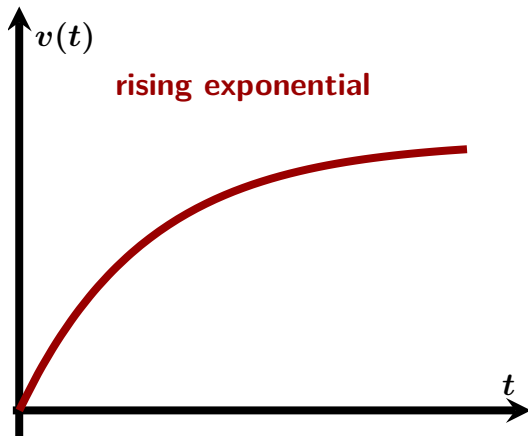
Non periodic signals



Application Example

- Discharge of a capacitor
- Discharge of a battery

Non periodic signals

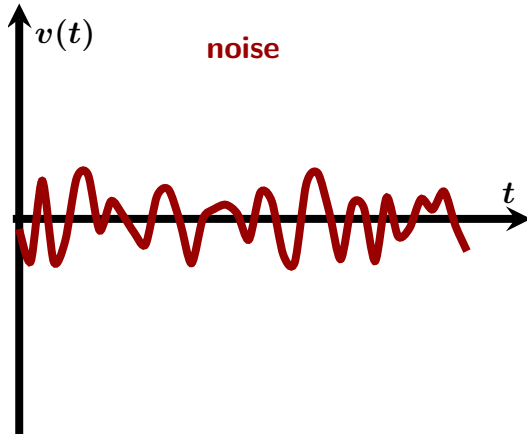


Application Example

- Charge of a capacitor
- Charge of a battery

Examples of signals

Non periodic signals

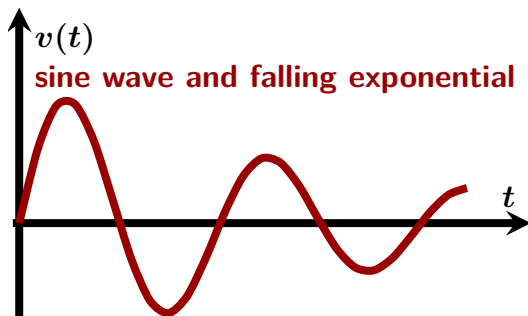


Application Example

- Noise of resistors
- Noise of amplifiers
- Wireless transmissions

Examples of signals

Superposition of periodic and non periodic signals

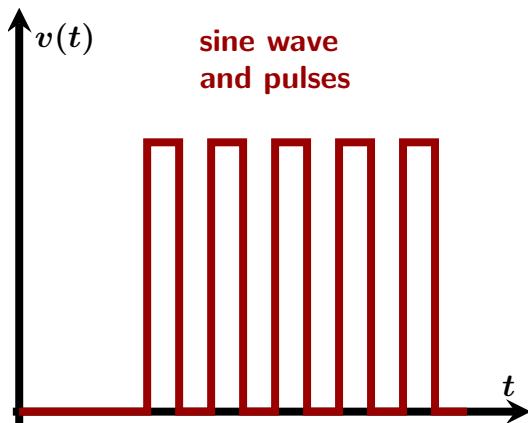


Application Example

- Dampened sine wave oscillation

(oscillation see Passive Filters and Oscillators)

Superposition of periodic and non periodic signals

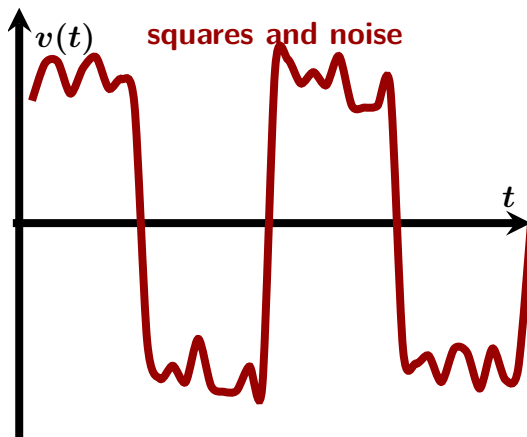


Application Example

- starting of a clock signal

A clock signal is providing a digital circuit with synchronization. The circuit executes its operations every time the clock transitions from low to high. Modern digital electronics operate up to the GHz range.

Superposition of periodic and non periodic signals



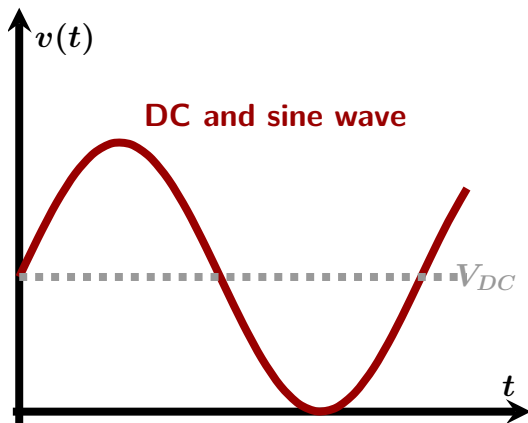
Application Example

- noise in a motor control
- noise in an audio amplifier

Note: mostly what is considered to be noise has its reason and is therefore a different kind of signal. The real noise level of electronics is typically as low as in the nV range and therefore well below the measuring capability of oscilloscopes.

Examples of signals

Superposition of periodic and non periodic signals



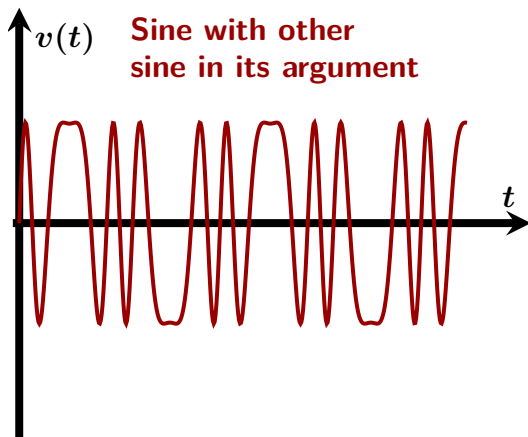
Application Example

- analog signals in battery driven applications

The audio signal coming out of the jackstick output of a mobil phone or tablet is centered around half of its battery voltage. It cannot swing negative, as there is no negative voltage in the device.

Examples of signals

Superposition of periodic and non periodic signals



Application Example

- Frequency modulation

This is used for radio broadcasting.

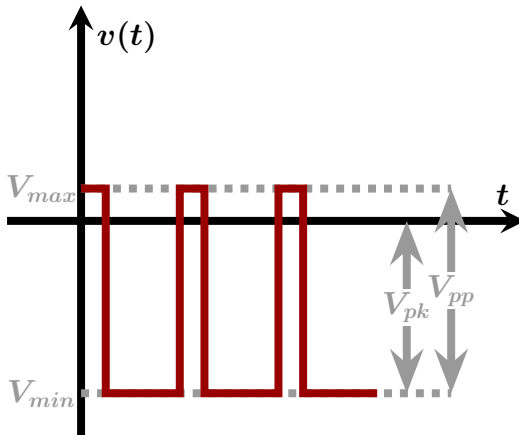
Amplitude

- Maximum value: V_{max} ,
- Minimum value: V_{min} ,
- Peak value: $V_{pk} = \max\{|V_{max}|, |V_{min}|\}$,
- Peak-to-peak value:
 $V_{pp} = V_{max} - V_{min}$

Amplitude

- Maximum value: V_{max} ,
- Minimum value: V_{min} ,
- Peak value: $V_{pk} = \max\{|V_{max}|, |V_{min}|\}$,
- Peak-to-peak value: $V_{pp} = V_{max} - V_{min}$

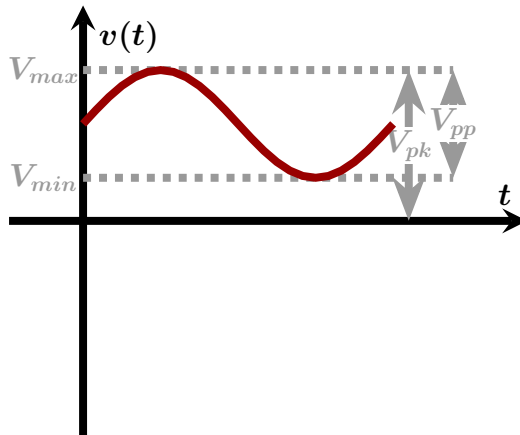
Examples



Amplitude

- Maximum value: V_{max} ,
- Minimum value: V_{min} ,
- Peak value: $V_{pk} = \max\{|V_{max}|, |V_{min}|\}$,
- Peak-to-peak value: $V_{pp} = V_{max} - V_{min}$

Examples



Phase φ

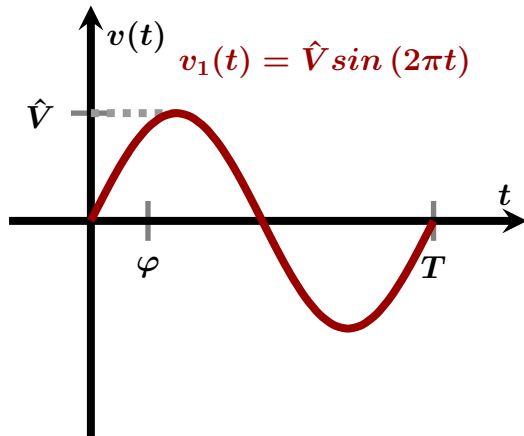
only defined for repetitive signals

- one signal defined as zero phase
- phases of all others are relative to the phase of this signal

Phase φ

only defined for repetitive signals

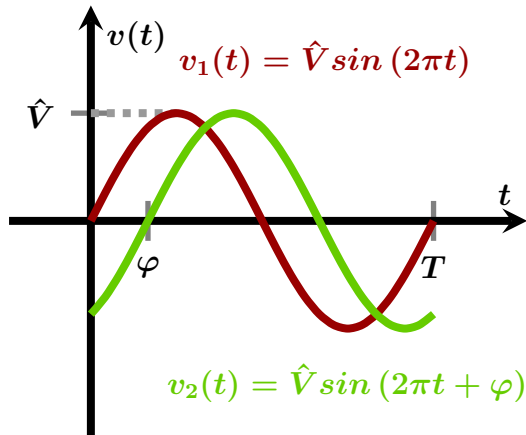
- one signal defined as zero phase
- phases of all others are relative to the phase of this signal

Examples

Phase φ

only defined for repetitive signals

- one signal defined as zero phase
- phases of all others are relative to the phase of this signal

Examples

Signal Parameters

Frequency f / period T

only defined for repetitive signals

- $f = \frac{1}{T}$

Angular frequency ω

only defined for repetitive signals

- $\omega = 2\pi f$

Signal Parameters

Frequency f / period T

only defined for repetitive signals

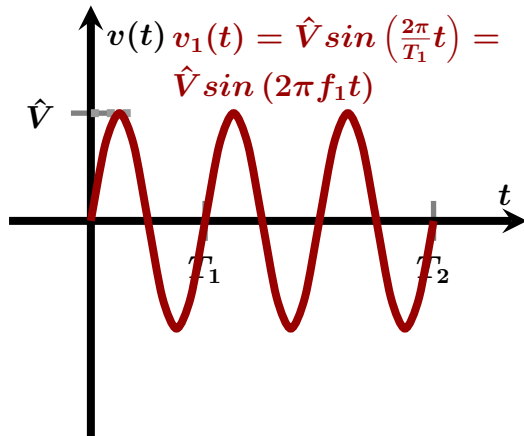
- $f = \frac{1}{T}$

Angular frequency ω

only defined for repetitive signals

- $\omega = 2\pi f$

Examples



Frequency f / period T

only defined for repetitive signals

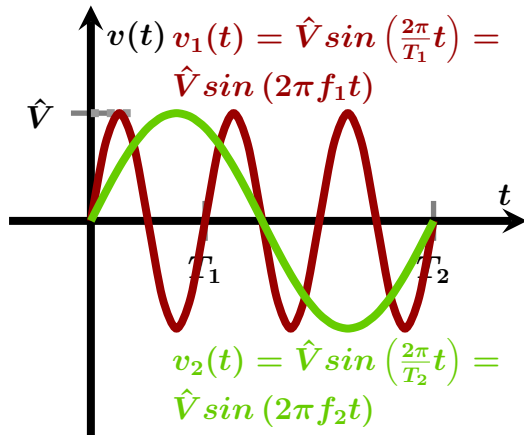
- $f = \frac{1}{T}$

Angular frequency ω

only defined for repetitive signals

- $\omega = 2\pi f$

Examples



Average Value

only defined for periodic signals, also called:

- Mean value
- DC (part of a signal)
- Average value
- Offset
- Operating point
- Bias point

IMPORTANT Equation for Average Value

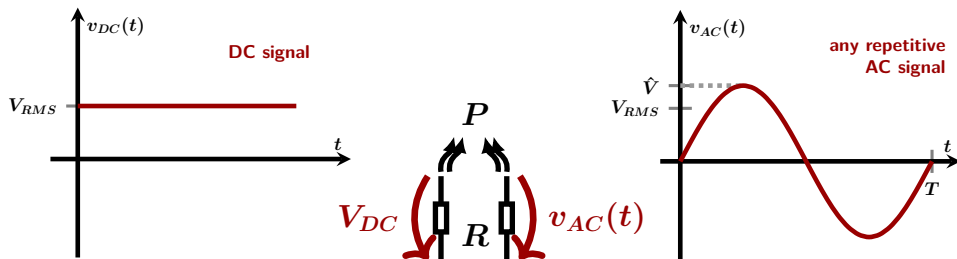
$$V_{avg} = \frac{1}{T} \int_0^T v(t) dt$$

Root Mean Square (RMS) Value - Meaning

only defined for periodic signals

Defines, how much DC voltage V_{DC} needs to be applied to a defined resistor R to get the exact same amount of power loss W as would be generated in this exact resistor, when the respective AC signal $v_{AC}(t)$ would be applied across it.

Derivation of Root Mean Square (RMS) Value

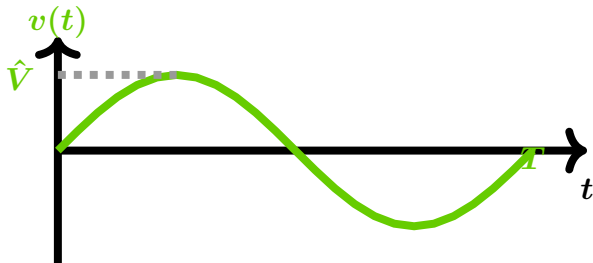
**IMPORTANT** Equation for Root-Mean-Square (RMS) Value

$$V_{RMS} = \underbrace{\sqrt{\underbrace{\frac{1}{T} \int_0^T \underbrace{v^2(t)}_S dt}_M}}_R$$

Average and RMS calculation

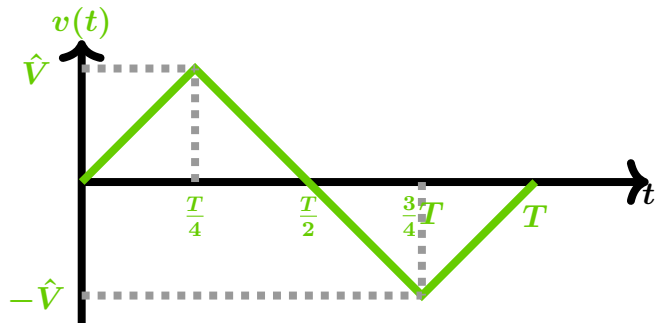
Exercise 1.1 Sine wave $v(t) = \hat{V} \sin(\omega t)$

What is the mean value V_{avg} , the RMS value V_{RMS} and the frequency f of these repetitive signals, with $v(t) = v(t + T)$?



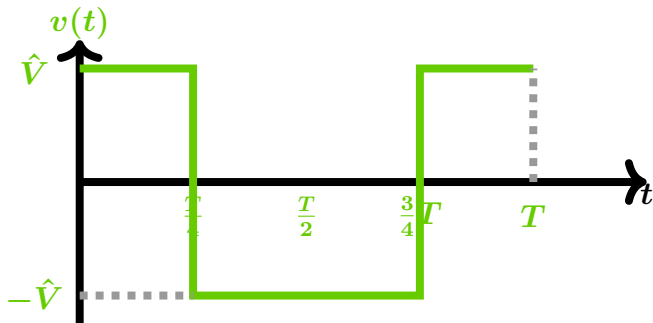
Average and RMS calculation

Exercise 1.2 A triangle



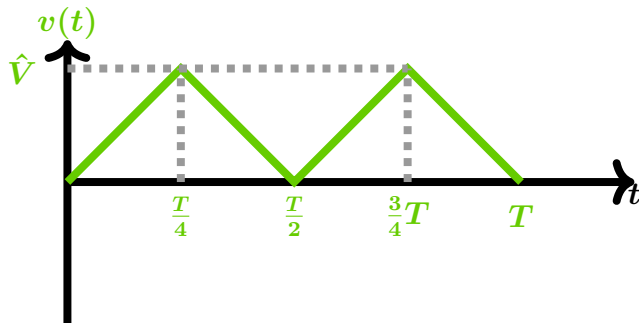
Average and RMS calculation

Exercise 1.3 A square wave



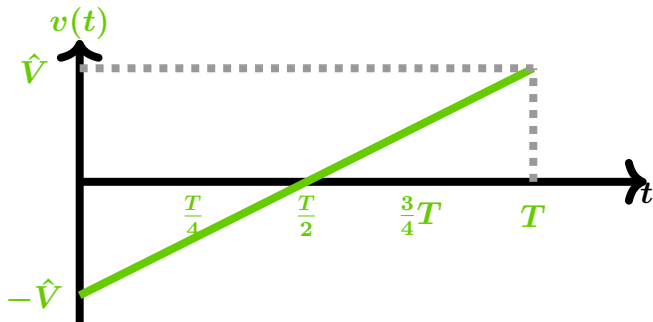
Average and RMS calculation

Exercise 1.4 Two triangles



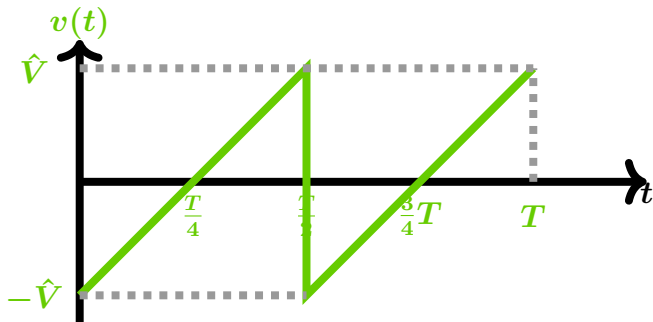
Average and RMS calculation

Exercise 1.5 Sawtooth



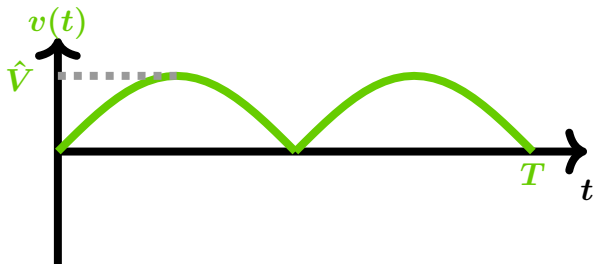
Average and RMS calculation

Exercise 1.6 Two sawteeth



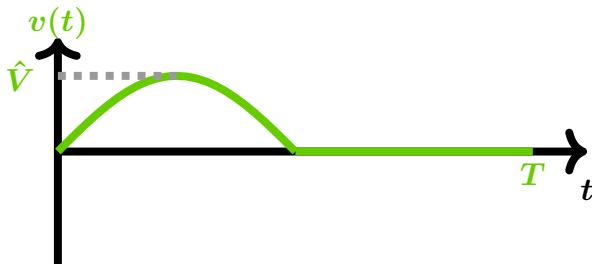
Average and RMS calculation

Exercise 1.7 A full wave rectified sine wave



Average and RMS calculation

Exercise 1.8 A half wave rectified sine wave



Derivation

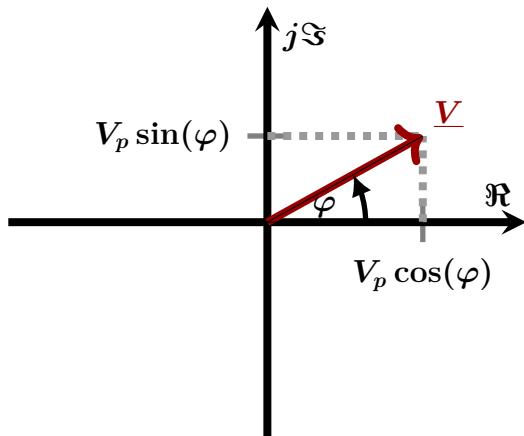
only valid for sinusoidal and repetitive signals

$$\begin{aligned}
 v_{AC}(t) &= V_p \cos(\omega t + \varphi) \\
 &\stackrel{\text{Euler } e^{j\alpha} = \cos \alpha + j \sin \alpha}{=} V_p \Re(e^{j(\omega t + \varphi)}) \\
 &= V_p \Re(e^{j\omega t} e^{j\varphi}) \\
 &\stackrel{V_p = \text{const}}{=} \underbrace{\Re(V_p e^{j\varphi})}_{\text{Phasor } \underline{V}} e^{j\omega t}
 \end{aligned} \tag{1}$$

IMPORTANT Definition of a Phasor

$$\underline{V} = V_p e^{j\varphi} = \underbrace{V_p \cos(\varphi)}_{\Re(\underline{V})} + j \underbrace{V_p \sin(\varphi)}_{\Im(\underline{V})}$$

Graphical Representation: Phasor Diagram



Validity

A phasor diagram is only valid for one frequency.

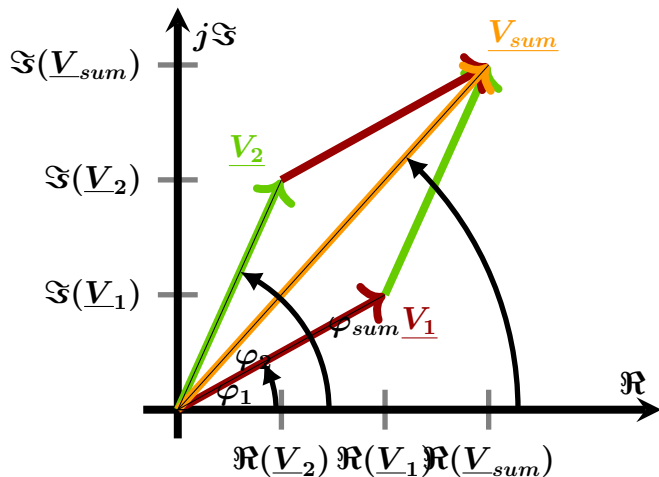
Derivation Superposition of Phasors

$$\begin{aligned}v_{sum}(t) &= v_1(t) + v_2(t) = \Re((\underline{V}_1 + \underline{V}_2)e^{j\omega t}) \\ \Rightarrow \underline{V}_{sum} &= \underline{V}_1 + \underline{V}_2 \\ &= V_{p1}e^{j\varphi_1} + V_{p2}e^{j\varphi_2} \\ &= \underbrace{V_{p1}\cos(\varphi_1) + V_{p2}\cos(\varphi_2)}_{\Re(\underline{V}_{sum})} + j \underbrace{(V_{p1}\sin(\varphi_1) + V_{p2}\sin(\varphi_2))}_{\Im(\underline{V}_{sum})}\end{aligned}\quad (2)$$

IMPORTANT Superposition of Phasors - Definition

$$\begin{aligned}|\underline{V}_{sum}| &= \sqrt{\Re^2(\underline{V}_{sum}) + \Im^2(\underline{V}_{sum})} \\ \angle \underline{V}_{sum} &= \arctan\left(\frac{\Im(\underline{V}_{sum})}{\Re(\underline{V}_{sum})}\right) \\ \underline{V}_{sum} &= |\underline{V}_{sum}|e^{j\angle \underline{V}_{sum}}\end{aligned}\quad (3)$$

Graphical Superposition of Phasors



Validity

Both signals need to have the same frequency.

Phasor calculation

Exercise 1.9 Converting sinusoids into phasors and add them

Add $v_1(t) = 100 \text{ V} \cos(\omega t)$ and $v_2(t) = 100 \text{ V} \sin(\omega t)$. Draw the phasor diagram and give the phase relation between the 3 signals.

Exercise 1.10 Converting sinusoids into phasors and add them

Calculate the resulting phasor for

$$5 \text{ A} \cos(\omega t + 75^\circ) - 3 \text{ A} \cos(\omega t - 75^\circ) + 4 \text{ A} \sin(\omega t).$$

Exercise 1.11 Converting sinusoids into phasors and add them

Add $v_1(t) = 90 \text{ V} \cos(\omega t - 15^\circ)$ and $v_2(t) = 50 \text{ V} \sin(\omega t - 60^\circ)$. Express the result as a sinusoidal waveform.

Phasor calculation

Exercise 1.12 Converting sinusoids into phasors and add them

Calculate the resulting phasor for

$$5 \text{ V} \sin(\omega t) + 15 \text{ V} \cos(\omega t - 30^\circ) + 20 \text{ V} \cos(\omega t - 120^\circ).$$

Exercise 1.13 Converting sinusoids into phasors and add them

Add $v_1(t) = 170 \text{ V} \cos(\omega t + 30^\circ)$ and $v_2(t) = 200 \text{ V} \sin(\omega t)$.

Exercise 1.14 Converting sinusoids into phasors and add them

What is $v_3(t)$ with $v_1(t) + v_2(t) + v_3(t) = 0$ and $v_1(t) = 50 \text{ V} \cos(\omega t + 45^\circ)$ and $v_2(t) = 25 \text{ V} \sin(\omega t)$.

Fourier Series

only defined for repetitive signals

IMPORTANT Definition of Fourier Series

$$v(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos(2\pi n f_0 t) + b_n \sin(2\pi n f_0 t))$$

IMPORTANT Fourier Coefficients

$$\begin{aligned} a_0 &= \frac{1}{T} \int_0^T v(t) dt \\ a_n &= \frac{2}{T} \int_0^T v(t) \cos(n\omega t) dt \\ b_n &= \frac{2}{T} \int_0^T v(t) \sin(n\omega t) dt \end{aligned} \tag{4}$$

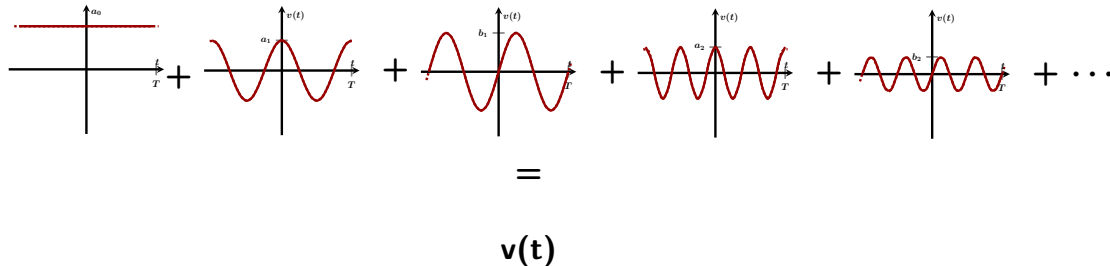
where $\omega = 2\pi f_0 = \frac{2\pi}{T}$ and $n \in \mathbb{N}$

Meaning of Fourier Series in words

Any repetitive signal $v(t)$ can be represented by a superposition (sum) of an offset a_0 , a fundamental and harmonics. The fundamental is a sinusoidal signal with the same frequency f_0 (and repetition rate $T = \frac{1}{f_0}$) as the signal $v(t)$. The harmonics are sinusoidal waveforms with multiples of the frequency of the fundamental.

- offset a_0 : DC, average, mean
- fundamental: $a_1 \cos(2\pi f_0 t) + b_1 \sin(2\pi f_0 t)$
- harmonics: $a_n \cos(2\pi n f_0 t) + b_n \sin(2\pi n f_0 t)$: n-th harmonic

Meaning of Fourier Series - Graphical Representation

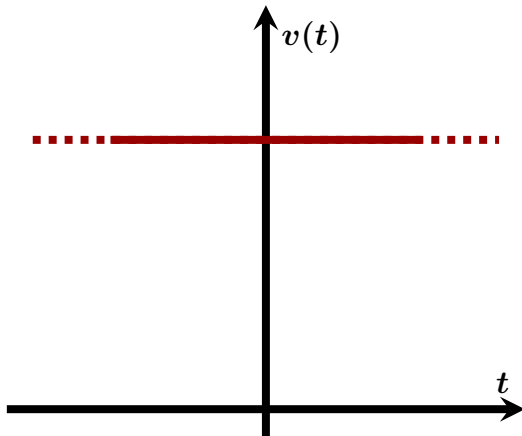


Further visualization

A good representation of a square wave signal as an animation is on wikipedia. Lean back and watch!

Examples of Fourier Series

DC

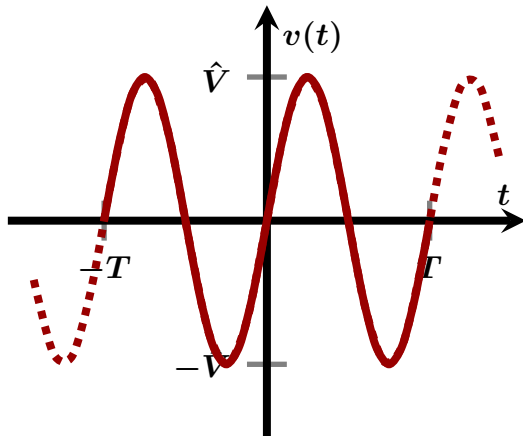


Fourier Coefficients

- $a_0 = V$
- $a_n = 0 \forall n$
- $b_n = 0 \forall n$

Examples of Fourier Series

Sine wave

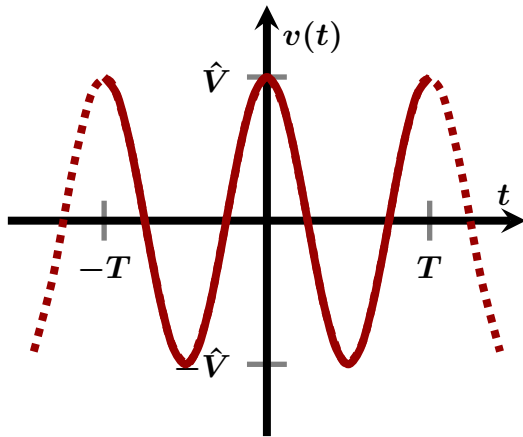


Fourier Coefficients

- $a_0 = 0$
- $a_n = 0 \forall n$
- $b_1 = \hat{V}$
- $b_n = 0 \forall n \neq 1$

Examples of Fourier Series

Cosine wave

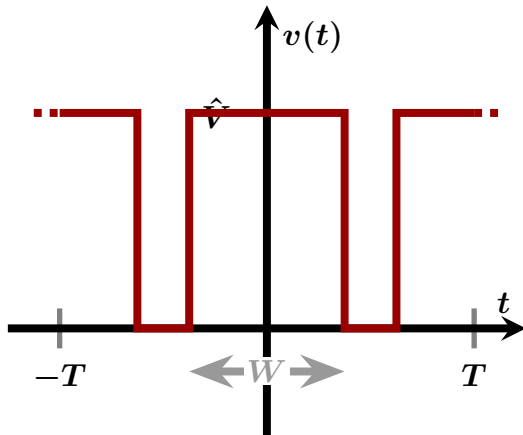


Fourier Coefficients

- $a_0 = 0$
- $a_1 = \hat{V}$
- $a_n = 0 \forall n \neq 1$
- $b_n = 0 \forall n$

Examples of Fourier Series

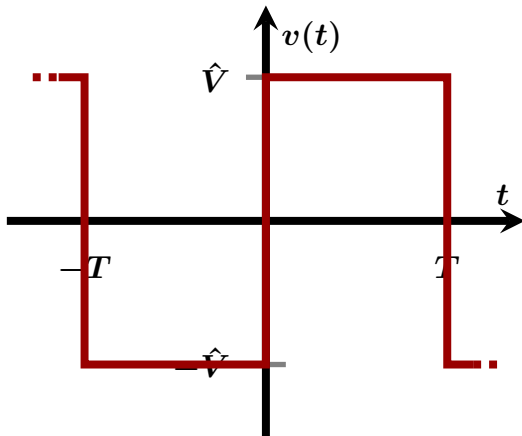
Pulses



Fourier Coefficients

- $a_0 = \frac{W}{T} \hat{V}$
- $a_n = \frac{2}{n\pi} \hat{V} \sin \left(n\pi \frac{W}{T} \right)$
- $b_n = 0 \quad \forall n$

Square wave

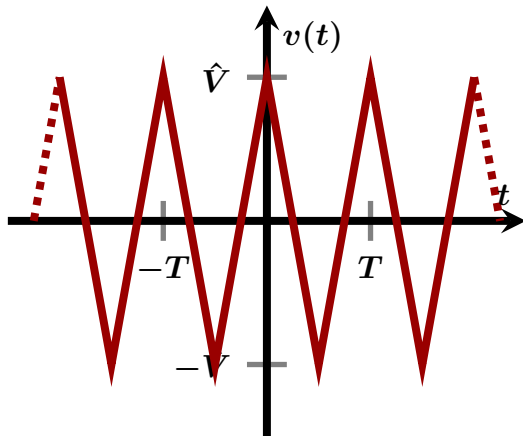


Fourier Coefficients

- $a_0 = 0$
- $a_n = 0 \quad \forall n$
- $b_n = \frac{4}{n\pi} \hat{V} \quad \forall n = 2m - 1$
where $m \in \mathbb{N}$
- $b_n = 0 \quad \forall n = 2m$
where $m \in \mathbb{N}$

Examples of Fourier Series

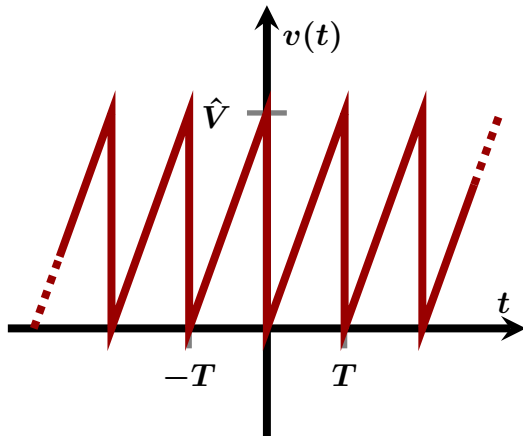
Triangle



Fourier Coefficients

- $a_0 = 0$
- $a_n = 0 \quad \forall n = 2m$
where $m \in \mathbb{N}$
- $a_n = -\frac{8}{(n\pi)^2} \hat{V}$
 $\forall n = 2m - 1$
where $m \in \mathbb{N}$
- $b_n = 0 \quad \forall n$

Sawtooth

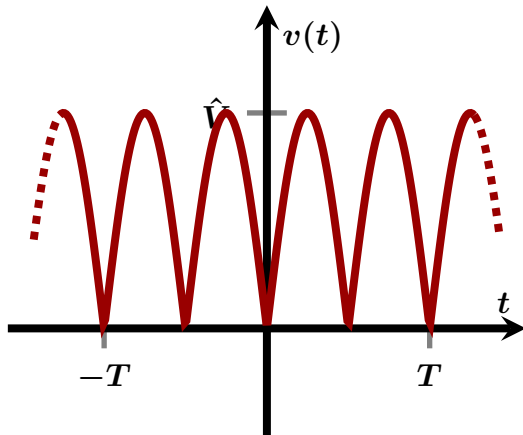


Fourier Coefficients

- $a_0 = \frac{\hat{V}}{2}$
- $a_n = 0 \quad \forall n$
- $b_n = -\frac{1}{n\pi} \hat{V} \quad \forall n$

Examples of Fourier Series

Full wave rectified Sine



Fourier Coefficients

- $a_0 = \frac{2}{\pi} \hat{V}$

- $a_n = \frac{4}{\pi(1-n^2)} \hat{V}$

$$\forall n = 2m$$

where $m \in \mathbb{N}$

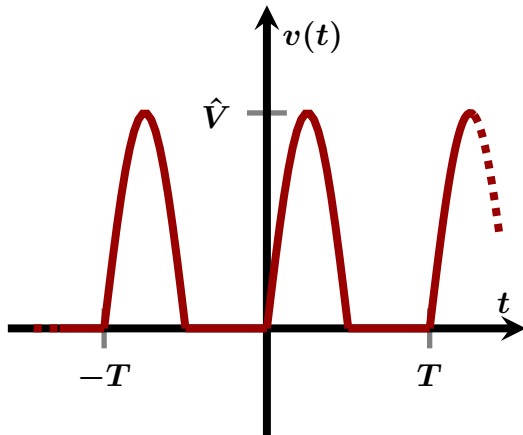
- $a_n = 0 \forall n = 2m - 1$

where $m \in \mathbb{N}$

- $b_n = 0 \forall n$

Examples of Fourier Series

Half wave rectified Sine



Fourier Coefficients

- $a_0 = \frac{1}{\pi} \hat{V}$

- $a_n = \frac{2}{\pi(1-n^2)} \hat{V}$

$$\forall n = 2m$$

where $m \in \mathbb{N}$

- $a_n = 0 \forall n = 2m - 1$

where $m \in \mathbb{N}$

- $b_1 = \frac{1}{2} \hat{V}$

- $b_n = 0 \forall n \neq 1$

EK2

IMPORTANT Definition

Using $a_i \cos(n\omega t) + b_i \sin(n\omega t) = A_i \cos(n\omega t + \varphi_i)$,

$$\Rightarrow v(t) = A_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega t + \varphi_n)$$

where

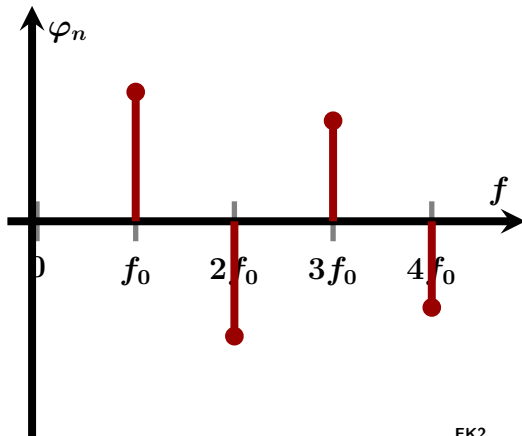
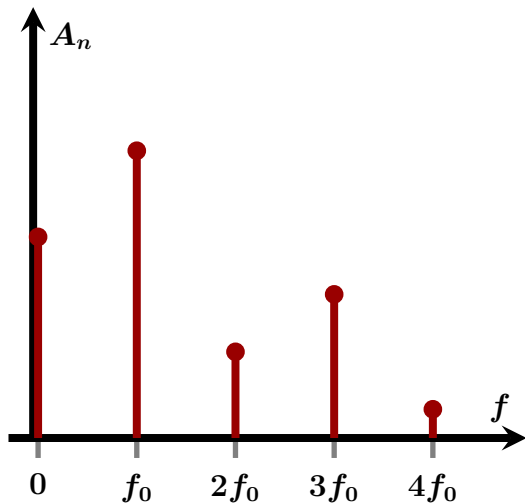
- the amplitudes $A_n = \sqrt{a_n^2 + b_n^2}$ and
- the phases $\varphi_n = \arctan\left(-\frac{b_n}{a_n}\right)$

are called the

- amplitude spectrum and
- phase spectrum

respectively.

Examples of a Spectrum



Useful plotting options

- The frequency axis for both the amplitude and the phase spectrum is plotted logarithmically.
- The amplitude (y-axis) is plotted in dB.
- The phase (y-axis) is plotted linear.

Doubling by approximately two

$$\begin{aligned} 0 \text{ dB V} &= 20 \log \left(\frac{1 \text{ V}}{1 \text{ V}} \right) \\ +6 \text{ dB V} \downarrow &\quad \downarrow \approx \cdot 2 \quad (5) \\ 6 \text{ dB V} &\approx 20 \log \left(\frac{2 \text{ V}}{1 \text{ V}} \right) \end{aligned}$$

Ten folding

$$\begin{aligned} 0 \text{ dB V} &= 20 \log \left(\frac{1 \text{ V}}{1 \text{ V}} \right) \\ +20 \text{ dB V} \downarrow &\quad \downarrow \cdot 10 \quad (6) \\ 20 \text{ dB V} &= 20 \log \left(\frac{10 \text{ V}}{1 \text{ V}} \right) \end{aligned}$$

Other metric prefixes

$$\begin{aligned} -60 \text{ dB V} &= 20 \log \left(\frac{1 \text{ mV}}{1 \text{ V}} \right) = \\ 0 \text{ dB mV} &= 20 \log \left(\frac{1 \text{ mV}}{1 \text{ mV}} \right) = \\ 60 \text{ dB } \mu\text{V} &= 20 \log \left(\frac{1 \text{ mV}}{1 \mu\text{V}} \right) = \\ 120 \text{ dB nV} &= 20 \log \left(\frac{1 \text{ mV}}{1 \text{ nV}} \right) \end{aligned} \tag{7}$$

Fourier Series

Exercise 1.15 Plot waveforms

Use your favourite math tool to plot the waveform as a function of time t of the waveforms in the above (DC, sine, cosine, pulses, square, triangle, sawtooth, half-wave rectified sine and full-wave rectified sine) as a superposition of sin and cos waves via Fourier Series. Wherever necessary make assumptions about values or normalize. Play around with the number of harmonics you take into account and observe how this affects the results.

Exercise 1.16 Plot spectrum

Use your favourite math tool to plot the spectrum (A_n and ϕ_n) as a function of frequency f from the Fourier coefficients of the above waveforms. Wherever necessary make assumptions about values or normalize. Play around with linear and logarithmic scaling of both axes and see what happens.

Exercise 1.17 How many dB μ V is

- 1 V,
- 2 V,
- 10 V,
- 1 mV,
- 2 mV,
- 10 mV,
- 1 μ V,
- 2 μ V &
- 10 μ V?

Exercise 1.18 How many V is

- 0 dB μ V,
- 12 dB μ V,
- 40 dB μ V,
- 46 dB μ V,
- 52 dB μ V,
- 60 dB μ V,
- 72 dB μ V,
- 80 dB μ V &
- 160 dB μ V?

Typical Symbols

generic

DC

AC

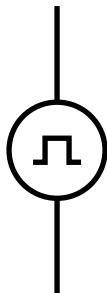
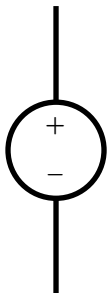
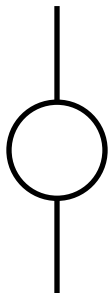
pulse

square

sawtooth

triangle

line



Types of Sources in Spice Simulators

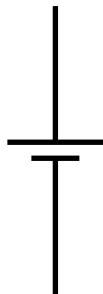
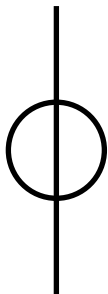
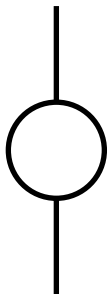
- DC
- Pulse
- Sine
- Exponential (EXP)
- Frequency modulation (SFFM)
- Arbitrary (PWL)
- Arbitrary from external file

Symbols for Voltage Sources

generic

voltage

battery

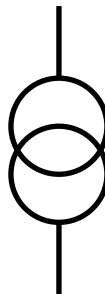
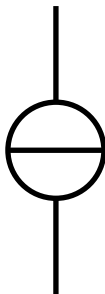


Symbols for Current Sources

current

current

current photovoltaic



Some Symbol Examples

generic

generic

current

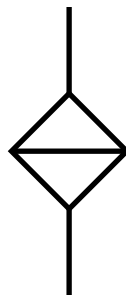
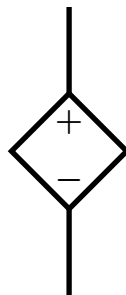
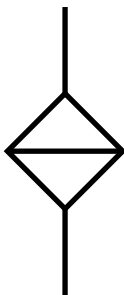
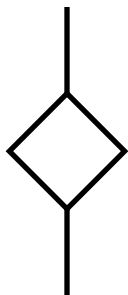
current

current

voltage

voltage

voltage



Dependent Voltage Sources

- voltage controlled
in Spice: "E"
application: operational amplifier
- current controlled
in Spice "H"
(application: transimpedance amplifier)

Dependent Current Sources

- voltage controlled
in Spice: "G"
application: Metal Oxid Semiconductor Field Effect Transistor (MOSFET),
transconductance amplifier
- current controlled
in Spice "F"
application: bipolar transistor (BJT)

IMPORTANT Definition

- DC: $P = VI$
- AC: $p(t) = v(t)i(t)$

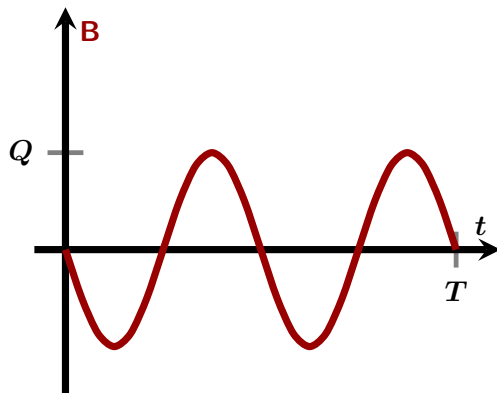
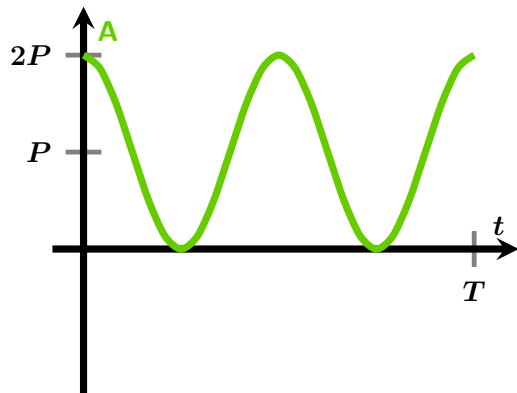
for sinusoidal signals

- $v(t) = V_p \cos(\omega t + \varphi)$
 - $i(t) = I_p \cos(\omega t)$
- $$p(t) = V_p I_p \cos(\omega t + \varphi) \cos(\omega t)$$

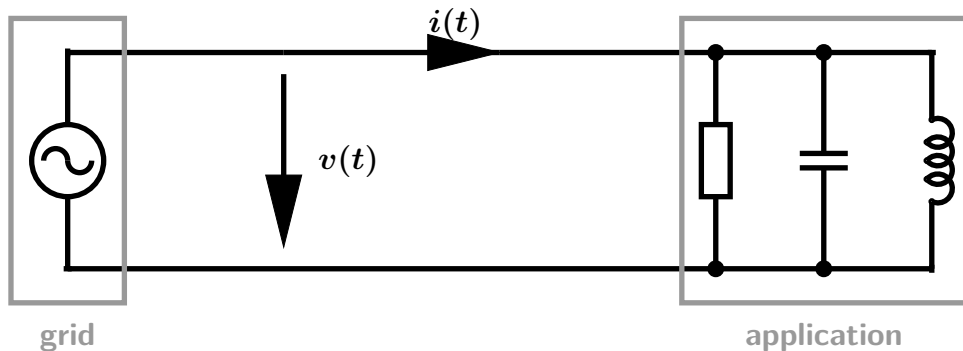
Derivation for sinusoidal signals

$$\begin{aligned}
 p(t) &= V_p I_p \cos(\omega t + \varphi) \cos(\omega t) \\
 &= \dots \text{see [1]} - \text{uses trigonometric equations} \\
 &= \underbrace{\frac{V_p I_p}{2} \cos(\varphi)}_{\text{DC-part of } p(t)} \\
 &\quad + \underbrace{\frac{V_p I_p}{2} [\cos(\varphi) \cos(2\omega t) - \sin(\varphi) \sin(2\omega t)]}_{\text{AC-part of } p(t)} \tag{8} \\
 &= \underbrace{P(1 + \cos(2\omega t))}_{\text{A}} - \underbrace{Q \sin(2\omega t)}_{\text{B}} \\
 &\text{with } P = \frac{V_p I_p}{2} \cos(\varphi) \text{ and } Q = \frac{V_p I_p}{2} \sin(\varphi)
 \end{aligned}$$

$p(t)$ split into parts A and B



Power transfer from source (generator) to sink (application/user)



IMPORTANT Definitions

- P is the part of the power, that is transferred from source to sink and is called active power or real power.
- Q is the part of the power, that is bouncing back and forth between source and sink and is called reactive power or imaginary power.
- $p(t)$ is called the instantaneous power or complex power.

Units for Power

$$[p(t)] = \text{VA}, [P] = \text{W}, [Q] = \text{Var}$$

Derivation of Power expressed through Phasors

$$\underline{V} = V_{RMS} e^{j\varphi_V}$$

$$\underline{I} = I_{RMS} e^{j\varphi_I}$$

$$\underline{S} = \underline{V} \underline{I}^* = V_{RMS} I_{RMS} e^{j(\varphi_V - \varphi_I)} \quad (9)$$

with $\varphi_I = 0 \Rightarrow \varphi_V = \varphi$

$\Downarrow$
 $\equiv$

$$V_{RMS} I_{RMS} e^{j\varphi}$$
$$= V_{RMS} I_{RMS} \cos(\varphi) + j V_{RMS} I_{RMS} \sin(\varphi)$$

for sinusoidal signals $V_{RMS} = \frac{V_p}{\sqrt{2}}$ and $I_{RMS} = \frac{I_p}{\sqrt{2}}$.

IMPORTANT Definition of Complex Power

$$\underline{S} = P + jQ \quad (10)$$

IMPORTANT Power Factor

$$\lambda = \frac{P}{|S|} = \cos(\varphi) \quad (11)$$

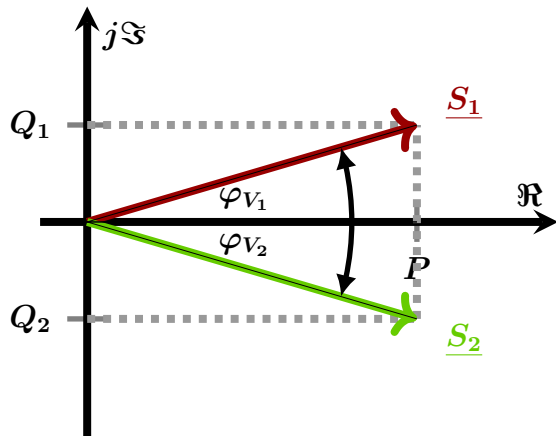
Meaning of Power Factor

The Power Factor is a measure for the amount of power, that is delivered to the load, with respect to the total amount of power, that the source is delivering.

Optimization of Power Factor

Ideally the Power Factor is as close to 1 as possible, i.e. P is maximized and Q is minimized, as Q is also loading the grid (e.g. cables) and generates losses in the resistances without contributing to the power transfer.

Power Factor - graphically



Two scenarios

- $\underline{S}_1$: $\varphi_{V_1} > 0$

Current / Power Factor is lagging

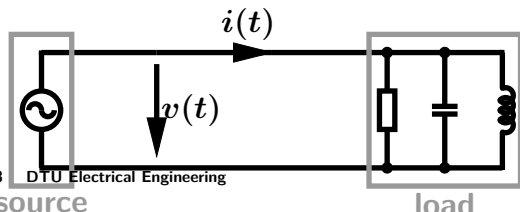
- $\underline{S}_2$: $\varphi_{V_2} < 0$

Current / Power Factor is leading

Exercise 1.19 Power in Time Domain

What is the average power, the reactive power and the instantaneous power delivered to the load for

- ❶ $v(t) = 1500 \text{ V} \cos(\omega t - 45^\circ)$, $i(t) = 2 \text{ A} \cos(\omega t + 50^\circ)$
- ❷ $v(t) = 90 \text{ V} \cos(\omega t + 60^\circ)$, $i(t) = 10.5 \text{ A} \cos(\omega t - 20^\circ)$
- ❸ $v(t) = 135 \text{ V} \cos(\omega t)$, $i(t) = 2 \text{ A} \cos(\omega t + 30^\circ)$
- ❹ $v(t) = 370 \text{ V} \sin(\omega t)$, $i(t) = 10 \text{ A} \cos(\omega t + 20^\circ)$



Exercise 1.20 Power in Frequency Domain

What is the average power and the reactive power for the following sets of voltage and current phasors? What is the power factor and is it lagging or leading?

① $\underline{V} = 250 \text{ V}_{\text{RMS}}, \underline{I} = 0.25 \text{ A}_{\text{RMS}} \angle -15^\circ$

② $\underline{V} = 120 \text{ V}_{\text{RMS}} e^{j135^\circ}, \underline{I} = 12.5 \text{ A}_{\text{RMS}} e^{j165^\circ}$

③ $\underline{V} = 120 \text{ V}_{\text{RMS}} e^{j30^\circ}, \underline{I} = 3.3 \text{ A}_{\text{RMS}} e^{j-15^\circ}$

④ $\underline{V} = 480 \text{ V}_{\text{RMS}} \angle 45^\circ, \underline{I} = 8.5 \text{ A}_{\text{RMS}} \angle 90^\circ$

[1]

Three Phase Circuits

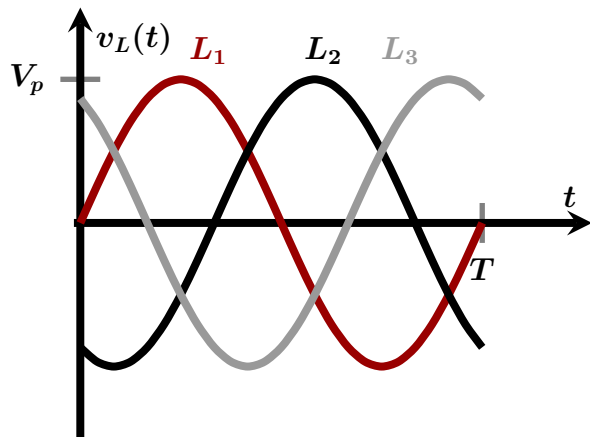
IMPORTANT Signals in Three Phase Systems

$$\begin{aligned}v_{L1} &= V_p \cos(\omega t) & \underline{V}_{L1} &= V_p \\v_{L2} &= V_p \cos(\omega t - 120^\circ) & \underline{V}_{L2} &= V_p e^{-j120^\circ} \\v_{L3} &= V_p \cos(\omega t + 120^\circ) & \underline{V}_{L3} &= V_p e^{j120^\circ}\end{aligned}\quad (12)$$

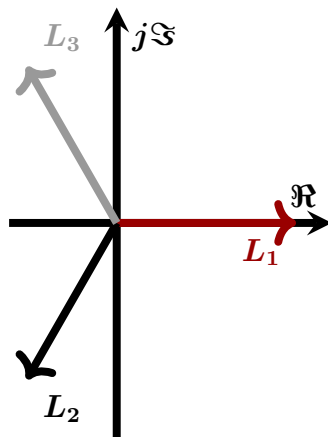
Typical Nomenclature

- L_1 , L_2 and L_3 : line 1, 2 and 3 (in electrical installations typically **brown**, **black** and **grey**)
- N : neutral (in electrical installations typically **blue**)
- PE : protection earth, ground (in electrical installations typically **yellow-green**)

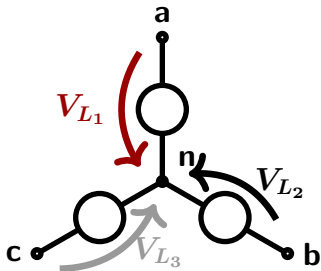
Three Phase Signals in Time Domain



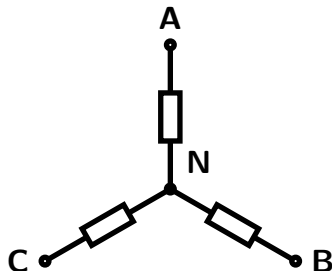
Three Phase Signals as Phasors



Star Connection of Sources



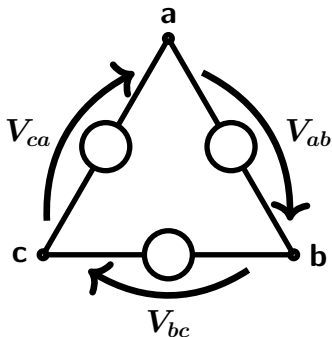
Star Connection of Loads



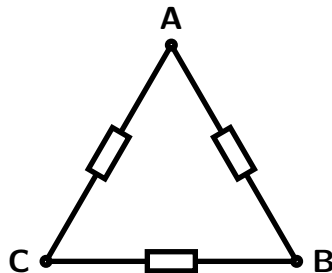
Connections

$a \rightarrow A, b \rightarrow B, c \rightarrow C$ and $n \rightarrow N$

Triangle Connection of Sources



Triangle Connection of Loads

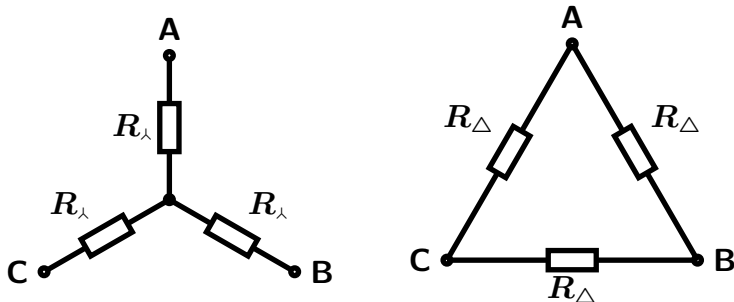


Connections

$a \rightarrow A$, $b \rightarrow B$ and $c \rightarrow C$

Conversion between Star and Triangle

for balanced loads:



$$R_\Delta = 3R_Y$$

(13)

Three Phase Circuits

Exercise 1.21 Phasors in Three Phase Systems

In a symmetrical star connected three phase circuit, the first line voltage is $v_{L_1}(t) = 200 \text{ V} \cos(\omega t + 120^\circ)$.

- 1 What are the expressions for $v_{L_2}(t)$, $v_{L_3}(t)$, $v_{ab}(t)$, $v_{bc}(t)$ and $v_{ca}(t)$?
- 2 Sketch all 6 voltages in a phasor diagram.

Exercise 1.22 Phasors in Three Phase Systems

In a symmetrical triangle connected three phase system, one voltage is $v_{an}(t) = 150 \text{ V} \cos(400\pi t)$.

- 1 What is the frequency in Hz?
- 2 What are the expressions for $v_{bn}(t)$ and $v_{cn}(t)$

Section 2

Components

- **Components**
 - Resistors
 - Inductors
 - Capacitors
 - Complex Impedances and Admittances
 - Power in Components
 - Energy in Components

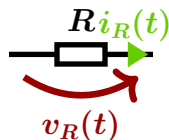
Physical background

$$R = \rho \frac{l}{A} \quad (14)$$

Ohms law

$$V = RI \quad (15)$$

Symbol



Resistors dependency on material and geometry

- The resistance R depends on the resistors

- material, where

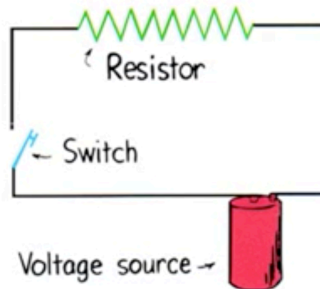
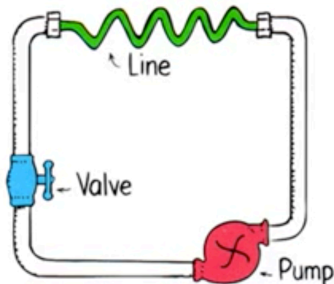
ρ is the resistivity of the material and

- its geometry, with

l being the length of the conductor and

A is the diameter of the conductor.

Analogy



hydraulic $\Leftrightarrow$ electric
water pressure $\Leftrightarrow$ voltage
sand in the water hose $\Leftrightarrow$ resistivity
(flow) current $\Leftrightarrow$ (electrical) current

Voltage across and Current through a Resistor in Time Domain - analytically

$$\begin{aligned} i_R(t) &\stackrel{!}{=} I_R \cos(\omega t + \varphi) \\ v_R(t) &= R i_R(t) = R I_R \cos(\omega t + \varphi) \end{aligned} \quad (16)$$

Voltage across and Current through a Resistor as Phasors - analytically

$$\begin{aligned} \underline{I}_R &= I_R e^{j\varphi} \\ \underline{V}_R &= R \underline{I}_R = R I_R e^{j\varphi} \end{aligned} \quad (17)$$

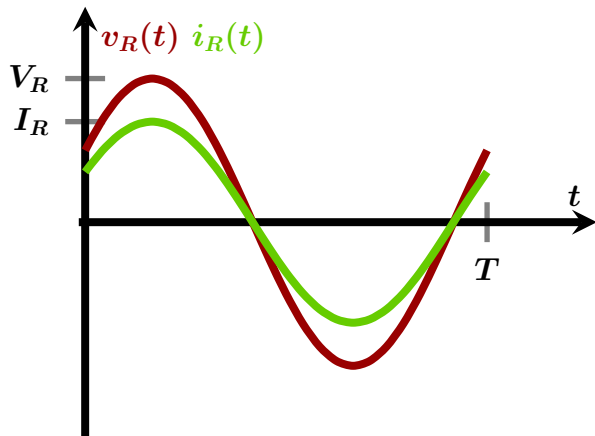
IMPORTANT Impedance of a Resistor

$$\underline{Z}_R = R \quad (18)$$

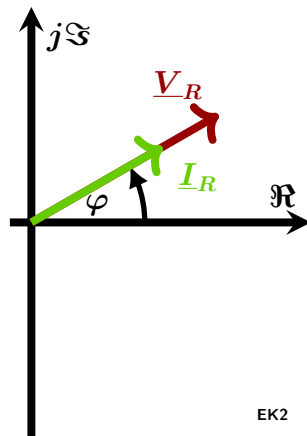
IMPORTANT Ohms Law applied to a Resistor

$$\underline{V}_R = R \underline{I}_R \quad (19)$$

Voltage across and Current through a Resistor in Time Domain - graphically



Voltage across and Current through a Resistor as Phasors - graphically



Analogy



- like the current in a watermill the current in an inductor can not abruptly change
- changing the pressure (voltage) changes the current flow gradually

Components

Inductors

Faradays induction law

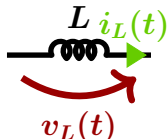
$$N\phi = \int V dt \quad (20)$$

$$\vec{B} = \frac{\phi}{A_{eff}} = \frac{\int V dt}{N A_{eff}} \Rightarrow V = N A_{eff} \frac{d\vec{B}}{dt} \quad (21)$$

Amperes law

$$NI = \int_0^{l_{eff}} \vec{H} dl \Rightarrow I = \frac{\int \vec{H} dl}{N} \quad (22)$$

Symbol



Nomenclature

- N number of turns
- ϕ magnetic flux
- $\vec{B}$ flux density
- A_{eff} effective cross section
- l_{eff} effective length
- $\vec{H}$ magnetic field

Ohms law applied to inductors

$$L = \frac{\int V dt}{I} = \frac{\int N A_{eff} \frac{d\vec{B}}{dt} dt}{\frac{\int_0^{l_{eff}} \vec{H} dl}{N}} = N^2 A_{eff} \frac{\vec{B}}{\int_0^{l_{eff}} \vec{H} dl} \quad (23)$$

$$L = \underbrace{\frac{\vec{B}}{\vec{H}}}_{\substack{\mu \\ A_L \text{ value}}} \frac{A_{eff}}{l_{eff}} N^2 \text{ (for homogeneous and linear materials)} \quad (24)$$

IMPORTANT Inductance dependency on material and geometry - or the engineering A_L value

$$L = \mu \frac{A_{eff}}{l_{eff}} N^2 = A_L N^2 \quad (25)$$

Voltage across and Current through an Inductor in Time Domain - analytically

$$\begin{aligned} i_L(t) &\stackrel{!}{=} I_L \cos(\omega t + \varphi) \\ v_L(t) &= L \frac{di_L(t)}{dt} = -\omega L I_L \sin(\omega t + \varphi) = \omega L I_L \cos(\omega t + \varphi + 90^\circ) \end{aligned} \quad (26)$$

Voltage across and Current through an Inductor as Phasors - analytically

$$\begin{aligned} i_L(t) &= \Re \left(\underbrace{I_L e^{j\varphi}}_{\underline{I}_L} e^{j\omega t} \right) \\ v_L(t) &= \Re \left(L \frac{d}{dt} I_L e^{j\varphi} e^{j\omega t} \right) = \Re \left(\underbrace{j\omega L I_L e^{j\varphi}}_{\underline{V}_L} e^{j\omega t} \right) = \Re \left(\underbrace{\omega L I_L e^{j(\varphi+90^\circ)}}_{\underline{V}_L} e^{j\omega t} \right) \end{aligned} \quad (27)$$

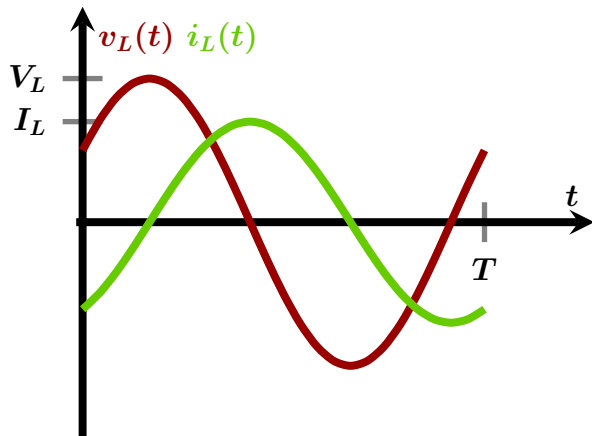
IMPORTANT Impedance of an Inductor

$$\underline{Z}_L = j\omega L \quad (28)$$

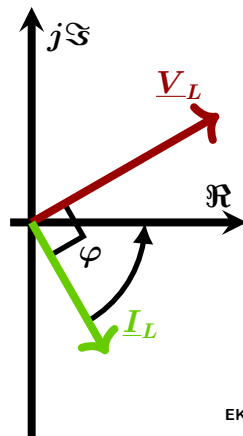
IMPORTANT Ohms Law applied to an Inductor

$$\underline{V}_L = j\omega L \underline{I}_L \quad (29)$$

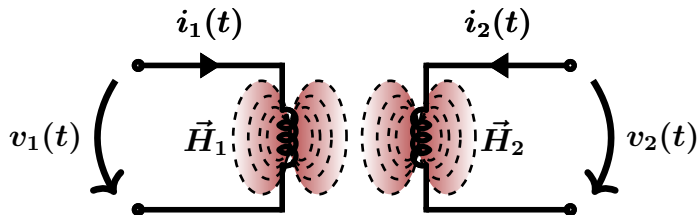
Voltage across and Current through an Inductor in Time Domain - graphically



Voltage across and Current through an Inductor as Phasors - graphically



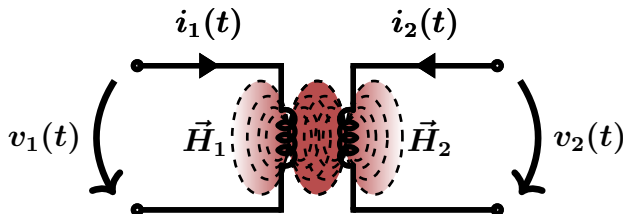
Fields of two distant inductors



The currents $i_1(t)$ and $i_2(t)$ through each inductor generate their own magnetic field H_1 and H_2 .

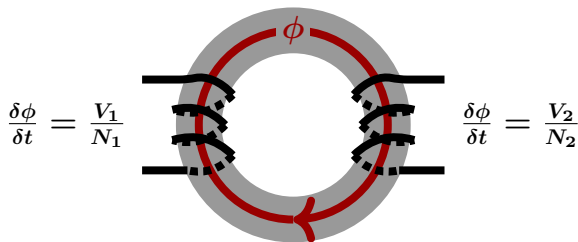
Coupled Inductors

Fields of two proximate inductors



The magnetic field H_1 of the left inductor, alters the magnetic field H_2 of the right inductor and therefore influences its corresponding current $i_2(t)$ and vice versa.

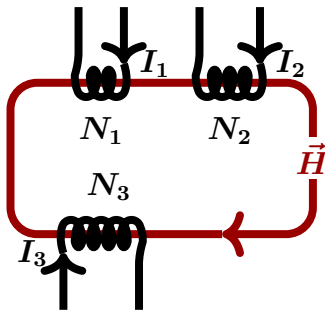
Faradays law



IMPORTANT Constant turns voltage

$$\frac{V_1}{N_1} = \frac{V_2}{N_2} = \dots = \frac{V_n}{N_n} = \frac{\delta\phi}{\delta t}$$

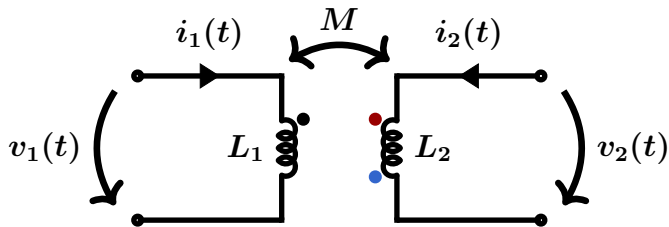
Amperes law



IMPORTANT Sum of all turns-current products is equal to externally provided magnetic flux (or zero if there is no external ϕ)

$$\sum_{n=1}^{\infty} N_n I_n = \vec{H} \delta l = \frac{\phi}{A_L}$$

Mutual Inductance



Note, the dot convention is defining the direction each winding is put across the core. Typically the dot indicates the start of a winding.

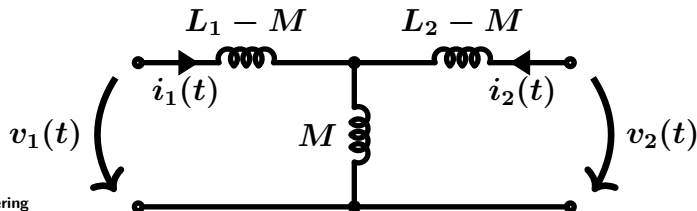
Transformer Equations in the Time Domain

$$\begin{aligned} v_1(t) &= L_1 \frac{di_1(t)}{dt} + / - M \frac{di_2(t)}{dt} \\ v_2(t) &= L_2 \frac{di_2(t)}{dt} + / - M \frac{di_1(t)}{dt} \end{aligned} \quad (30)$$

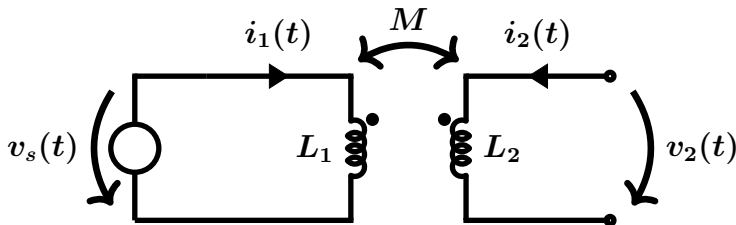
Transformer Equations in the Frequency Domain

$$\begin{aligned} \underline{V}_1 &= j\omega L_1 \underline{I}_1 + / - j\omega M \underline{I}_2 \\ \underline{V}_2 &= j\omega L_2 \underline{I}_2 + / - j\omega M \underline{I}_1 \end{aligned} \quad (31)$$

T-Network Model of a Transformer



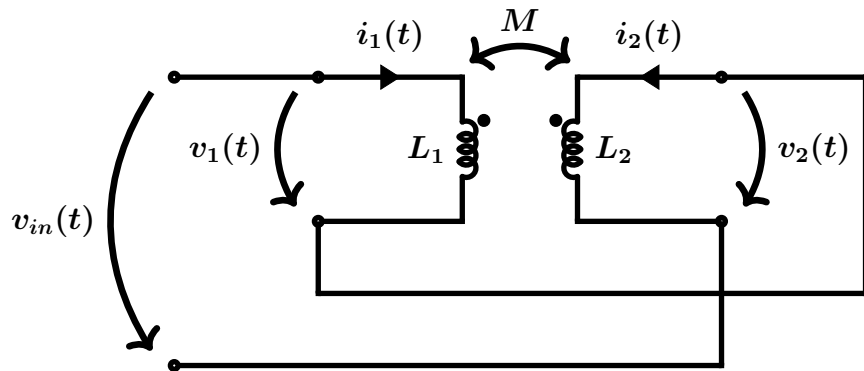
Exercise 2.1 Transformer Equations



In the above circuit $L_1 = 10 \text{ mH}$, $L_2 = 5 \text{ mH}$, $M = 7 \text{ mH}$ and $v_s(t) = 200 \text{ V} \sin(100t)$.

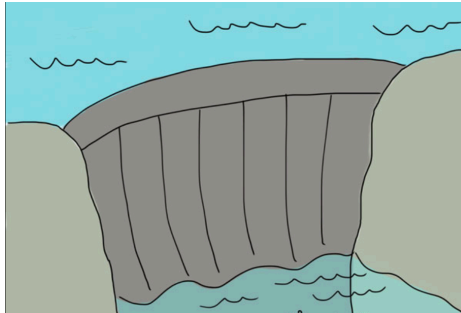
- a** What are the $i - v$ relationships for the coupled inductors?
- b** What is $v_2(t)$, when the terminal on the secondary side is open?

Exercise 2.2 Connecting Primary and Secondary



In the above circuit $L_1 = L_2 = 6 \text{ mH}$, $M = 4 \text{ mH}$ and $i_1(t) = 5 \text{ A} \sin(1000t)$. What is $v_{in}(t)$?

Analogy



- like a dam withstands the pressure from the water, a capacitor sustains voltage
- a current flow changes the water level (voltage)

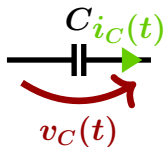
Gaus law

$$q = \oint_A \vec{D} d\sigma \quad (32)$$

Faradays law

$$V = \oint_l \vec{E} ds \quad (33)$$

Symbol



Nomenclature

- q charge
- $\vec{D}$ displacement field
- A surface
- σ 2D path along surface
- $\vec{E}$ electric field
- l length of a loop
- s path along loop

Ohms law applied to capacitors

$$C = \frac{\int I dt}{V} = \frac{\int \frac{dq}{dt} dt}{V} = \frac{q}{V} = \frac{\oint_A \vec{D} d\sigma}{\oint_l \vec{E} ds} \quad (34)$$

which gives for homogeneous and linear materials:

$$C = \underbrace{\frac{\vec{D}}{\vec{E}}}_{\epsilon} \frac{A}{d} \quad (35)$$

IMPORTANT Capacitance dependency on material and geometry

$$C = \frac{q}{V} = \epsilon \frac{A}{d} \quad (36)$$

where A is the area of each plate and d is the distance between the plates.

Voltage across and Current through a Capacitor in Time Domain - analytically

$$\begin{aligned} v_C(t) &\stackrel{!}{=} V_C \cos(\omega t + \varphi) \\ i_C(t) &= C \frac{dv_C(t)}{dt} = -\omega C V_C \sin(\omega t + \varphi) = \omega C V_C \cos(\omega t + \varphi + 90^\circ) \end{aligned} \quad (37)$$

Voltage across and Current through a Capacitor as Phasors - analytically

$$\begin{aligned} v_C(t) &= \Re \left(\underbrace{V_C e^{j\varphi}}_{\underline{V}_C} e^{j\omega t} \right) \\ i_C(t) &= \Re \left(C \frac{d}{dt} V_C e^{j\varphi} e^{j\omega t} \right) = \Re \left(\underbrace{j\omega C V_C e^{j\varphi}}_{\underline{I}_C} e^{j\omega t} \right) = \Re \left(\underbrace{\omega C V_C e^{j(\varphi+90^\circ)}}_{\underline{I}_C} e^{j\omega t} \right) \end{aligned} \quad (38)$$

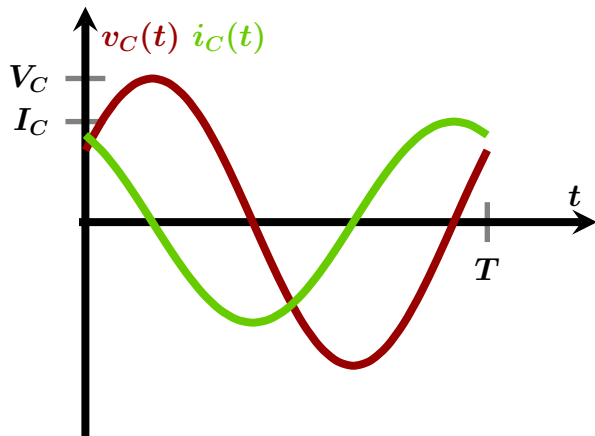
IMPORTANT Impedance of a Capacitor

$$\underline{Z}_C = \frac{1}{j\omega C} \quad (39)$$

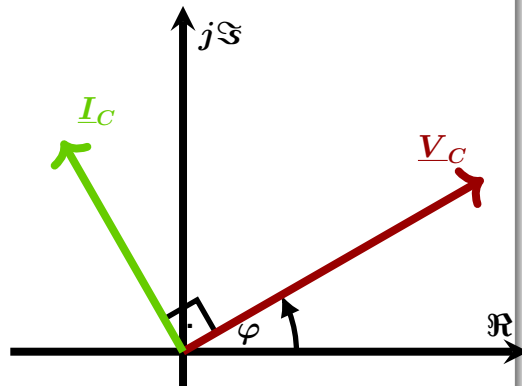
IMPORTANT Ohms Law applied to a Capacitor

$$\underline{V}_C = \frac{1}{j\omega C} \underline{I}_C \quad (40)$$

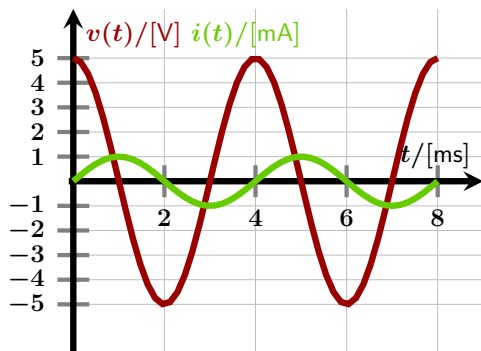
Voltage across and Current through a Capacitor in Time Domain - graphically



Voltage across and Current through a Capacitor as Phasors - graphically

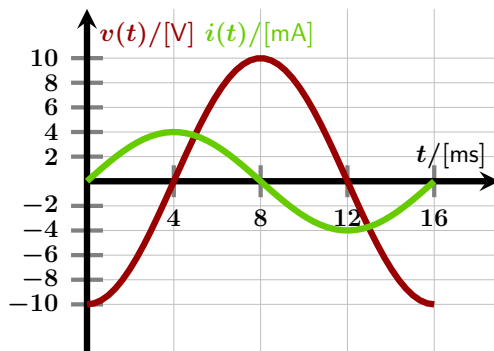


Exercise 2.3 Voltage across and Current through a Component



What is the value of the component?

Exercise 2.4 Voltage across and Current through a Component



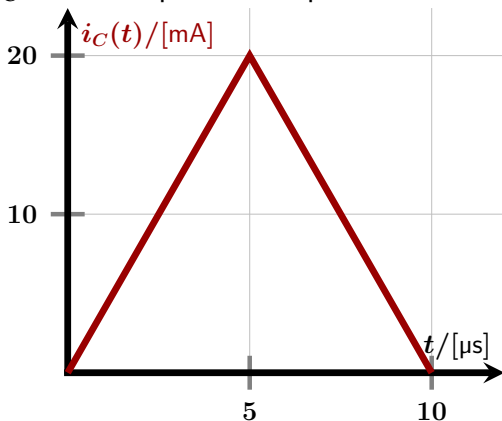
What is the value of the component?

Exercise 2.5 Capacitance

For $t \geq 0$ the current through a capacitor is $i_C(t) = 10 \text{ mA}r(t)$, where $r(t)$ is a unity ramp. At $t_0 = 0$, the capacitor voltage $v_C(t_0) = 3 \text{ V}$ and at $t_1 = 1 \text{ ms}$, it is $v_C(t_1) = 8 \text{ V}$. What is the capacitance of the component?

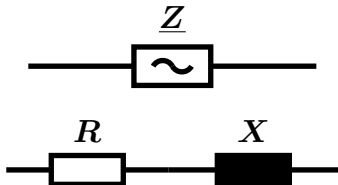
Exercise 2.6 Capacitor Voltage

The waveform below shows the current through a $C = 10 \text{ nF}$ capacitor. At time t_0 , the voltage across the capacitor is $v_C(t) = -5 \text{ V}$. What is the voltage across the capacitor v_C at $t = 5 \mu\text{s}$, $t = 10 \mu\text{s}$ and $t = 20 \mu\text{s}$?



Complex Impedances and Admittances

Symbol



IMPORTANT Impedance

$$\underline{Z} = R + jX = |\underline{Z}|e^{j\angle \underline{Z}} \quad (41)$$

Nomenclature

- $\underline{Z}$ Impedance
- R Resistance
- X Reactance
- $|\underline{Z}|$ Amplitude of Impedance
- $\angle \underline{Z}$ Phase of Impedance

Complex Impedances and Admittances

IMPORTANT Admittance

$$\underline{Y} = G + jB = |\underline{Y}|e^{j\angle Y} \quad (42)$$

IMPORTANT Impedance and Admittance

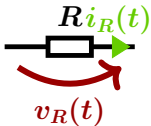
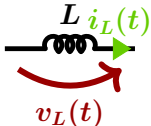
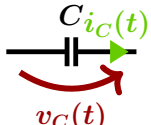
$$\underline{Z} = \frac{1}{\underline{Y}} \quad (43)$$

Nomenclature

- $\underline{Y}$ Admittance
- G Conductance
- B Susceptance
- $|\underline{Y}|$ Amplitude of Admittance
- $\angle Y$ Phase of Admittance

Complex Impedances and Admittances

Components, Signals, Impedances and Admittances

Component	Ohms law	Impedance	Admittance
	$\underline{V}_R = R \underline{I}_R$	$\underline{Z}_R = R$	$\underline{Y}_R = \frac{1}{R} = G$
	$\underline{V}_L = j\omega L \underline{I}_L$	$\underline{Z}_L = j\omega L = jX_L$	$\underline{Y}_L = \frac{1}{j\omega L} = jB_L$ with $B_L = -\frac{1}{\omega L}$
	$\underline{V}_C = \frac{1}{j\omega C} \underline{I}_C$	$\underline{Z}_C = \frac{1}{j\omega C} = jX_C$ with $X_C = -\frac{1}{\omega C}$	$\underline{Y}_C = j\omega C = jB_C$

Complex Impedances and Admittances

Exercise 2.7 Impedance of an Inductor

 /  15 min

The voltage $v_L(t) = 10 \text{ V} \cos(2000\pi t)$ is applied across an $L = 100 \text{ mH}$ inductance.

- a** What is the complex impedance of the inductor?
- b** What are the voltage and current phasors? How does the phasor diagram look like?
- c** Which analytical function describes the current? How do the voltage and current waveforms look like?
- d** What is the phase relationship between current and voltage?

Complex Impedances and Admittances

Exercise 2.8 Impedance of a Capacitor

 /  15 min

The voltage $v_C(t) = 10 \text{ V} \cos(2000\pi t)$ is applied across a $C = 10 \mu\text{F}$ capacitance.

- a** What is the complex impedance of the capacitor?
- b** What are the voltage and current phasors? How does the phasor diagram look like?
- c** Which analytical function describes the current? How do the voltage and current waveforms look like?
- d** What is the phase relationship between current and voltage?

Power in Resistors

Derivation

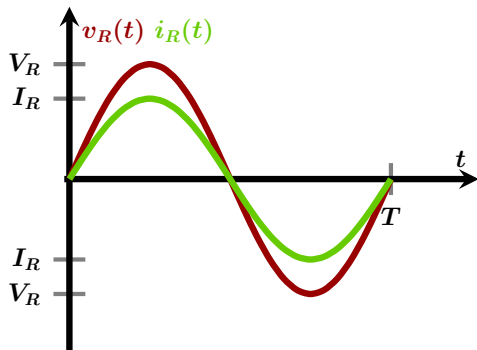
$$\begin{aligned}
 i_R(t) &= I_R \cos(\omega t) \\
 p_R(t) &= v_R(t)i_R(t) = \frac{v_R^2(t)}{R} = Ri_R^2(t) \\
 &= RI_R^2 \cos^2(\omega t) \stackrel{\cos^2 \alpha = \frac{1}{2}(1 + \cos(2\alpha))}{=} R \underbrace{\frac{I_R^2}{2}}_{I_{RMS}^2} (1 + \cos(2\omega t)) \\
 &= \frac{V_{RMS}^2}{R} (1 + \cos(2\omega t))
 \end{aligned} \tag{44}$$

IMPORTANT Average Power in Resistors

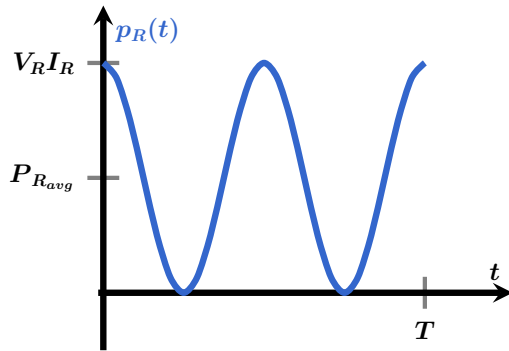
$$P_{R_{avg}} = \frac{1}{T} \int_0^T p_R(t) dt = RI_{RMS}^2 = \frac{V_{RMS}^2}{R} \quad Q_R = 0 \tag{45}$$

Power in Resistors

Voltage and Current



Instantaneous Power



Power in Inductors

Derivation

$$v_L(t) = V_L \cos(\omega t) \quad \Rightarrow \quad i_L(t) = I_L \sin(\omega t) \quad (46)$$

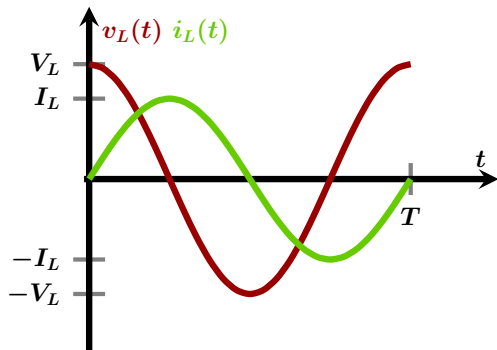
$$p_L(t) = v_L(t)i_L(t) = V_L I_L \cos(\omega t) \sin(\omega t)$$
$$\begin{array}{c} \cos(\alpha) \sin(\alpha) = \frac{1}{2} \sin(2\alpha) \\ \Downarrow \\ \frac{V_L I_L}{2} \sin(2\omega t) \end{array} \quad (47)$$

IMPORTANT Average Power in Inductors

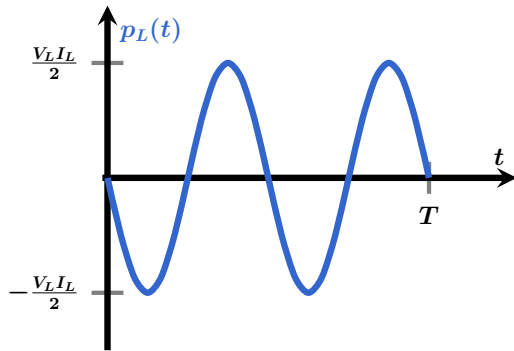
$$P_{L_{avg}} = \frac{1}{T} \int_0^T p_L(t) dt = 0 \quad (48)$$

Power in Inductors

Voltage and Current



Instantaneous Power



Derivation

$$v_C(t) = V_C \cos(\omega t) \quad \Rightarrow \quad i_C(t) = -I_C \sin(\omega t) \quad (49)$$

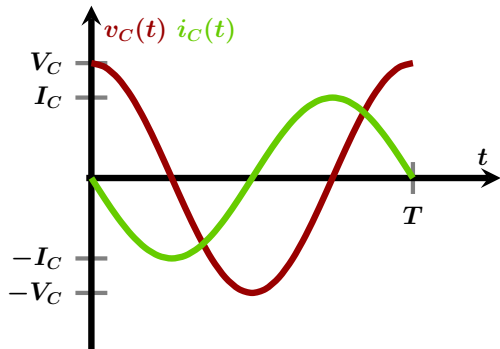
$$p_C(t) = v_C(t)i_C(t) = -V_C I_C \cos(\omega t) \sin(\omega t)$$
$$\begin{aligned} \cos(\alpha) \sin(\alpha) &= \frac{1}{2} \sin(2\alpha) \\ \Downarrow \\ &= -\frac{V_C I_C}{2} \sin(2\omega t) \end{aligned} \quad (50)$$

IMPORTANT Average Power in Capacitors

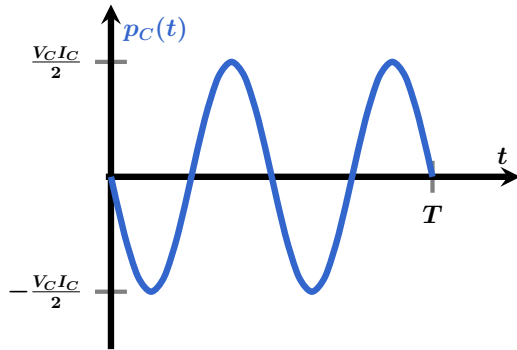
$$P_{C_{avg}} = \frac{1}{T} \int_0^T p_C(t) dt = 0 \quad (51)$$

Power in Capacitors

Voltage and Current



Instantaneous Power



Complex Power in Components

Complex Power in Impedances

$$\underline{S} = \underline{V}\underline{I}^* = \frac{\underline{V}\underline{V}^*}{\underline{Z}} = \underline{Z}\underline{I}\underline{I}^* \text{ with } \underline{V} = \hat{V}e^{j\varphi_V} \text{ and } \underline{I} = \hat{I}e^{j\varphi_I} \quad (52)$$

IMPORTANT Complex Power in Resistors $\underline{Z}_R = R$

$$\begin{aligned}
 \underline{S}_R &= \underline{V}_R \underline{I}_R^* \stackrel{\substack{\text{Phasors are always sinusoidal} \\ \downarrow}}{=} \frac{\hat{V}\hat{I}}{2} e^{j(\varphi_{V_R} - \varphi_{I_R})} \stackrel{\substack{\varphi_V = \varphi_R \\ \downarrow}}{=} \frac{\hat{V}\hat{I}}{2} \\
 &= P_R + jQ_R = \frac{\hat{V}\hat{I}}{2} + j0 \\
 \Rightarrow P_R &= \frac{\hat{V}\hat{I}}{2} \text{ and } Q_R = 0
 \end{aligned} \quad (53)$$

Complex Power in Components

IMPORTANT Complex Power in Inductors $\underline{Z}_L = j\omega L$

$$\begin{aligned}\underline{S}_L &= \underline{V}_L \underline{I}_L^* \stackrel{\underline{V}_L = j\omega L \underline{I}_L}{=} j\omega L \underline{I}_L \underline{I}_L^* = j\omega L I_L^2 e^{j(\varphi_{I_L} - \varphi_{I_L})} = j\omega L I_L^2 \\ &= P_L + jQ_L = 0 + j\omega L I_L^2 \\ &\Rightarrow P_L = 0 \text{ and } Q_L = \omega L I_L^2\end{aligned}\tag{54}$$

IMPORTANT Complex Power in capacitors $\underline{Y}_C = j\omega C \Rightarrow \underline{Z}_C = \frac{1}{j\omega C}$

$$\begin{aligned}\underline{S}_C &= \underline{V}_C \underline{I}_C^* = \underline{V}_C j\omega C \underline{V}_C^* = j\omega C V_C^2 \\ &= P_C + jQ_C = 0 + j\omega C V_C^2 \\ &\Rightarrow P_C = 0 \text{ and } Q_C = \omega C V_C^2\end{aligned}\tag{55}$$

IMPORTANT Definition

$$E = \int p(t) dt \quad (56)$$

Unit of Energy

$$[E] = \text{J} = \text{W s} = \frac{1}{1000 \cdot 3600} \text{kW h} \quad (57)$$

Energy in Resistors

Derivation of Energy in a Resistor

$$\begin{aligned}
 i_R(t) &= I_R \cos(\omega t) \Rightarrow p_R(t) = R \frac{I_R^2}{2} (1 + \cos(2\omega t)) \\
 e_R(t) &= \int_0^T p_R(t) dt = \frac{RI_R^2}{2} t + \frac{RI_R^2}{4\omega} \sin(2\omega t)
 \end{aligned}
 \tag{58}$$

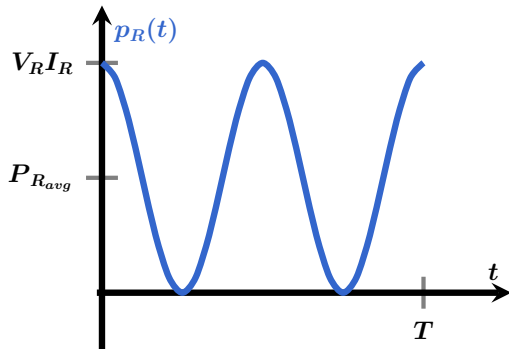
Meaning of Energy in a Resistor

Energy in a resistor R accumulates linearly with sinusoidal ripple, when excited with a sinusoidal current $i_R(t)$.

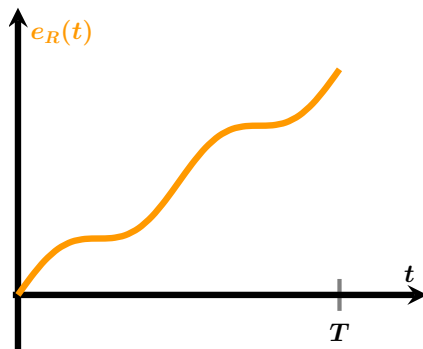
The energy in a resistor can only rise, that is all energy is converted into heat (thermal energy).

Electrically, this is called "losses".

Power in a Resistor



Energy in a Resistor



Energy in Inductors

Derivation of Energy in an Inductor

$$E_L = \int v_L(t) i_L(t) dt = L \int i_L(t) \frac{di_L}{dt} dt \quad (59)$$

Derivation using the Rule for Partial Integration ...

$$\int x \frac{dx}{dt} dt = x^2 - \int x \frac{dx}{dt} dt \Rightarrow \int x \frac{dx}{dt} dt = \frac{x^2}{2} \quad (60)$$

IMPORTANT ...gives the Energy stored in an Inductor at a given Time

$$e_L(t) = \frac{1}{2} L i_L^2(t) \quad (61)$$

Energy in Inductors

Derivation of Energy in Inductors for sinusoidal Signals

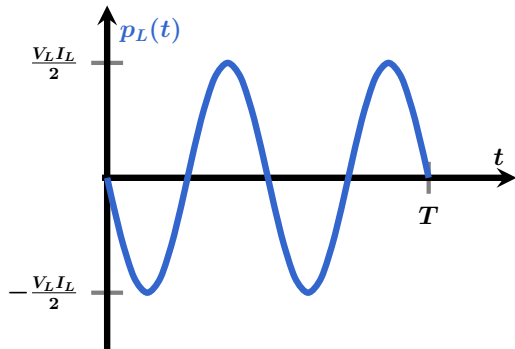
$$\begin{aligned}
 i_L(t) &= I_L \cos(\omega t) \\
 \Rightarrow e_L(t) &= \frac{1}{2} L I_L^2 \cos^2(\omega t) = \frac{1}{4} L I_L^2 (1 + \cos(2\omega t))
 \end{aligned} \tag{62}$$

Meaning of Energy in an Inductor

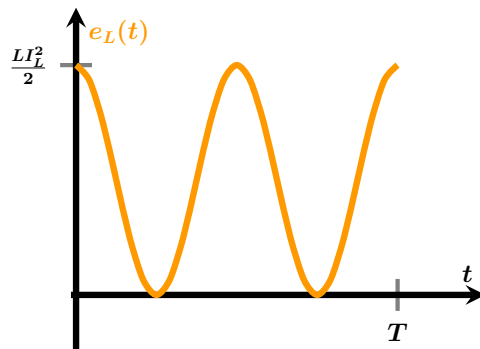
The energy in an inductor L is sinusoidally cycling between 0 and its peak energy. The inductor is "magnetized". At the peak all energy is stored in the magnetic field $\vec{H}$ and afterwards, the inductor gets "demagnetized", i.e. the energy is collected and is used in other parts of the circuit. An inductor "stores" energy in a magnetic field $\vec{H}$.

Energy in Inductors

Power in an Inductor



Energy in an Inductor



Energy in Capacitors



Derivation of Energy in a Capacitor

$$E_C = \int v_C(t) i_C(t) dt = C \int v_C(t) \frac{dv_C}{dt} dt \quad (63)$$

Derivation using the Rule for Partial Integration ...

$$\int x \frac{dx}{dt} dt = x^2 - \int x \frac{dx}{dt} dt \Rightarrow \int x \frac{dx}{dt} dt = \frac{x^2}{2} \quad (64)$$

IMPORTANT ...gives the Energy stored in a Capacitor at a given Time

$$e_C(t) = \frac{1}{2} C v_C^2(t) \quad (65)$$

Energy in Capacitors

Derivation of Energy in Capacitors for sinusoidal Signals

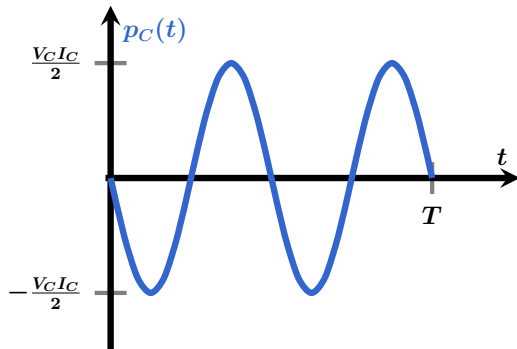
$$\begin{aligned}
 v_C(t) &= V_C \cos(\omega t) \\
 \Rightarrow e_C(t) &= \frac{1}{2} C V_C^2 \cos^2(\omega t) = \frac{1}{4} C V_C^2 (1 + \cos(2\omega t))
 \end{aligned} \tag{66}$$

Meaning of Energy in a Capacitor

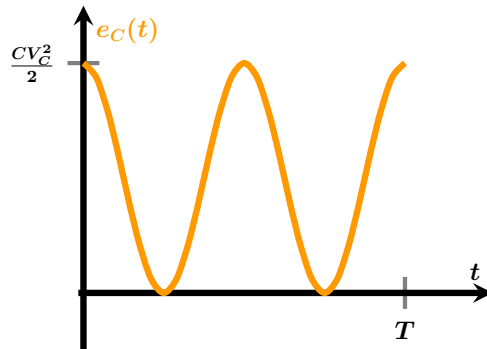
The energy in a capacitor C is sinusoidally cycling between 0 and its peak energy. The capacitor is "charged". At the peak, all energy is stored in the electric field $\vec{E}$ and afterwards, the capacitor gets "discharged", i.e. the energy is collected and is used in other parts of the circuit. A capacitor "stores" energy in a electric field $\vec{E}$. A prominent example of a capacitor is a battery. The maximum energy, that can be stored in a battery is typically given in Ah, which is the charge at a given voltage (the rated voltage of a battery). Note that $e_C(t) = q(t)V_C(t)$, where the charge $q(t) = \int i_C(t)dt$.

Energy in Capacitors

Power in a Capacitor



Energy in a Capacitor

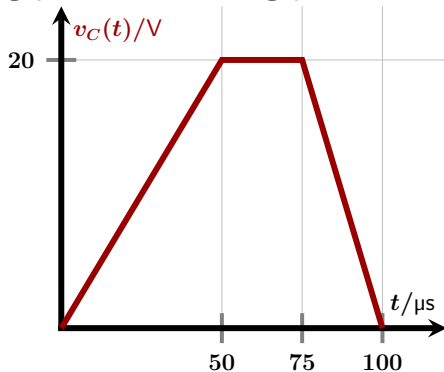


Power and Energy in Components

Exercise 2.9 Capacitor: Waveforms, Power and Energy

☕☕☕ / ⌚ 20 min

The voltage in the following figure is applied across a capacitor $C = 0.01 \mu\text{F}$. How do the current $i_C(t)$, power $p_C(t)$ and energy $e_C(t)$ waveforms look like? Is the capacitor absorbing power, delivering power or both?



Power and Energy in Components

Exercise 2.10 Inductor: Power and Energy

 /  10 min

For $t \geq 0$ the current $i_L(t) = 30 At + 60$ A (consider t without unit here) is flowing through an inductor $L = 100$ mH. How do the power $p_L(t)$ and energy $e_L(t)$ waveforms look like? Is the inductor absorbing power, delivering power or both?

Section 3

Circuits

- **Circuits**
 - Series and Parallel Connections
 - Norton & Thevenin Equivalent Circuits
 - Impedance Matching
 - LaPlace Transformations
 - Transfer Functions
 - Network Analysis
 - Bode Plots

Recap

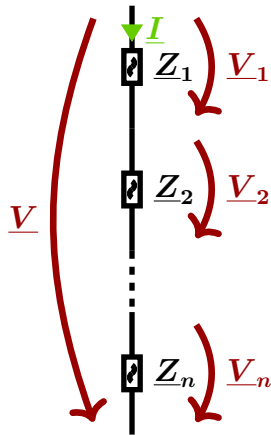
So far:

- Section **Signals** provided through **Sources** as voltages and currents, which together result in **Power**
- Section **Components** provided the building blocks **Resistors**, **Inductors** and **Capacitors**, with their respective **Power in Components** and **Energy in Components**.

This section

Combining those building blocks creates **Circuits**.

Series Circuit

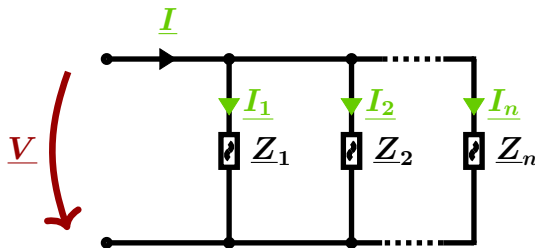
**IMPORTANT** Kirchhoffs voltage law

$$\underline{V} = \underline{V}_1 + \underline{V}_2 + \dots + \underline{V}_n \quad (67)$$

IMPORTANT Impedances in series

$$\underline{Z} = \frac{\underline{V}}{\underline{I}} = \underline{Z}_1 + \underline{Z}_2 + \dots + \underline{Z}_n \quad (68)$$

Parallel Circuit

**IMPORTANT** Kirchhoffs current law

$$\underline{I} = \underline{I}_1 + \underline{I}_2 + \dots + \underline{I}_n \quad (69)$$

IMPORTANT Admittances in parallel

$$\underline{Y} = \frac{\underline{I}}{\underline{V}} = \underline{Y}_1 + \underline{Y}_2 + \dots + \underline{Y}_n \quad (70)$$

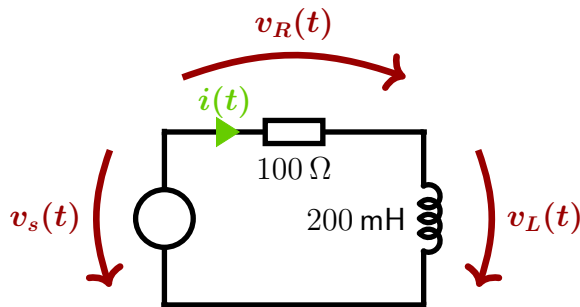
with $\underline{Y}_x = \frac{1}{\underline{Z}_x}$

Series & Parallel Connections

Exercise 3.1 Phasors in an RL series connection

 / ⌚ 15 min

What are the phasors for all voltages and currents in the following circuit?
How does the phasor diagram look like? What is the phase relationship between the source voltage $v_s(t) = 10 \text{ V} \cos(500t)$ and $i(t)$?

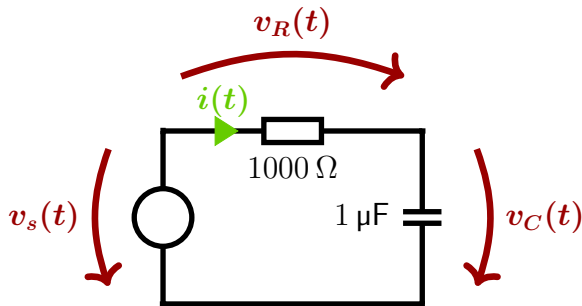


Series & Parallel Connections

Exercise 3.2 Phasors in an RC series connection

 / ⌚ 15 min

What are the phasors for all voltages and currents in the following circuit?
How does the phasor diagram look like? What is the phase relationship between the source voltage $v_s(t) = 10 \text{ V} \cos(500t)$ and $i(t)$?

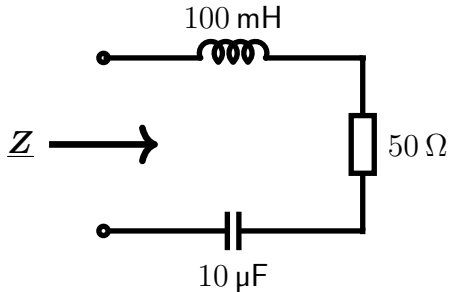


Series & Parallel Connections

Exercise 3.3 Resulting impedance of a series connection

    / ⌚ 10 min

What is the complex impedance in phasors for the shown circuit at $\omega_1 = 500 \text{ rad s}^{-1}$, $\omega_2 = 1000 \text{ rad s}^{-1}$ and $\omega_3 = 2000 \text{ rad s}^{-1}$?



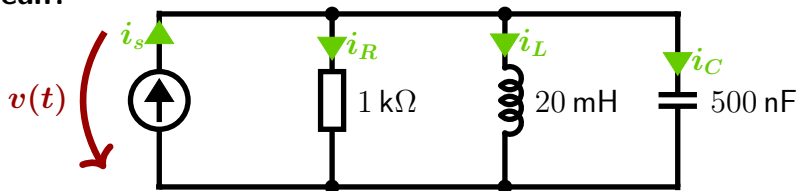
Series & Parallel Connections

Exercise 3.4 Phasors in a Parallel connection

 /  15 min

What are the phasors $\underline{I}_S$, $\underline{V}$, $\underline{I}_R$, $\underline{I}_L$ and $\underline{I}_C$ for $i_s(t) = 10 \text{ mA} \cos(10^4 t)$?

What is the relationship between the peak values of $i_L(t)$ and $i_s(t)$? What does this mean?

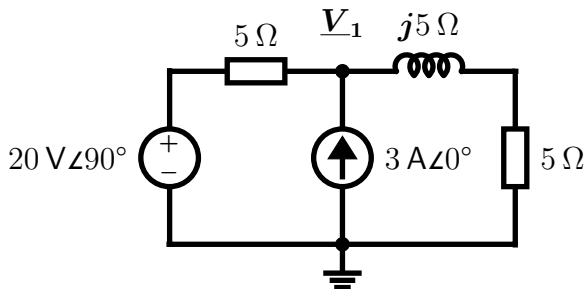


Series & Parallel Connections

Exercise 3.5 Superposition of a voltage and a current source

 /  10 min

What is $\underline{V}_1$?

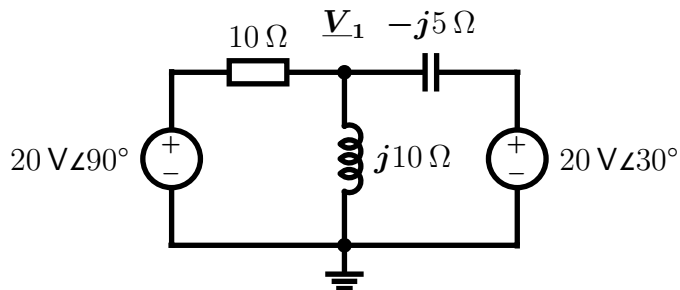


Series & Parallel Connections

Exercise 3.6 Superposition of two voltage sources

 /  10 min

What is $\underline{V}_1$?

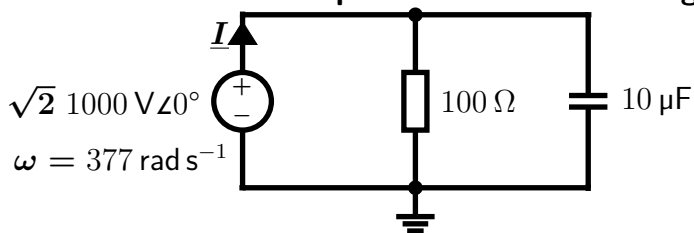


Series & Parallel Connections

Exercise 3.7 Power in a parallel connection

     / ⌚ 15 min

What is the phasor $\underline{I}$? What is the power, reactive power and apparent power delivered by the source? What is the power factor and is it lagging or leading?



Series & Parallel Connections

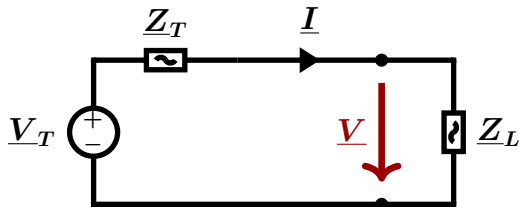
Exercise 3.8 Power in a series connection



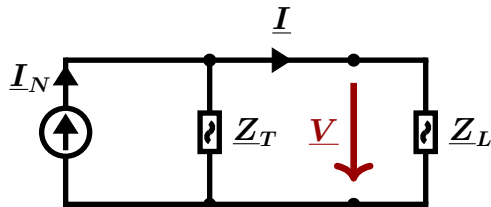
A 60 Hz, 240 V_{RMS} source supplies energy to a load consisting of a resistor in series with an inductance. The real power is 1500 W and the apparent power is 2500 VA . What is the resistance of the resistor, the inductance of the inductor and the resulting power factor?

Norton & Thevenin Equivalent Circuits

Thevenin Circuit



Norton Circuit



Meaning

The electrical conditions V and I for the load impedance Z_L are the same in both cases.

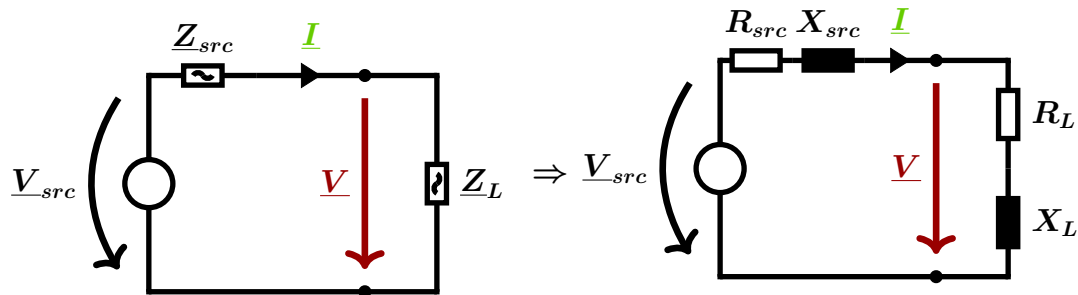
IMPORTANT Norton & Thevenin Transformations

$$V_T = V_{open-load} = I_N Z_T$$

$$I_N = I_{short-circuit} = \frac{V_T}{Z_T} \quad (71)$$

$$\text{with } Z_T = \frac{1}{Y_N} = \frac{V_{open-load}}{I_{short-circuit}} \quad \text{EK2}$$

Circuit



Parameters

$$\begin{aligned} \underline{V}_{ol} &= \underline{V}_{src} \\ \underline{I}_{sc} &= \frac{\underline{V}_{src}}{\underline{Z}_{src}} \end{aligned} \quad (72)$$

Impedance Matching

Meaning of Matching

Matching means to adjust a load impedance $\underline{Z}_L = R_L + jX_L$ to a given source - here voltage source $\underline{V}_{src}$ - and a given output impedance (Thevenin impedance) $\underline{Z}_{src} = R_{src} + jX_{src}$ to maximize another signal.

Maximum Voltage Matching

$$\frac{\underline{V}}{\underline{V}_{ol}} = \frac{\underline{Z}_L}{\underline{Z}_{src} + \underline{Z}_L} = \frac{\frac{\underline{Z}_L}{\underline{Z}_{src}}}{1 + \frac{\underline{Z}_L}{\underline{Z}_{src}}} \quad (73)$$

achieves maximum voltage $\underline{V}$ for

$$\frac{\underline{Z}_L}{\underline{Z}_{src}} \rightarrow \infty \Rightarrow \underline{Z}_L \rightarrow \infty \quad (74)$$

Maximum Current Matching

$$\frac{\underline{I}}{\underline{I}_{sc}} = \frac{\frac{\underline{V}_{src}}{\underline{Z}_{src} + \underline{Z}_L}}{\frac{\underline{V}_{src}}{\underline{Z}_{src}}} = \frac{\underline{Z}_{src}}{\underline{Z}_{src} + \underline{Z}_L} = \frac{1}{1 + \frac{\underline{Z}_L}{\underline{Z}_{src}}} \quad (75)$$

achieves maximum current $\underline{I}$ for

$$\frac{\underline{Z}_L}{\underline{Z}_{src}} \rightarrow 0 \Rightarrow \underline{Z}_L \rightarrow 0 \quad (76)$$

Impedance Matching



Derivation Sum of Source and Load Impedance

$$\underline{Z}_{sum} = R_{src} + R_L + j(X_{src} + X_L) = |\underline{Z}_{sum}| e^{j\angle \underline{Z}_{sum}} \quad (77)$$

Derivation Loop Current

$$\underline{I} = \frac{\underline{V}_{src}}{\underline{Z}_{sum}} = \underbrace{\frac{|\underline{V}_{src}|}{|\underline{Z}_{sum}|}}_{\underline{I}} e^{j(\underbrace{\angle \underline{V}_{src} - \angle \underline{Z}_{sum}}_{\angle \underline{I}})} \quad (78)$$

Derivation Complex Power across Load Impedance

$$\underline{S}_L = \underline{V}\underline{I}^* = \underline{Z}_L \underline{I}\underline{I}^* = \underline{Z}_L |\underline{I}|^2 e^{j(\underbrace{\angle \underline{I} - \angle \underline{I}}_{=0})} = \underbrace{R_L |\underline{I}|^2}_{P_L} + j \underbrace{X_L |\underline{I}|^2}_{Q_L \text{ EK2}} \quad (79)$$

Matching for maximum Power Transfer

$$P_L = \frac{R_L |V_{src}|^2}{(R_{src} + R_L)^2 + (X_{src} + X_L)^2} \quad (80)$$

achieves maximum power for $X_L = -X_{src}$ and at $\frac{dP_L}{dR_L} = 0$

Derivation Extrema of Power, where its Derivative is Zero

$$\frac{dP_L}{dR_L} = \frac{(R_{src} + R_L)^2 - 2R_L(R_{src} + R_L)}{(R_{src} + R_L)^4} |V_{src}|^2 = \frac{R_{src} - R_L}{(R_{src} + R_L)^3} \stackrel{!}{=} 0 \quad (81)$$

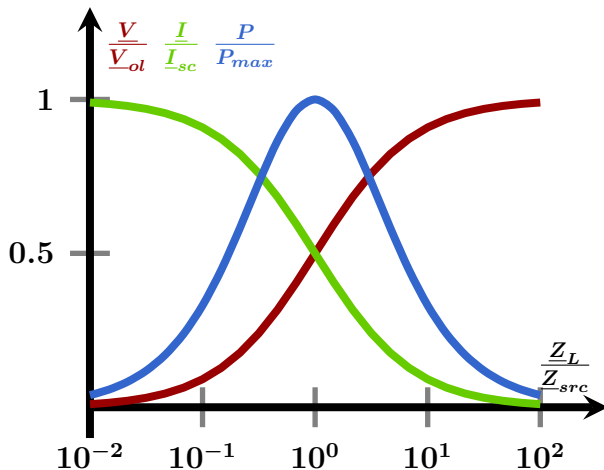
$\frac{dP_L}{dR_L} = 0$ at $R_L = R_{src}$

Maximum Power

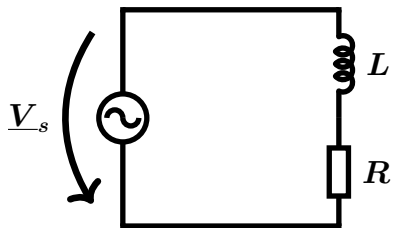
$$P_{L_{max}} = \frac{R_L}{(2R_L)^2} V_{src}^2 = \frac{V_{src}^2}{4R_L} \quad (82)$$

Impedance Matching

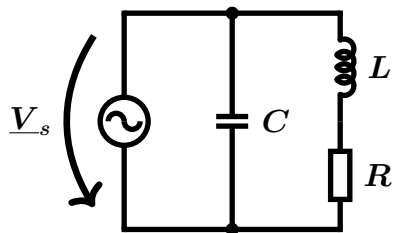
Voltage, Current and Power as a Function of Load Variation



Reactive Power due to Inductor L



Compensation with a Capacitor C



Impedance Matching

Matching for optimum Power Factor = Power Factor Correction

$$\begin{aligned}\underline{Z} &= \frac{\frac{1}{j\omega C} (R + j\omega L)}{R + j\omega L + \frac{1}{j\omega C}} = \frac{R + j\omega L}{1 - \omega^2 LC + j\omega RC} \\ &= \frac{(R + j\omega L)(1 - \omega^2 LC - j\omega RC)}{(1 - \omega^2 LC)^2 + \omega^2 R^2 C^2} \\ \Rightarrow \Im\{\underline{Z}\} &= j\omega L - j\omega^3 L^2 C - j\omega R^2 C \stackrel{!}{=} 0 \\ \Rightarrow L - \omega^2 L^2 C - R^2 C &= 0 \\ \Rightarrow C(\omega^2 L^2 + R^2) &= L \\ \Rightarrow C &= \frac{L}{R^2 + \omega^2 L^2}\end{aligned}\tag{83}$$

Impedance Matching

Example

$$\begin{aligned} L &= 100 \mu\text{H} \\ R &= 20 \Omega \\ f &= 60 \text{ Hz} \end{aligned} \quad (84)$$

Size of compensation Capacitor C

$$C = \frac{100 \mu\text{H}}{(20 \Omega)^2 + (2\pi 60 \text{ Hz} 100 \mu\text{H})^2} \approx 250 \text{ nF} \quad (85)$$

Choosing from E12-series: $C \rightarrow 270 \text{ nF}$.

Checking Units of C

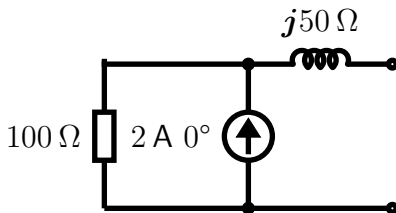
$$[C] = \frac{\frac{V_s}{A}}{\frac{V^2}{A^2} + \frac{V^2 s^2}{s^2 A^2}} = \frac{As}{V} = F \quad (86)$$

Norton & Thevenin Equivalents and Matching

Exercise 3.9 Current Source Equivalent and Matching

 /  10 min

- What is the Thevenin and Norton equivalent of the following circuit?
- b What is the maximum power, this circuit can deliver for any complex load impedance?
- c What is the maximum power, this circuit can deliver for a resistive load?



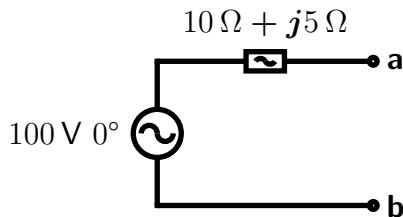
Norton & Thevenin Equivalents and Matching

Exercise 3.10 Matching of Voltage Source with series Impedance



10 min

What parallel connection of a resistor R and a capacitor C to the terminals a and b of the Thevenin equivalent source in the diagram allows maximum power delivery to this load resistance R at 60 Hz?



LaPlace Transformation - Math Background

Definition of the LaPlace Transformation

$$\underline{F}(\underline{s}) = \int_0^{\infty} f(t)e^{-st}dt \Rightarrow f(t) \circ\bullet \underline{F}(\underline{s}) \quad (87)$$

with the complex frequency $\underline{s} = \sigma + j\omega$

Note that the real part of the complex frequency σ is usually zero.

Suggested extra reading material: [6].

Definition of the inverse LaPlace Transformation

$$f(t) = \frac{1}{2\pi j} \int_{\alpha-j\infty}^{\alpha+j\infty} \underline{F}(\underline{s})e^{st}d\underline{s} \quad (88)$$

Initial and final Values

$$\lim_{t \rightarrow 0^+} f(t) = \lim_{\underline{s} \rightarrow \infty} \underline{sF}(\underline{s})$$

$$\lim_{t \rightarrow \infty} f(t) = \lim_{\underline{s} \rightarrow 0} \underline{sF}(\underline{s}) \quad (89)$$

LaPlace Transformation - Math Background

IMPORTANT Properties

Properties	Time (t) Domain	Frequency ($\underline{s}$) Domain
independent variable	t	$\underline{s}$
signal representation	$f(t)$	$\underline{F}(\underline{s})$
linearity	$Af_1(t) + Bf_2(t)$	$AF_1(\underline{s}) + BF_2(\underline{s})$
integration	$\int_0^t f(\tau) d\tau$	$\frac{\underline{F}(\underline{s})}{\underline{s}}$
differentiation	$\frac{df(t)}{dt}$ $\frac{d^2f(t)}{dt^2}$	$\underline{s}\underline{F}(\underline{s}) - f(0s)\delta(t)$ $\underline{s}^2\underline{F}(\underline{s}) - \underline{s}f(0s)\delta(t) - \frac{df(0s)}{dt}\delta(t)$
offset in frequency	$e^{-\alpha t}f(t)$	$\underline{F}(\underline{s} + \alpha)$
offset in time	$f(t - \beta)u(t - \beta)$	$e^{-\beta\underline{s}}\underline{F}(\underline{s})$

with the unity step function $u(t)$, which is 1 for $t \geq 0s$ and 0 for $t < 0s$
and the Dirac impulse $\delta(t)$, which is 1Hz at $t = 0s$ and $0 \forall t \neq 0s$.

LaPlace Transformation - Application in Circuits

General Application of LaPlace Transformations in Circuit Analysis

- Describe circuit analytically by Kirchhoffs Rules and Ohms Law (perhaps also Thevenin and/or Norton equivalents) in the s -domain.
- ② Apply the relevant signals - coming from the present sources - to the circuit description
- ③ Either use the results in the frequency domain (such as **Transfer Functions**) or
- ④ Apply an inverse LaPlace Transformation and interpret the system response to a specific stimulation.

LaPlace Transformation - Application in Circuits

Specific Application of LaPlace Transformations to Sources

Uncontrolled (independent) sources:

$$v(t) \circ \bullet \underline{V}(\underline{s}) \quad (90)$$

for voltage sources and

$$i(t) \circ \bullet \underline{I}(\underline{s}) \quad (91)$$

for current sources.

Controlled (dependent) sources:

$$v_2(t) = \mu v_1(t) \circ \bullet \underline{V}_2(\underline{s}) = \mu \underline{V}_1(\underline{s}) \quad (92)$$

e.g.: operational amplifier,

$$i_2(t) = \beta i_1(t) \circ \bullet \underline{I}_2(\underline{s}) = \beta \underline{I}_1(\underline{s}) \quad (93)$$

e.g.: bipolar transistor,

$$v_2(t) = r i_1(t) \circ \bullet \underline{V}_2(\underline{s}) = r \underline{I}_1(\underline{s}) \quad (94)$$

e.g.: transimpedance amplifier,

$$i_2(t) = g v_1(t) \circ \bullet \underline{I}_2(\underline{s}) = g \underline{V}_1(\underline{s}) \quad (95)$$

e.g.: transconductance amplifier, MOSFET.

Signals

Signal	Time Domain	Frequency Domain
dirac impulse	$\delta(t)$	1Hz
unity step	$u(t)$	$\frac{1}{s}$
unity ramp	$r(t) = tu(t)$	$\frac{1}{s^2}$
exponential	$e^{-\alpha t}u(t)$	$\frac{1}{s+\alpha}$
damped unity ramp	$te^{-\alpha t}u(t)$	$\frac{1}{(s+\alpha)^2}$
sine wave	$\sin(\beta t)u(t)$	$\frac{\beta}{s^2+\beta^2}$
cosine wave	$\cos(\beta t)u(t)$	$\frac{s}{s^2+\beta^2}$
damped sine	$e^{-\alpha t} \sin(\beta t)u(t)$	$\frac{\beta}{(s+\alpha)^2+\beta^2}$
damped cosine	$e^{-\alpha t} \cos(\beta t)u(t)$	$\frac{s+\alpha}{(s+\alpha)^2+\beta^2}$

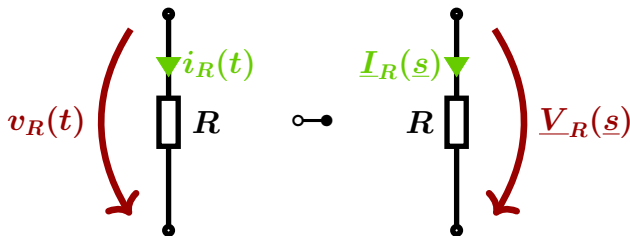
with the unity step function $u(t)$, which is 1 for $t \geq 0$ and 0 for $t < 0$
 and the Dirac impulse $\delta(t)$, which is 1Hz at $t = 0$ s and 0 $\forall t \neq 0$ s.

LaPlace Transformation - Application in Circuits

Specific Application of LaPlace Transformations to Resistors

$$v_R(t) = Ri_R(t) \quad \longleftrightarrow \quad \underline{V}_R(\underline{s}) = R\underline{I}_R(\underline{s}) \quad (96)$$

Modelling of a Resistor in Time and Frequency Domain

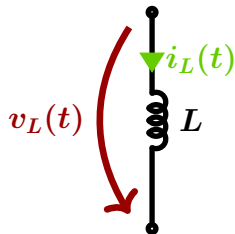


LaPlace Transformation - Application in Circuits

Specific Application of LaPlace Transformations to Inductors

$$v_L(t) = L \frac{di_L(t)}{dt} \quad (97)$$

Modelling of an Inductor in Time and Frequency Domain

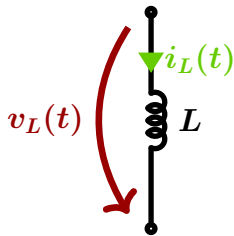


LaPlace Transformation - Application in Circuits

Specific Application of LaPlace Transformations to Inductors

$$v_L(t) = L \frac{di_L(t)}{dt} \longleftrightarrow \underline{V}_L(\underline{s}) = L (\underline{s} \underline{I}_L(\underline{s}) - i_L(0)\delta(t)) \quad (97)$$

Modelling of an Inductor in Time and Frequency Domain

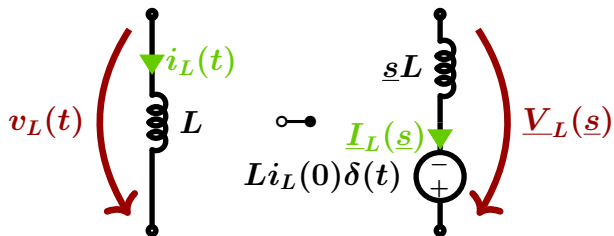


LaPlace Transformation - Application in Circuits

Specific Application of LaPlace Transformations to Inductors

$$v_L(t) = L \frac{di_L(t)}{dt} \longleftrightarrow \underline{V}_L(\underline{s}) = L (\underline{s}\underline{I}_L(\underline{s}) - i_L(0)\delta(t)) \quad (97)$$

Modelling of an Inductor in Time and Frequency Domain

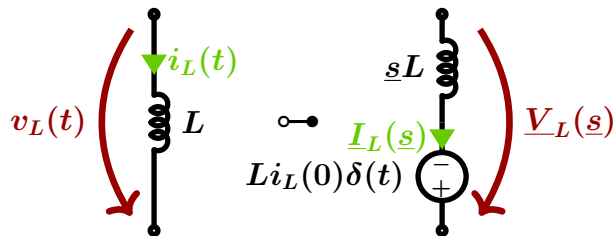


LaPlace Transformation - Application in Circuits

Specific Application of LaPlace Transformations to Inductors

$$v_L(t) = L \frac{di_L(t)}{dt} \quad \bullet \quad \underline{V_L(s)} = L (s \underline{I_L(s)} - i_L(0)\delta(t)) \Rightarrow \underline{I_L(s)} = \frac{\underline{V_L(s)}}{sL} + \frac{i_L(0)\delta(t)}{s} \quad (97)$$

Modelling of an Inductor in Time and Frequency Domain

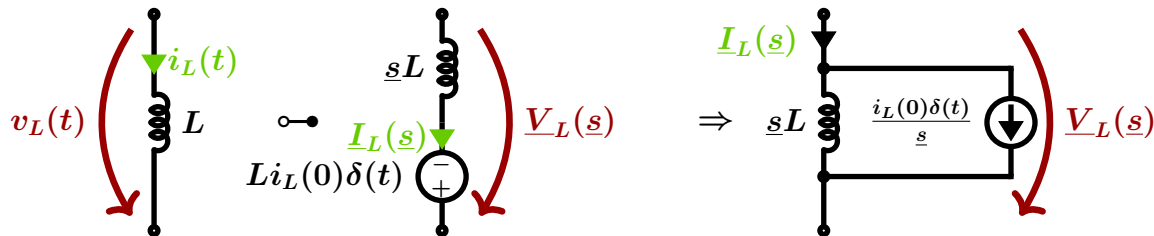


LaPlace Transformation - Application in Circuits

Specific Application of LaPlace Transformations to Inductors

$$v_L(t) = L \frac{di_L(t)}{dt} \quad \bullet \quad \underline{V}_L(\underline{s}) = L (\underline{s} \underline{I}_L(\underline{s}) - i_L(0)\delta(t)) \Rightarrow \underline{I}_L(\underline{s}) = \frac{\underline{V}_L(\underline{s})}{\underline{s}L} + \frac{i_L(0)\delta(t)}{\underline{s}} \quad (97)$$

Modelling of an Inductor in Time and Frequency Domain



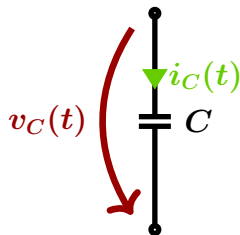
LaPlace Transformation - Application in Circuits

Specific Application of LaPlace Transformations to Capacitors

$$i_C(t) = C \frac{dv_C(t)}{dt}$$

(98)

Modelling of a Capacitor in Time and Frequency Domain

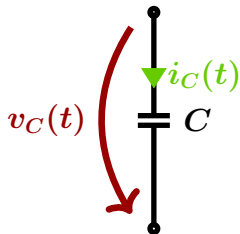


LaPlace Transformation - Application in Circuits

Specific Application of LaPlace Transformations to Capacitors

$$i_C(t) = C \frac{dv_C(t)}{dt} \quad \text{---} \quad \underline{I}_C(\underline{s}) = C (\underline{s} \underline{V}_C(\underline{s}) - v_C(0)\delta(t)) \quad (98)$$

Modelling of a Capacitor in Time and Frequency Domain



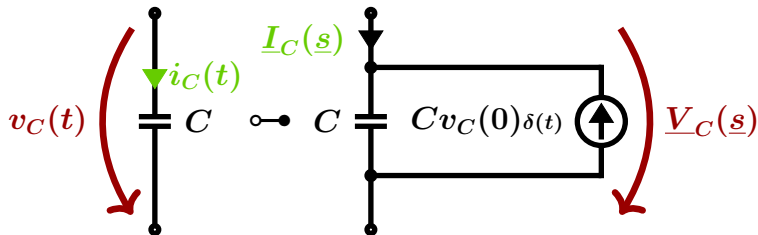
LaPlace Transformation - Application in Circuits

Specific Application of LaPlace Transformations to Capacitors

$$i_C(t) = C \frac{dv_C(t)}{dt} \quad \longleftrightarrow \quad \underline{I}_C(\underline{s}) = C (\underline{s} \underline{V}_C(\underline{s}) - v_C(0)\delta(t))$$

(98)

Modelling of a Capacitor in Time and Frequency Domain

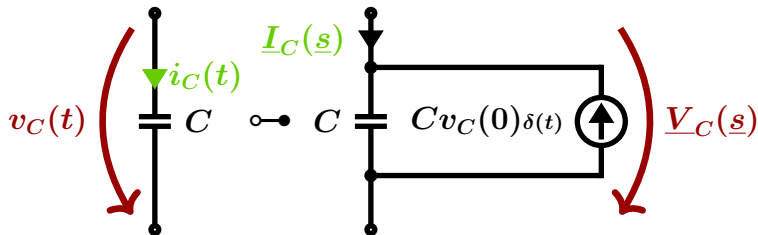


LaPlace Transformation - Application in Circuits

Specific Application of LaPlace Transformations to Capacitors

$$i_C(t) = C \frac{dv_C(t)}{dt} \quad \text{---} \quad \underline{I_C(s)} = C (\underline{sV_C(s)} - v_C(0)\delta(t)) \Rightarrow \underline{V_C(s)} = \frac{\underline{I_C(s)}}{\underline{sC}} + \frac{v_C(0)\delta(t)}{\underline{s}} \quad (98)$$

Modelling of a Capacitor in Time and Frequency Domain

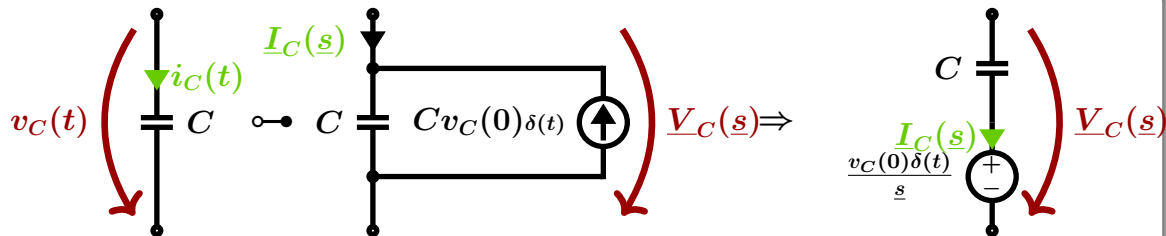


LaPlace Transformation - Application in Circuits

Specific Application of LaPlace Transformations to Capacitors

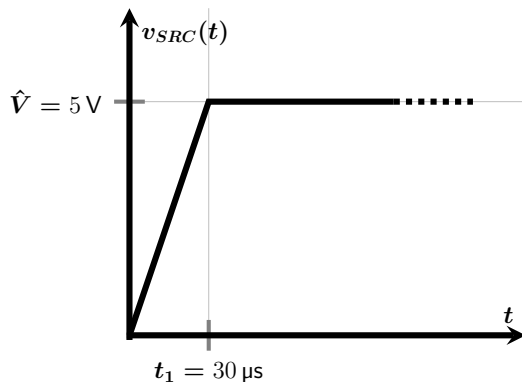
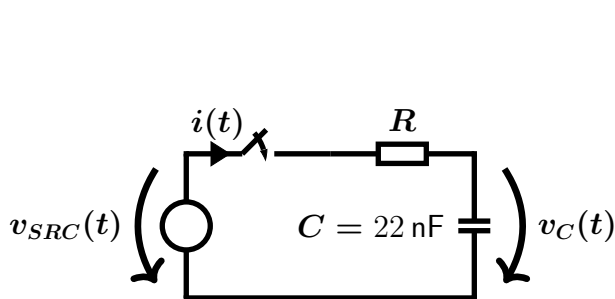
$$i_C(t) = C \frac{dv_C(t)}{dt} \quad \text{---} \quad \underline{I}_C(\underline{s}) = C (\underline{s} \underline{V}_C(\underline{s}) - v_C(0)\delta(t)) \Rightarrow \underline{V}_C(\underline{s}) = \frac{\underline{I}_C(\underline{s})}{\underline{s}C} + \frac{v_C(0)\delta(t)}{\underline{s}} \quad (98)$$

Modelling of a Capacitor in Time and Frequency Domain



LaPlace Transformation - Application in Circuits

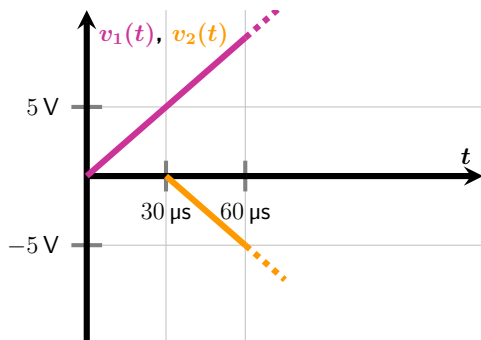
Example of an RC circuit with a ramp that turns into a constant



What is $v_C(t)$ for $R = 1 \text{ k}\Omega$, $R = 100 \Omega$ and $R = 10 \text{ k}\Omega$ with $v_C(0) = 2 \text{ V}$, when the switch is closed at $t = 0$?

LaPlace Transformation - Application in Circuits

Describing the source voltage as two superimposed functions,...



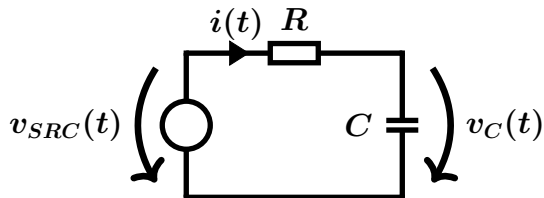
$$v_{src}(t) = v_1(t) + v_2(t)$$

$$v_1(t) = \frac{\hat{V}}{t_1}t$$

$$v_2(t) = -v_1(t - t_1) = -\frac{\hat{V}}{t_1}(t - t_1) \quad (99)$$

LaPlace Transformation - Application in Circuits

... analyzing the circuit for $t \geq 0$, ...



$$v_{src}(t) = Ri(t) + v_C(t) \text{ with}$$

$$i(t) = C \frac{dv_C(t)}{dt} \quad (100)$$

$$\Rightarrow v_{SRC}(t) = RC \frac{dv_C(t)}{dt} + v_C(t)$$

Inserting the superimposed functions into the equations from the circuits analysis ...

$$v_1(t) - v_1(t-t_1) = RC \frac{dv_C(t)}{dt} + v_C(t) \Rightarrow \frac{\hat{V}}{t_1} t - \frac{\hat{V}}{t_1} (t-t_1) = RC \frac{dv_C(t)}{dt} + v_C(t) \quad (101)$$

LaPlace Transformation - Application in Circuits

... and transferring the results in the $\underline{s}$ -domain.

$$\bullet \frac{\hat{V}}{t_1} \frac{1}{\underline{s}^2} - \frac{\hat{V}}{t_1} \frac{1}{\underline{s}^2} e^{-t_1 \underline{s}} = RC (s \underline{V}_C(s) - v_C(t=0)) + \underline{V}_C(s) \quad (102)$$

Then solving the equation in the $\underline{s}$ -domain for $\underline{V}_C$, ...

$$\begin{aligned} \underline{V}_C(s) &= \frac{1}{1 + sRC} \left(\frac{\hat{V}}{t_1} \frac{1}{\underline{s}^2} - \frac{\hat{V}}{t_1} \frac{1}{\underline{s}^2} e^{-t_1 \underline{s}} + RC v_C(0) \right) \\ &= \underbrace{\frac{\hat{V}}{t_1 RC}}_{\text{const}} \underbrace{\frac{1}{\underline{s}^2 (s + \frac{1}{RC})}}_{\frac{\underline{F}(s)}{\underline{s}^2}} - \underbrace{\frac{\hat{V}}{t_1 RC}}_{\text{const}} \underbrace{\frac{e^{-t_1 \underline{s}}}{\underline{s}^2 (s + \frac{1}{RC})}}_{\frac{\underline{F}(s)}{\underline{s}^2} e^{-t_1 \underline{s}}} + \underbrace{v_C(0)}_{\text{const}} \underbrace{\frac{1}{s + \frac{1}{RC}}}_{\underline{F}(s)} \end{aligned} \quad (103)$$

LaPlace Transformation - Application in Circuits

... and transferring each part back into the time domain: the first two here, ...

$$\underline{F}(\underline{s}) \stackrel{!}{=} \frac{1}{\underline{s} + \frac{1}{RC}} \bullet \circ e^{-\frac{t}{RC}} = f(t)$$

$$\begin{aligned} \frac{\underline{F}(\underline{s})}{\underline{s}^2} \bullet \circ \int_0^t \int_0^{\vartheta} f(\tau) d\tau d\vartheta &= \int_0^t \int_0^{\vartheta} e^{-\frac{\tau}{RC}} d\tau d\vartheta = \int_0^t [-RC e^{-\frac{\tau}{RC}}]_0^{\vartheta} d\vartheta = (104) \\ &= \int_0^t (-RC) e^{-\frac{\vartheta}{RC}} d\vartheta + \int_0^t RC d\vartheta = \\ &= -RC [-RC e^{-\frac{\vartheta}{RC}}]_0^t + RCt = R^2 C^2 e^{-\frac{t}{RC}} - R^2 C^2 + RCt \end{aligned}$$

LaPlace Transformation - Application in Circuits

... and the last one here:

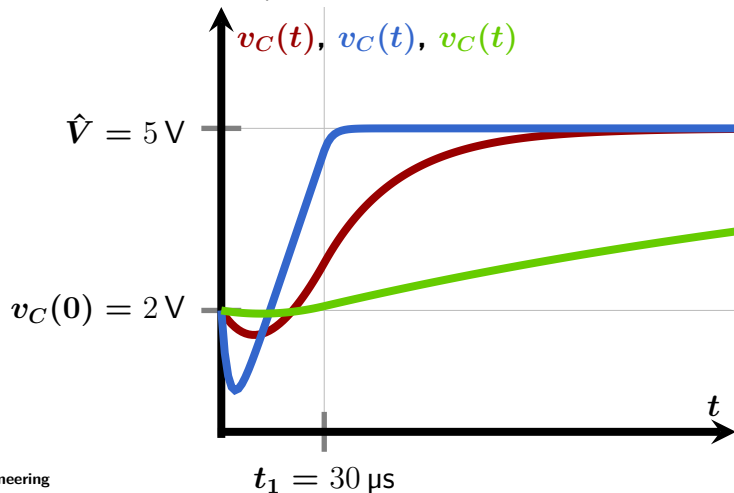
$$\begin{aligned}
 \frac{\underline{F}(\underline{s})}{\underline{s}^2} e^{-t_1 \underline{s}} &\bullet \circ \int_0^{t-t_1} \int_0^{\vartheta} f(\tau) d\tau d\vartheta u(t-t_1) = \\
 &= \begin{cases} 0 & \forall t < t_1 \\ R^2 C^2 e^{-\frac{(t-t_1)}{RC}} - R^2 C^2 + RC(t-t_1) & \forall t \geq t_1 \end{cases} \quad (105)
 \end{aligned}$$

LaPlace Transformation - Application in Circuits

Finally putting all the pieces together gives the resulting $v_C(t)$:

$$\begin{aligned}\Rightarrow v_C(t) &= \frac{\hat{V}}{t_1} \left(t + RCe^{-\frac{t}{RC}} - RC \right) \\ &\quad + v_C(0)e^{-\frac{t}{RC}} \\ &\quad - \begin{cases} 0 \quad \forall t < t_1 \\ \frac{\hat{V}}{t_1} \left(t - t_1 + RCe^{-\frac{t-t_1}{RC}} - RC \right) \quad \forall t \geq t_1 \end{cases} \\ &= v_C(0)e^{-\frac{t}{RC}} + \begin{cases} \frac{\hat{V}}{t_1}t - \frac{\hat{V}}{t_1}RC \left(1 - e^{-\frac{t}{RC}} \right) \quad \forall t < t_1 \\ \hat{V} - \frac{\hat{V}}{t_1}RC \left(e^{-\frac{t-t_1}{RC}} - e^{-\frac{t}{RC}} \right) \quad \forall t \geq t_1 \end{cases}\end{aligned}\tag{106}$$

LaPlace Transformation - Application in Circuits

Result - visualization of $v_C(t)$ for $R = 1\text{ k}\Omega$, $R = 100\text{ }\Omega$ and $R = 10\text{ k}\Omega$:

LaPlace Transformation - Exercises

Exercise 3.11 Offset and Exponential Function

 / ⌚ 10 min

What is the LaPlace transformation of $v_1(t) = 500 \text{ V} (1 - e^{-100 \text{ Hz}t}) u(t)$ and where are its poles and zeros?

Exercise 3.12 Sine Wave

 / ⌚ 5 min

What is the LaPlace transformation of $v_2(t) = 20 \text{ V} \sin(377 \text{ Hz}t)u(t)$ and where are its poles and zeros?

Exercise 3.13 Impulse and Step

 / ⌚ 15 min

How does $v_3(t) = -10 \text{ V}\delta(t) + 10 \text{ V}u(t)$ look like in the time domain? What is its the LaPlace transformation of and where are its poles and zeros?

LaPlace Transformation - Exercises

Exercise 3.14 Offset and Exponential Function

 / ⌚ 10 min

What is the LaPlace transformation of $v_4(t) = 20 \text{ V} (e^{-200 \text{ Hz}t} - 2e^{-100 \text{ Hz}t}) u(t)$ and where are its poles and zeros?

Exercise 3.15 Step and Sine Wave

 / ⌚ 10 min

How does $i_5(t) = 0.005 \text{ A} (10 - 10 \sin(1000 \text{ Hz}t)) u(t)$ look like in the time domain? What is its the LaPlace transformation of and where are its poles and zeros?

Transfer Functions

Circuit analysis tools

In the s domain, the same circuit analysis principles are valid as in the time and phasor domain:

- Ohms law,
- Kirchhofs laws,
- Thevenin & Norton,
- series and parallel connections, and
- superposition of sources.

However the math is simpler:

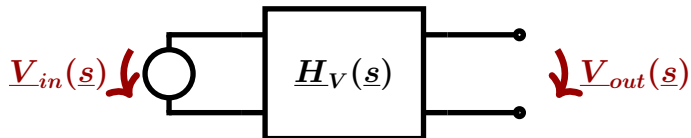
Multiplication and division (linear math) instead of differentiation and integration (differential equations).

Two-Port Representation of a linear, time-invariant Circuit (Network)



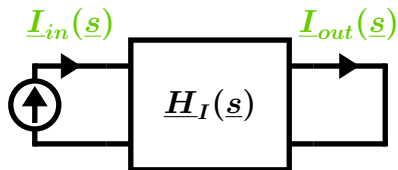
Transfer Functions

Voltage Transfer Function



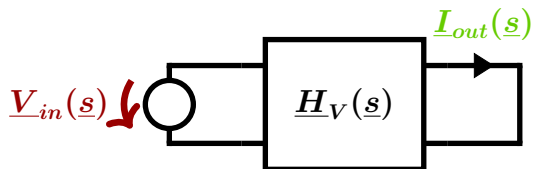
$$\underline{H}_V(\underline{s}) = \frac{\underline{V}_{out}(\underline{s})}{\underline{V}_{in}(\underline{s})} \quad (107)$$

Current Transfer Function



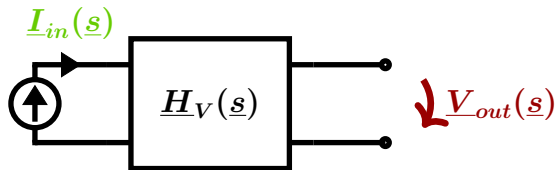
$$\underline{H}_I(\underline{s}) = \frac{\underline{I}_{out}(\underline{s})}{\underline{I}_{in}(\underline{s})} \quad (108)$$

Transconductance Transfer Function



$$\underline{H}_Y(\underline{s}) = \frac{\underline{I}_{out}(\underline{s})}{\underline{V}_{in}(\underline{s})} \quad (109)$$

Transimpedance Transfer Function



$$\underline{H}_Z(\underline{s}) = \frac{\underline{V}_{out}(\underline{s})}{\underline{I}_{in}(\underline{s})} \quad (110)$$

Poles and Zeros

$$\underline{H}(\underline{s}) = \frac{b_0 + b_1\underline{s} + b_2\underline{s}^2 + \dots + b_n\underline{s}^n}{a_0 + a_1\underline{s} + a_2\underline{s}^2 + \dots + a_m\underline{s}^m} \quad (111)$$

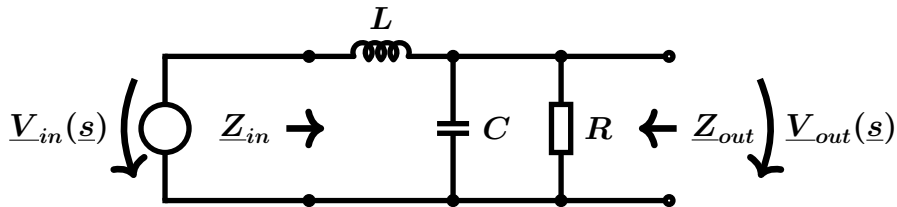
can be rewritten as

$$\underline{H}(\underline{s}) = \underbrace{K}_{\text{DC-gain}} \frac{\overbrace{(\underline{s} - z_1)}^{\text{1st zero}} \overbrace{(\underline{s} - z_2)}^{\text{2nd zero}} \dots \overbrace{(\underline{s} - z_n)}^{\text{n-th zero}}}{\underbrace{(\underline{s} - p_1)}_{\text{1st pole}} \underbrace{(\underline{s} - p_2)}_{\text{2nd pole}} \dots \underbrace{(\underline{s} - p_m)}_{\text{m-th pole}}} \quad (112)$$

Visualization of Poles and Zeros

as Pole-Zero-Diagrams → chapter 9.3 in [1] and [Bode Plots](#)

Example of Transfer Functions on an LCR-network:



What are the input and output impedances $\underline{Z}_{in}(s)$ and $\underline{Z}_{out}(s)$ and the voltage transfer function $\underline{H}_V(s) = \frac{\underline{V}_{out}(s)}{\underline{V}_{in}(s)}$?

Calculating the input impedance $\underline{Z}_{in}(s)$...

$$\begin{aligned}\underline{Z}_{in}(s) &= \underline{s}L + \left(R \parallel \frac{1}{\underline{s}C} \right) = \underline{s}L + \frac{R \frac{1}{\underline{s}C}}{R + \frac{1}{\underline{s}C}} = \\ &= \underline{s}L + \frac{R}{1 + \underline{s}RC} = \frac{R + \underline{s}L + \underline{s}^2 RLC}{1 + \underline{s}RC}\end{aligned}\quad (113)$$

... checking the units ...

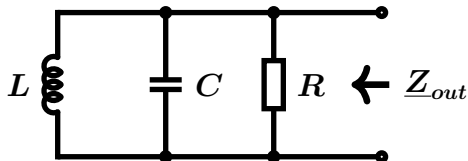
Numerator:

- $[R] = \Omega$
- $[\underline{s}L] = \frac{1}{s} \Omega s = \Omega$
- $[\underline{s}^2 RLC] = \frac{1}{s^2} \Omega \Omega s \frac{s}{\Omega} = \Omega$

Denominator:

- $[1] = 1$
 - $[\underline{s}RC] = \frac{1}{s} \Omega \frac{s}{\Omega} = 1$
- $$\Rightarrow [\underline{Z}_{in}] = \frac{\Omega + \Omega + \Omega}{1 + 1} = \Omega \checkmark$$

... shorting the voltage source to calculate the output impedance $\underline{Z}_{out}(\underline{s})$...



... calculating the output impedance...

$$\begin{aligned}\underline{Z}_{out}(\underline{s}) &= \underline{s}L || R || \frac{1}{\underline{s}C} = \frac{\underline{s}L \frac{R}{1+\underline{s}RC}}{\underline{s}L + \frac{R}{1+\underline{s}RC}} = \\ &= \frac{\underline{s}RL}{R + \underline{s}L + \underline{s}^2RLC}\end{aligned}\quad (114)$$

... and checking the units:

$$[\underline{Z}_{out}(\underline{s})] = \frac{\Omega^2}{\Omega} = \Omega \quad \checkmark$$

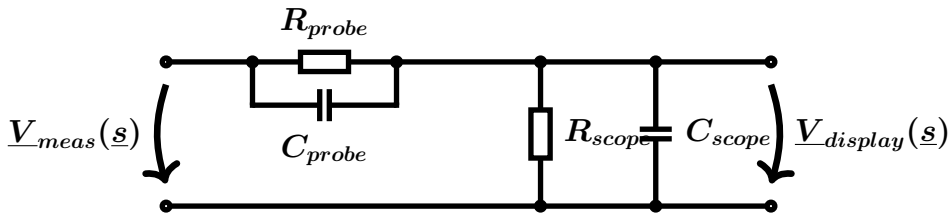
Finally calculating the voltage transfer function $\underline{H}_V(\underline{s})...$

$$\begin{aligned}\underline{H}_V(\underline{s}) &= \frac{R \parallel \frac{1}{\underline{s}C}}{\underline{s}L + R \parallel \frac{1}{\underline{s}C}} = \\ &= \frac{\frac{R}{1+\underline{s}RC}}{\underline{s}L + \frac{R}{1+\underline{s}RC}} = \\ &= \frac{R}{R + \underline{s}L + \underline{s}^2RLC}\end{aligned}\tag{115}$$

... and checking it's units:

$$[\underline{H}_V(\underline{s})] = \frac{\Omega}{\Omega} = 1 \checkmark$$

Example of Transfer Functions in an Oscilloscope Probe



Requirements & constraints for oscilloscope measurements:

- The displayed voltage $V_{display}(s)$ is supposed to be a linear representation of the actual measured voltage $V_{meas}(s)$ in the device under test (DUT) for any given frequency.
- Widely used values for oscilloscopes: $R_{scope} = 1 \text{ M}\Omega$ and $C_{scope} = 5 \text{ pF}$.
- The data processing in digital storage oscilloscopes (DSO) typically expects a voltage division from $V_{meas}(s)$ to $V_{display}(s)$ by a factor of 10.

Transferring the constraints into equations

- "linear representation" $\rightarrow \underline{H}_V(\underline{s}) = \frac{V_{display}}{V_{meas}}$ must be real $\Rightarrow \Im(\underline{H}_V(\underline{s})) \stackrel{!}{=} 0$.
- "voltage division by a factor of 10" $\rightarrow \underline{H}_V(\underline{s}) \stackrel{!}{=} \frac{1}{10}$.

Setting up the voltage transfer function from measured voltage to displayed voltage

$$\begin{aligned}
 \underline{H}_V(\underline{s}) &= \frac{\frac{R_{scope}}{1 + \underline{s}R_{scope}C_{scope}}}{\frac{R_{scope}}{1 + \underline{s}R_{scope}C_{scope}} + \frac{R_{probe}}{1 + \underline{s}R_{probe}C_{probe}}} \\
 &= \frac{R_{scope} + \underline{s}R_{scope}R_{probe}C_{probe}}{R_{scope} + R_{probe} + \underline{s}R_{scope}R_{probe}(C_{scope} + C_{probe})}
 \end{aligned} \tag{116}$$

Satisfying the requirement for voltage division by 10:

$$\Rightarrow \underline{H}_V(0) = \frac{R_{scope}}{R_{scope} + R_{probe}} \stackrel{!}{=} \frac{1}{10}$$

with $R_{scope} = 1 \text{ M}\Omega$

(117)

$$\Rightarrow R_{probe} = 10R_{scope}\left(1 - \frac{1}{10}\right) = 9R_{scope} = 9 \text{ M}\Omega$$

Satisfying the requirement for linear scaling from measured to displayed voltage:

$$\begin{aligned}
 \underline{H}_V(\underline{s}) &= \frac{(R_{scope} + \underline{s} R_{scope} R_{probe} C_{probe})(R_{scope} + R_{probe} - \underline{s} R_{scope} R_{probe} (C_{scope} + C_{probe}))}{(R_{scope} + R_{probe})^2 - \underline{s}^2 (R_{scope} R_{probe} (C_{scope} + C_{probe}))^2} \\
 &= \frac{\Re \left\{ \overbrace{R_{scope} (R_{scope} + R_{probe}) + \underline{s}^2 R_{scope}^2 R_{probe}^2 C_{probe} (C_{scope} + C_{probe})} \right\}}{(R_{scope} + R_{probe})^2 - \underline{s}^2 (R_{scope} R_{probe} (C_{scope} + C_{probe}))^2} \quad (118)
 \end{aligned}$$

$$\begin{aligned}
 &+ \frac{\Im \left\{ \overbrace{\underline{s} (R_{scope} R_{probe} (R_{scope} + R_{probe}) C_{probe} - R_{scope}^2 R_{probe} (C_{scope} + C_{probe}))} \right\}}{(R_{scope} + R_{probe})^2 - \underline{s}^2 (R_{scope} R_{probe} (C_{scope} + C_{probe}))^2} \\
 \Rightarrow \Im(\underline{H}_V(\underline{s})) &\stackrel{!}{=} 0 \Rightarrow (R_{scope} + R_{probe}) C_{probe} = R_{scope} (C_{scope} + C_{probe}) \\
 \Rightarrow R_{probe} C_{probe} &= R_{scope} C_{scope} \Rightarrow C_{probe} = \frac{R_{scope}}{R_{probe}} C_{scope} = \frac{1}{9} 5 \text{ pF} \approx 556 \text{ fF} \quad (119)
 \end{aligned}$$

LaPlace and Transfer Functions

Exercise 3.16 Equivalent Impedance of series RC

 / ⌚ 15 min

What is the equivalent impedance of a series RC connection? Select a resistor value R . What capacitance C gives a pole at 0 rad s^{-1} and a zero at -1 krad s^{-1} ?

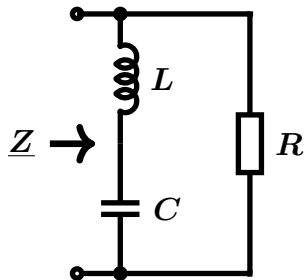
Exercise 3.17 Equivalent Impedance of parallel RC

 / ⌚ 10 min

What is the equivalent impedance of a parallel RC connection? Select a resistor value R . What capacitance C gives a pole at -10 krad s^{-1} ?

LaPlace and Transfer Functions

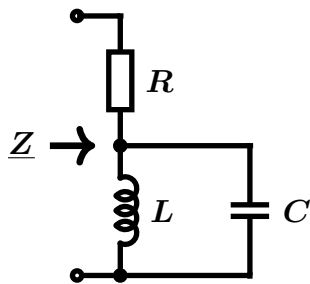
Exercise 3.18 Equivalent impedance of LCR circuit

 /  10 min

- What is the equivalent input impedance $\underline{Z}$ (as a rational function) of the circuit?
- What is the equivalent input admittance $\underline{Y}$ of the circuit?
- For $R = 2 \text{ k}\Omega$ and $C = 100 \text{ nF}$: what inductance L puts the zeros of Z at $\pm j5000 \text{ rad s}^{-1}$?
- Where are the poles for those component values?

LaPlace and Transfer Functions

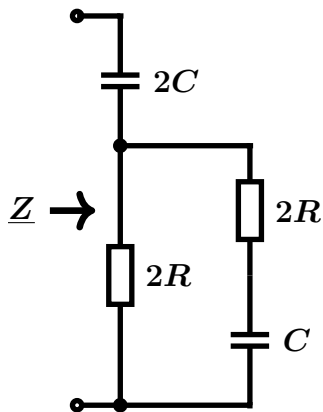
Exercise 3.19 Equivalent impedance of RLC circuit

 /  10 min

- a** What is the equivalent input impedance $\underline{Z}$ (as a rational function) of the circuit?
- b** What is the equivalent input admittance $\underline{Y}$ (as a rational function) of the circuit?
- c** For $R = 10 \text{ k}\Omega$: How can the inductance L and capacitance C be selected to locate the poles of Z at $\pm j200 \text{ krad s}^{-1}$?
- d** Where are the zeros of $\underline{Z}$ for those component values?

LaPlace and Transfer Functions

Exercise 3.20 Equivalent impedance of CRRC circuit

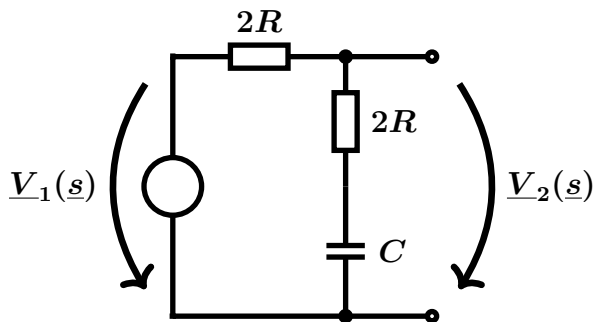
 /  10 min

- a What is the equivalent input impedance $\underline{Z}$ (as a rational function) of the circuit?
- b Where are its poles and zeros?
- c What is the equivalent input admittance $\underline{Y}$ (as a rational function) of the circuit?
- d Assuming that the components were dimensioned in a way, so that the poles of $\underline{Z}$ were originally located at -20 krad s^{-1} and afterwards the resistance R is reduced to 25% of its original value, where would the poles move to after the change?

LaPlace and Transfer Functions

Exercise 3.21 Equivalent input impedance and voltage transfer function /

🕒 10 min

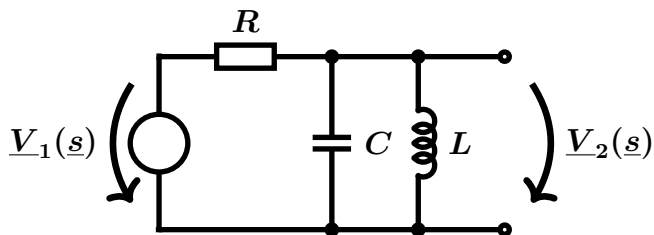


- a What is the impedance, the source $\underline{V}_1(\underline{s})$ is loaded with?
- b What is the voltage transfer function $\underline{H}_V(\underline{s}) = \frac{\underline{V}_2(\underline{s})}{\underline{V}_1(\underline{s})}$?

LaPlace and Transfer Functions

Exercise 3.22 Equivalent input impedance and voltage transfer function 🍺🍺🍺/

🕒 10 min



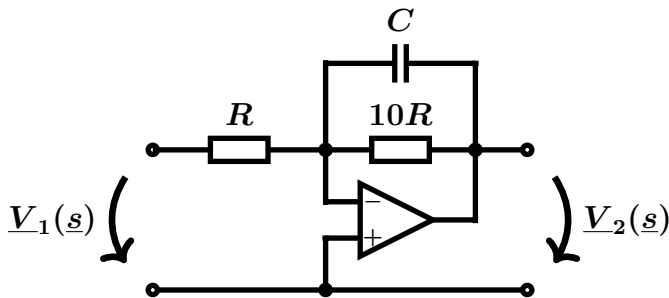
- a What is the impedance, the source $\underline{V}_1(\underline{s})$ is loaded with?
- b What is the voltage transfer function $\underline{H}_V(\underline{s}) = \frac{\underline{V}_2(\underline{s})}{\underline{V}_1(\underline{s})}$?

LaPlace and Transfer Functions

Exercise 3.23 Equivalent input impedance and voltage transfer function



15 min

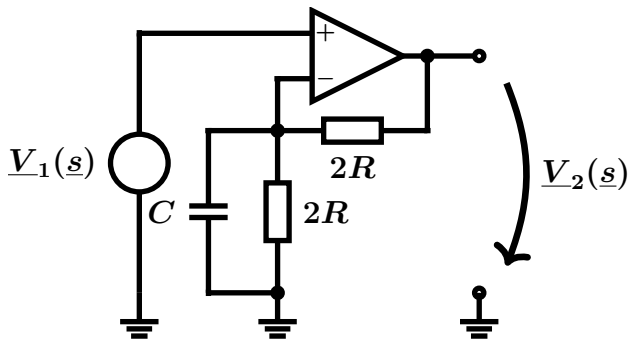


- a What is the impedance, the source $\underline{V}_1(\underline{s})$ is loaded with?
- b What is the voltage transfer function $\underline{H}_V(\underline{s}) = \frac{\underline{V}_2(\underline{s})}{\underline{V}_1(\underline{s})}$?

LaPlace and Transfer Functions

Exercise 3.24 Equivalent input impedance and voltage transfer function /

10 min



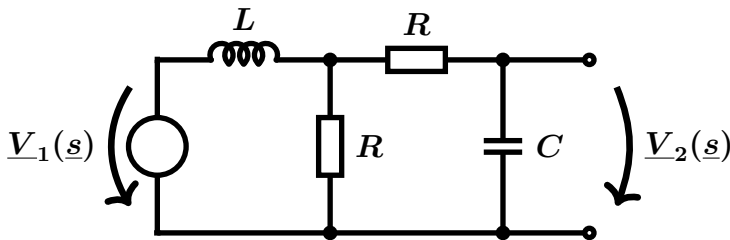
- a What is the impedance, the source $\underline{V}_1(s)$ is loaded with?
- b What is the voltage transfer function $\underline{H}_V(s) = \frac{\underline{V}_2(s)}{\underline{V}_1(s)}$?

LaPlace and Transfer Functions

Exercise 3.25 Voltage transfer function

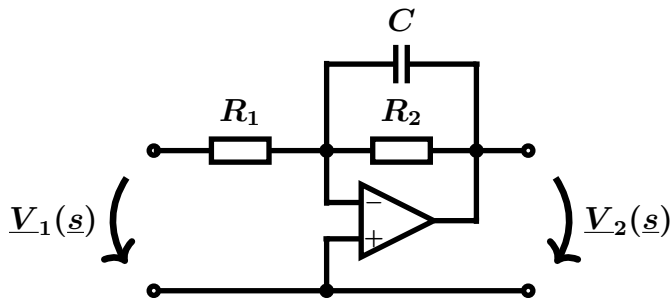
 / ⌚ 10 min

What is the voltage transfer function $\underline{H}_V(\underline{s}) = \frac{\underline{V}_2(\underline{s})}{\underline{V}_1(\underline{s})}$?



LaPlace and Transfer Functions

Exercise 3.26 Voltage transfer function and pole placement

 /  10 min

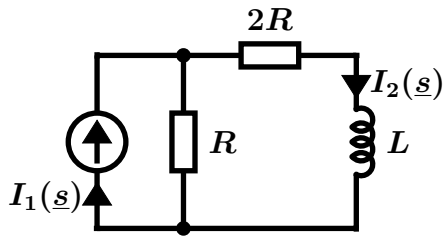
What values for R_1 , R_2 and C place a pole of the voltage transfer function

$$\underline{H}_V(\underline{s}) = \frac{V_2(\underline{s})}{V_1(\underline{s})} \text{ at}$$

$\underline{s} = -250 \text{ krad s}^{-1}$ and its DC gain $\frac{R_2}{R_1} = 100$?

LaPlace and Transfer Functions

Exercise 3.27 Current transfer function and pole placement

 / ⌚ 10 min

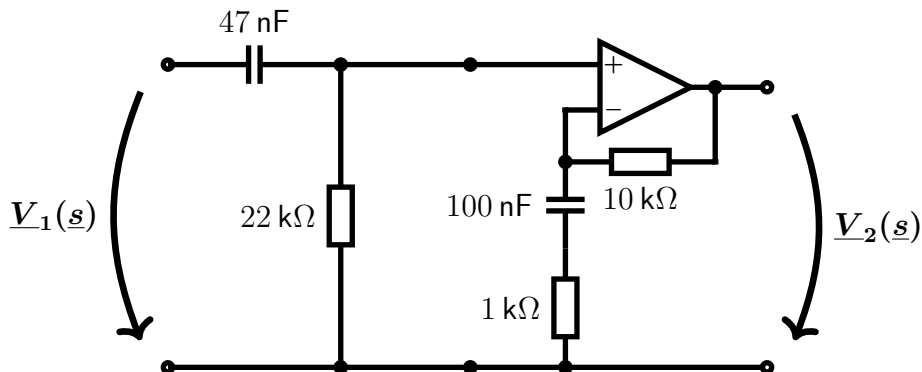
- What is the current transfer function $\underline{H}_I(\underline{s}) = \frac{I_2(\underline{s})}{I_1(\underline{s})}$?
- b What values of R and L place a pole $\underline{H}_I(\underline{s})$ at $\underline{s} = -377 \text{ rad s}^{-1}$?

LaPlace and Transfer Functions

Exercise 3.28 Voltage transfer function and cascaded networks

 /  15 min

- a What is the voltage transfer function $\underline{H}_V(\underline{s}) = \frac{V_2(\underline{s})}{V_1(\underline{s})}$?
- b Where are its poles and zeros located?



System Responses

with $\underline{Hs} = \frac{V_{out}(s)}{V_{in}(s)}$, i.e. $\underline{H}$ is a voltage transfer function

- impulse response: $v_{in}(t) = \delta(t) \circ \bullet \underline{V}_{in}(s) = 1$

$$\Rightarrow \underline{V}_{out} = \underline{H}(s)$$

- step response: $v_{in}(t) = 0 \forall t < 0, 1 \forall t \geq 0 \circ \bullet \underline{V}_{in}(s) = \frac{1}{s}$

$$\Rightarrow \underline{V}_{out} = \frac{\underline{H}(s)}{s}$$

- sinusoidal response: $v_{in}(t) = 0 \forall t < 0, \sin(\omega t) \forall t \geq 0 \circ \bullet \underline{V}_{in}(s) = \frac{\omega}{s^2 + \omega^2}$

$$\Rightarrow \underline{V}_{out} = \underline{H}(s) \frac{\omega}{s^2 + \omega^2}$$

Transfer Functions

For a given transfer function $\underline{H}(\underline{s})$:

$$\underline{H}(\underline{s}) = k \frac{(\underline{s} - z_1)(\underline{s} - z_2) \dots (\underline{s} - z_n)}{(\underline{s} - p_1)(\underline{s} - p_2) \dots (\underline{s} - p_m)} \quad (120)$$

Partial Fraction Decomposition for $n < m$

$$\underline{H}(\underline{s}) = \frac{k_1}{\underline{s} - p_1} + \frac{k_2}{\underline{s} - p_2} + \dots + \frac{k_m}{\underline{s} - p_m} \quad (121)$$

where $k_i = (\underline{s} - p_i) \underline{H}(\underline{s})|_{\underline{s}=p_i}$

Example of Partial Fraction Decomposition for $n < m$:

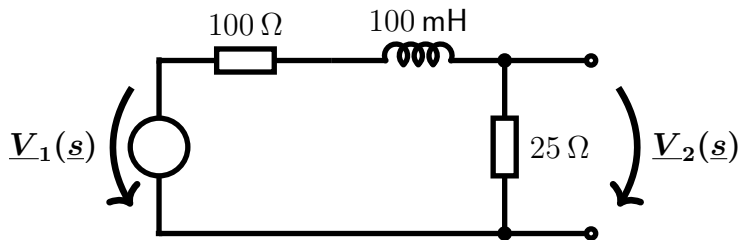
$$\begin{aligned}
 \underline{H}(\underline{s}) &= 2 \frac{(\underline{s} + 3)}{\underline{s} (\underline{s} + 1) (\underline{s} + 2)} \\
 k_1 &= \underline{s} H|_{\underline{s}=0} = 2 \frac{(\underline{s} + 3)}{(\underline{s} + 1) (\underline{s} + 2)} \Big|_{\underline{s}=0} = 3 \\
 k_2 &= (\underline{s} + 1) H|_{\underline{s}=-1} = 2 \frac{(\underline{s} + 3)}{\underline{s} (\underline{s} + 2)} \Big|_{\underline{s}=-1} = -4 \\
 k_3 &= (\underline{s} + 2) H|_{\underline{s}=-2} = 2 \frac{(\underline{s} + 3)}{\underline{s} (\underline{s} + 1)} \Big|_{\underline{s}=-2} = 1 \\
 \Rightarrow \underline{H}(\underline{s}) &= \frac{3}{\underline{s}} - \frac{4}{\underline{s} + 1} + \frac{1}{\underline{s} + 2} \\
 \bullet \circ h(t) &= 3u(t) + (e^{-2t} - 4e^{-t}) u(t)
 \end{aligned} \tag{122}$$

Example of Partial Fraction Decomposition for $n \geq m$:

$$\begin{aligned}\underline{H}(\underline{s}) &= \frac{\underline{s}^2 - 4}{\underline{s}^2 + 4} = \\&= \frac{\underline{s}^2 + 4 - 8}{\underline{s}^2 + 4} = \\&= \frac{\underline{s}^2 + 4}{\underline{s}^2 + 4} - 8 \frac{1}{\underline{s}^2 + 4} = \\&= 1 - 8 \frac{1}{\underline{s}^2 + 4} = \\&= 1 - 4 \frac{2}{\underline{s}^2 + 4} \\&\bullet \rightarrow h(t) = \delta(t) - 4 \sin(2t)u(t)\end{aligned}\tag{123}$$

Transfer Functions & System Responses

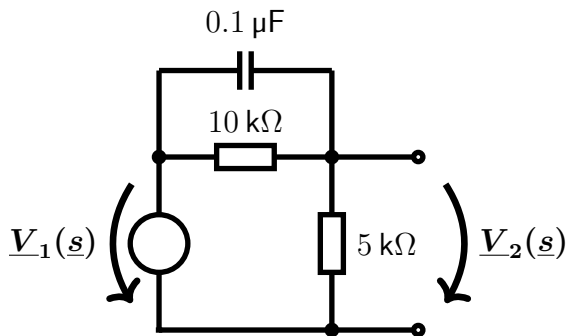
Exercise 3.29 Impulse & Step Response of RLR circuit

 /  15 min

- What is $v_2(t)$ for $v_1(t) = \delta(t)$?
- What is $v_2(t)$ for $v_1(t) = u(t)$?

Transfer Functions & System Responses

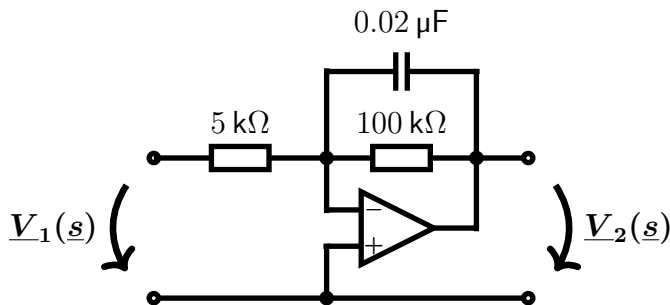
Exercise 3.30 Impulse & Step Response of a passive RCR circuit 📱📱📱 / ⌚ 10 min



- a What is $v_2(t)$ for $v_1(t) = \delta(t)$?
- b What is $v_2(t)$ for $v_1(t) = u(t)$?

Transfer Functions & System Responses

Exercise 3.31 Impulse & Step Response of an active RCR circuit  / ⌚ 10 min



- a What is the impulse response of the circuit?
- b What is the step response of the circuit?

Network Analysis

Node-Voltage-Analysis

- 1 convert all sources into current sources (Norton)
- 2 define reference node (ground)
- 3 apply Kirchhoffs current law to all other nodes and write down as $\sum YV = I_{src}$.
- 4 write down in matrix form
- 5 apply Cramer's rule
 - $\Rightarrow$ all voltages are known
 - $\rightarrow$ circuit is calculated!

Node-Voltage-Analysis

- 1 convert all sources into current sources (Norton)
- 2 define reference node (ground)
- 3 apply Kirchhoffs current law to all other nodes and write down as $\sum YV = I_{src}$.
- 4 write down in matrix form
- 5 apply Cramer's rule
 $\Rightarrow$ all voltages are known
 $\rightarrow$ circuit is calculated!

Mesh-Current-Analysis

- 1 convert all sources into voltage sources (Thevenin)
- 2 choose a connection of all but one nodes and define all loops
- 3 apply Kirchhoffs voltage law to all loops and write down as $\sum ZI = V_{src}$.
- 4 write down in matrix form
- 5 apply Cramer's rule
 $\Rightarrow$ all currents are known
 $\rightarrow$ circuit is calculated!

Node-Voltage-Analysis

- 1 convert all sources into current sources (Norton)
- 2 define reference node (ground)
- 3 apply Kirchhoffs current law to all other nodes and write down as $\sum YV = I_{src}$.
- 4 write down in matrix form
- 5 apply Cramer's rule
 $\Rightarrow$ all voltages are known
 $\rightarrow$ circuit is calculated!

Mesh-Current-Analysis

- 1 convert all sources into voltage sources (Thevenin)
- 2 choose a connection of all but one nodes and define all loops
- 3 apply Kirchhoffs voltage law to all loops and write down as $\sum ZI = V_{src}$.
- 4 write down in matrix form
- 5 apply Cramer's rule
 $\Rightarrow$ all currents are known
 $\rightarrow$ circuit is calculated!

Applying Ohms law to each component gives the remaining signals.

Derivation Mathematical Background: Cramers Rule

Any set of linear equations

$$\begin{aligned}a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= y_1 \\a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= y_2 \\&\vdots \\a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= y_m\end{aligned}\tag{124}$$

can be rewritten in matrix form

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \ddots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{pmatrix}\tag{125}$$

Derivation Solving for x through

$$x_i = \frac{\det(A_i)}{\det(A)} \quad (126)$$

where

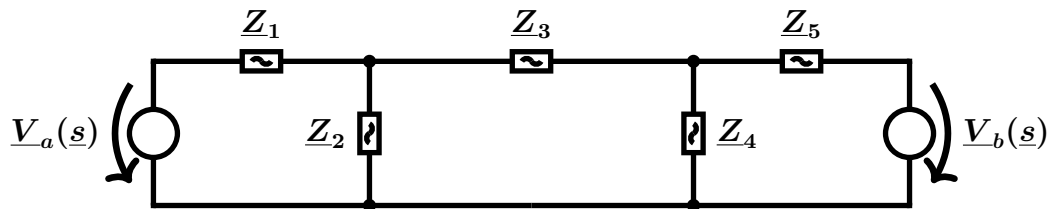
$\det(A)$ is the determinant of A and

$\det(A_i)$ is the determinant of A with the i -th column replaced by vector y

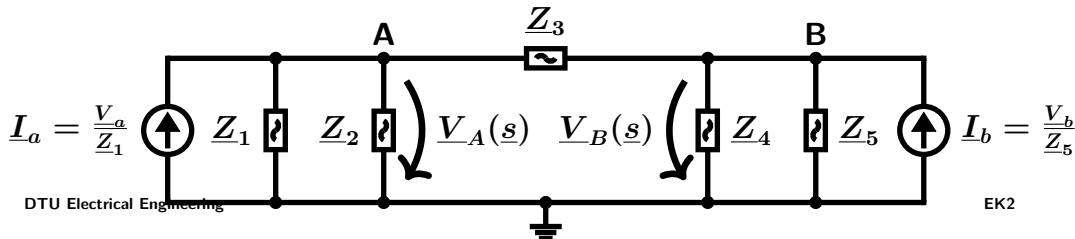
Circuit Simulators

Circuit simulators like Spice (and its derivatives) use these procedures (node voltage analysis and/or mesh current analysis) and solve the matrices with the selected math solvers to calculate the circuit, that the user entered under the given constraints from simulation parameters.

Example of Node Voltage Analysis - Circuit



Applying step 1 (converting all sources into current sources (Norton)) and step 2 defining a reference (ground) node:



Example of Node Voltage Analysis - Equations

Step 3: applying Kirchhoffs current law and analytically describe the circuit:

$$\text{A: } (\underline{Y}_1 + \underline{Y}_2)\underline{V}_A + \underline{Y}_3(\underline{V}_A - \underline{V}_B) = \underline{I}_a \quad (127)$$

$$\text{B: } (\underline{Y}_4 + \underline{Y}_5)\underline{V}_B + \underline{Y}_3(\underline{V}_B - \underline{V}_A) = \underline{I}_b$$

with $\underline{Y}_i = \frac{1}{\underline{Z}_i}$

Example of Node Voltage Analysis - Matrices

Step 4: write the equations in matrix form:

$$\begin{pmatrix} \underline{Y}_1 + \underline{Y}_2 + \underline{Y}_3 & -\underline{Y}_3 \\ -\underline{Y}_3 & \underline{Y}_3 + \underline{Y}_4 + \underline{Y}_5 \end{pmatrix} \begin{pmatrix} \underline{V}_A \\ \underline{V}_B \end{pmatrix} = \begin{pmatrix} \underline{I}_a \\ \underline{I}_b \end{pmatrix} \quad (128)$$

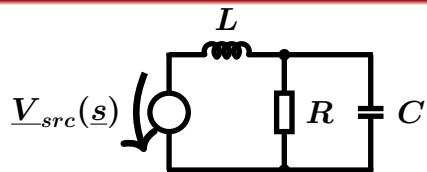
Example of Node Voltage Analysis - Solving for node voltages $\underline{V}_A$ & $\underline{V}_B$

$$\begin{aligned}
 \underline{V}_A &= \frac{\begin{vmatrix} \underline{I}_a & -\underline{Y}_3 \\ \underline{I}_b & \underline{Y}_3 + \underline{Y}_4 + \underline{Y}_5 \end{vmatrix}}{\begin{vmatrix} \underline{Y}_1 + \underline{Y}_2 + \underline{Y}_3 & -\underline{Y}_3 \\ -\underline{Y}_3 & \underline{Y}_3 + \underline{Y}_4 + \underline{Y}_5 \end{vmatrix}} \\
 &= \frac{(\underline{Y}_3 + \underline{Y}_4 + \underline{Y}_5)\underline{I}_a + \underline{Y}_3\underline{I}_b}{(\underline{Y}_1 + \underline{Y}_2 + \underline{Y}_3)(\underline{Y}_3 + \underline{Y}_4 + \underline{Y}_5) - \underline{Y}_3\underline{Y}_3} \quad (129)
 \end{aligned}$$

$$\begin{aligned}
 \underline{V}_B &= \frac{\begin{vmatrix} \underline{Y}_1 + \underline{Y}_2 + \underline{Y}_3 & \underline{I}_a \\ -\underline{Y}_3 & \underline{I}_b \end{vmatrix}}{\underline{Y}_{13} + \underline{Y}_{14} + \underline{Y}_{15} + \underline{Y}_{23} + \underline{Y}_{24} + \underline{Y}_{25} + \underline{Y}_{34} + \underline{Y}_{35}} \\
 &= \frac{(\underline{Y}_1 + \underline{Y}_2 + \underline{Y}_3)\underline{I}_b + \underline{Y}_3\underline{I}_a}{\underline{Y}_{13} + \underline{Y}_{14} + \underline{Y}_{15} + \underline{Y}_{23} + \underline{Y}_{24} + \underline{Y}_{25} + \underline{Y}_{34} + \underline{Y}_{35}}
 \end{aligned}$$

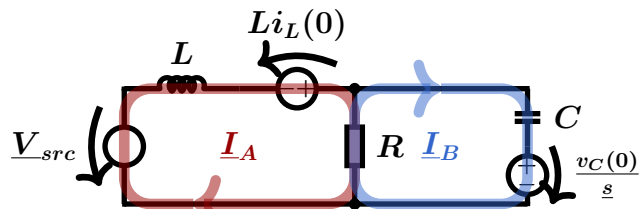
with $\underline{Y}_{\alpha\beta} = \sum \underline{Y}_{\alpha\beta\epsilon\kappa\zeta}$

Example of Mesh Current Analysis - Circuit



$$i_L(0) \neq 0 \text{ \& } v_C(0) \neq 0$$

Applying step 1 (drawing the circuit with voltage sources only) & step 2 (choosing R as connection of 2 / 3 nodes & define the loop currents $\underline{I}_A$ & $\underline{I}_B$):



Example of Mesh Current Analysis - Equations

Step 3: applying Kirchhoffs voltage law and analytically describe the circuit:

$$\begin{aligned} \text{A: } \underline{sL}\underline{I}_A + R(\underline{I}_A - \underline{I}_B) &= \underline{V}_{src} + Li_L(0) \\ \text{B: } R(\underline{I}_B - \underline{I}_A) + \frac{1}{\underline{sC}}\underline{I}_B &= -\frac{v_C(0)}{\underline{s}} \end{aligned} \quad (130)$$

Example of Mesh Current Analysis - Matrices

Step 4: write the equations in matrix form:

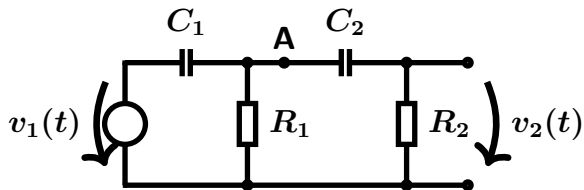
$$\begin{pmatrix} R + \underline{sL} & -R \\ -R & R + \frac{1}{\underline{sC}} \end{pmatrix} \begin{pmatrix} \underline{I}_A \\ \underline{I}_B \end{pmatrix} = \begin{pmatrix} \underline{V}_{src} + Li_L(0) \\ -\frac{v_C(0)}{\underline{s}} \end{pmatrix} \quad (131)$$

Example of Node Voltage Analysis - Solving for node voltages $\underline{V}_A$ & $\underline{V}_B$

Step 5: calculating the relevant determinants and dividing them:

$$\begin{aligned}\Rightarrow \underline{I}_A &= \frac{(R + \frac{1}{\underline{s}C})(\underline{V}_{src} + Li_L(0)) - R \frac{v_C(0)}{\underline{s}}}{\frac{L}{C} + \frac{R}{\underline{s}C} + \underline{s}RL} \\ \Rightarrow \underline{I}_B &= \frac{R(\underline{V}_{src} + Li_L(0)) - (R + \underline{s}L) \frac{v_C(0)}{\underline{s}}}{\frac{L}{C} + \frac{R}{\underline{s}C} + \underline{s}RL}\end{aligned}\tag{132}$$

Exercise 3.32 Node Voltage Analysis

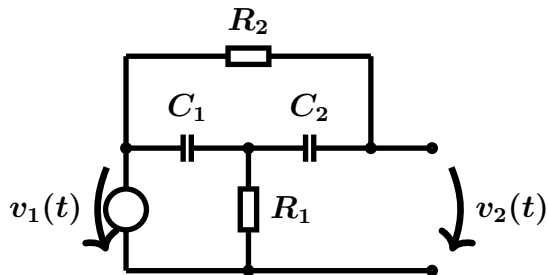
 /  20 min

All initial conditions are zero.

- What are the node voltage equations in the $\underline{s}$ -domain?

- Solve the equations for $\underline{V}_2(\underline{s})$?
- Insert an operational amplifier buffer at point A . How did this change the denominator and where are the poles now?
- What values for R_1 , R_2 , C_1 and C_2 place one pole at -10 krad s^{-1} and another one at -100 krad s^{-1} for one of the circuits?

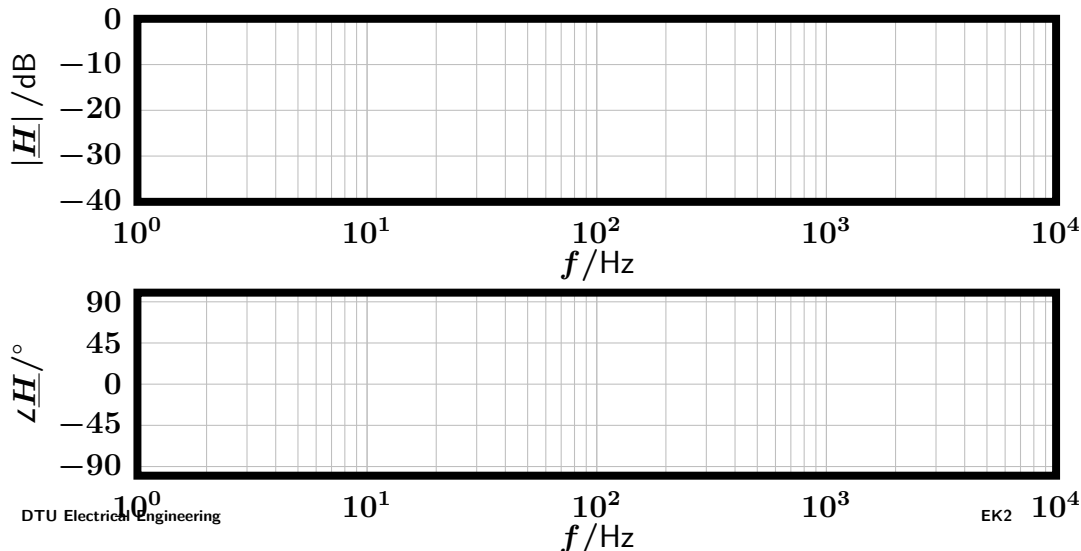
Exercise 3.33 Mesh Current Analysis

 /  15 min

All initial conditions are zero.

- a What are the mesh current equations in the $\underline{s}$ -domain?
- b With the help of the mesh current equations, what is the relationship between input voltage $\underline{V}_1(\underline{s})$ and output voltage $\underline{V}_2(\underline{s})$?

Visualization of Transfer Functions



Impact of poles and zeros on Bode plot

	zero at $\underline{s} = z$	pole at $\underline{s} = p$
amplitude	rises by $+20$ dB per decade	falls with -20 dB per decade
phase	rises by $+90^\circ$ for $\underline{s} \rightarrow \infty$ and gives $+45^\circ$ at $\underline{s} = z$	falls with -90° for $\underline{s} \rightarrow \infty$ and gives -45° at $\underline{s} = p$

Section 4

Applications of Circuits

- **Applications of Circuits**

- Passive Filters
- Active Filters
- Motor Drives
- Amplifiers
- Oscillators

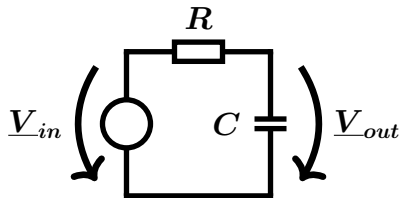
Passive Filters - First Order

IMPORTANT Terminology

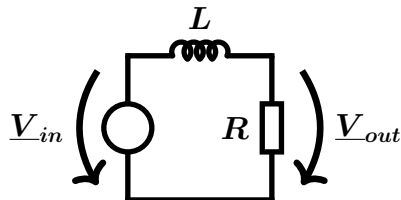
- τ is the "(characteristic) time constant" of the filter,
- f_c is the "corner frequency", where $f_c = \frac{1}{2\pi\tau}$ and the "angular corner frequency" $\omega_c = 2\pi f_c$,
- H_{-3dB} is defined at the frequency, where the output signal (e.g. voltage) is -3 dB , i.e. a factor of about $\sqrt{2}$ less, than the input signal (voltage). This means only $1/2$ the power left (at a given resistance - in communications technology typically $50\ \Omega$).
- "passband" is the frequency range, where the filter lets the input signal pass to the output, e.g. in case of the lowpass filter all frequencies $f < f_c$,
- "stopband" is the frequency range, where the filter suppresses the input signal, e.g. in case of the lowpass filter all frequencies $f \geq f_c$.
- "asymptotes" are a simplified representation of the filter characteristics.

Passive Filters - First Order

Lowpass - RC implementation



Lowpass - LR implementation

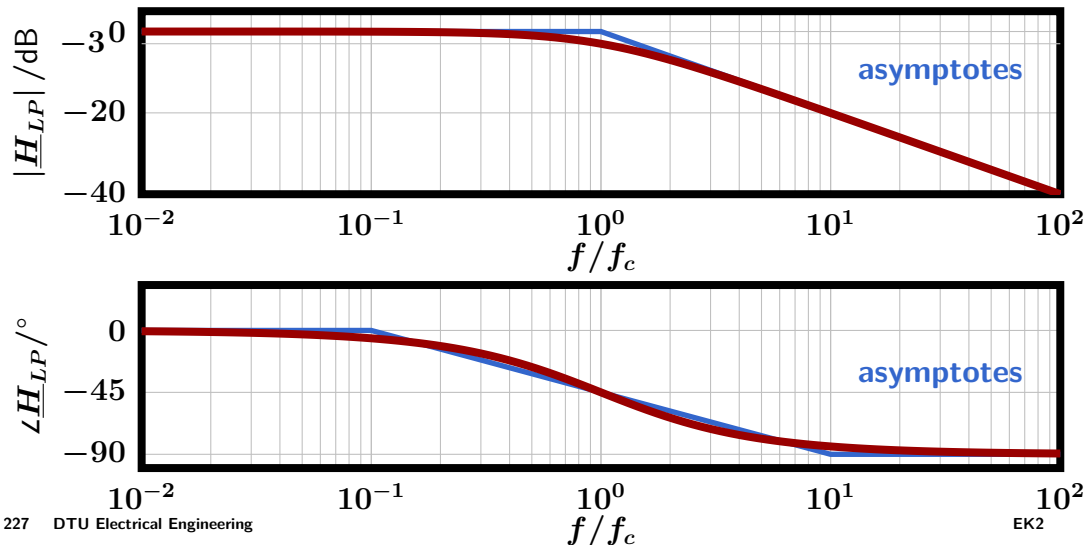


Lowpass - Transfer Function

$$\underline{H}_{LP}(\underline{s}) = \frac{\underline{V}_{out}(\underline{s})}{\underline{V}_{in}(\underline{s})} = \frac{1}{1 + \underline{s}\tau} \text{ with } \tau_{RC} = RC \text{ and } \tau_{LR} = \frac{L}{R} \quad (133)$$

Passive Filters - First Order

Lowpass - Bode Plot



Passive Filters - First Order

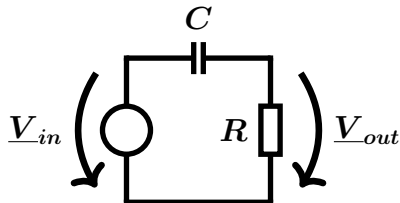


IMPORTANT Meaning - Lowpass

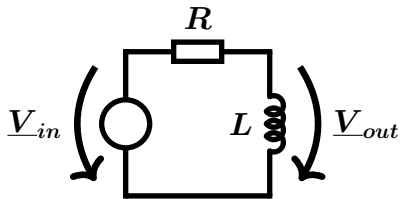
- At low frequencies $f \ll f_c$, the impedance of the capacitor is approaching ∞ and therefore can be neglected. The resistors behaviours is dominating the voltage divider. (Equivalent for the LR implementation)
 - The amplitude is frequency independent and
 - the filter has no influence on the phase.
- Above the corner frequency f_c , the impedance of the capacitance dominates.
 - The amplitude is falling with -20 dB per decade caused by the single pole f_c and
 - the phase is following an arctan shaped curve towards -90° .
 - At f_c , the frequency is half way at -45° .

Passive Filters - First Order

Highpass - CR implementation



Highpass - RL implementation

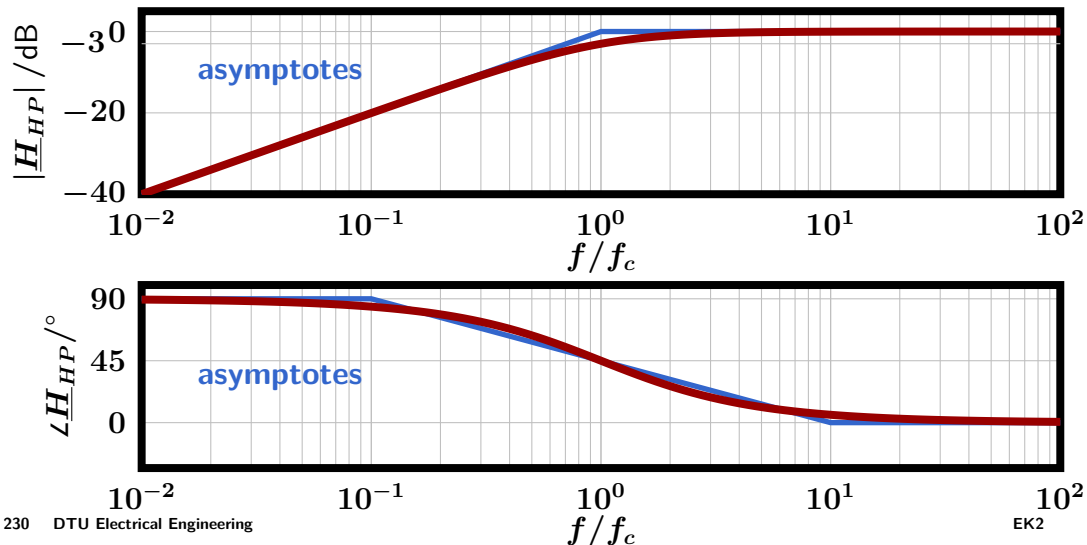


Highpass - Transfer Function

$$\underline{H}_{HP}(\underline{s}) = \frac{V_{out}(\underline{s})}{V_{in}(\underline{s})} = \frac{\underline{s}\tau}{1 + \underline{s}\tau} \quad \text{with } \tau_{RC} = RC \text{ and } \tau_{LR} = \frac{L}{R} \quad (134)$$

Passive Filters - First Order

Highpass - Bode Plot



Passive Filters - First Order

IMPORTANT Meaning - Highpass

- At low frequencies $f \ll f_c$, the impedance of the inductor is approaching 0 and therefore shorts the output. This short is dominating the voltage divider. (Equivalent for the CR implementation.)
 - The amplitude is raising with the impedance of the inductor by 20 dB per decade which is caused by the single zero at $f = 0$ in the transfer function and
 - the phase is following an arctan shaped curve falling from 90° .
 - At f_c , the frequency is half way at 45° .
- Above the corner frequency f_c , the impedance of the inductor approaches ∞ and the resistor dominates (limits the current delivered towards the output).
 - The amplitude is frequency independent and
 - the filter has no influence on the phase.

Passive Filters - Second Order

IMPORTANT General Form of Denominator in Transfer Function

$$1 + s2dT_0 + s^2T_0^2$$

with $T_0 = \frac{1}{2\pi f_c} = \frac{1}{\omega_c}$

and $d = \frac{1}{2Q}$

defines $B = \frac{\omega_c}{Q}$

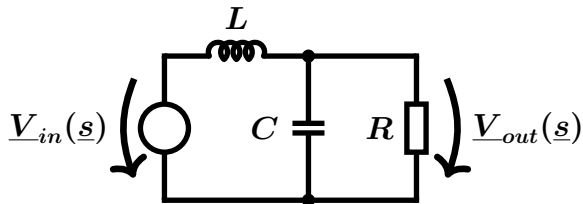
(135)

IMPORTANT Variables

- T_0 : characterizing time constant
- ω_c : angular cut-off frequency or "angular resonance frequency"
- f_c : cut-off frequency or "resonance frequency"
- d : damping factor
- Q : quality
- B : bandwidth

Passive Filters - Second Order

Lowpass - Circuit Diagram



Lowpass - Transfer Function

$$\begin{aligned}\underline{H}_V(s) &= \frac{\frac{R}{1+sRC}}{sL + \frac{R}{1+sRC}} = \\ &= \frac{1}{1 + s\frac{L}{R} + s^2LC}\end{aligned}\quad (136)$$

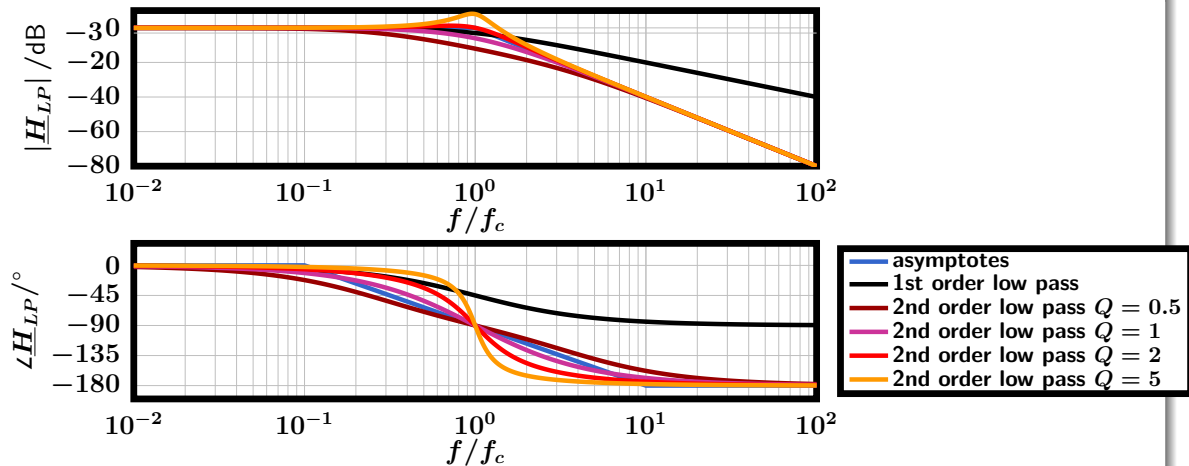
Passive Filters - Second Order

IMPORTANT Lowpass - Meaning

- At low frequencies, an inductor is a short and a capacitor is an open.
- At high frequencies, a capacitor is a short and an inductor is an open.
- The resistor is frequency neutral.
- Therefore in the lowpass, the low frequencies $f \ll f_c$
 - get shorted through to the output by the inductor and
 - the capacitor is high impedance, therefore has no influence.
- The high frequencies $f \gg f_c$
 - get blocked by the series inductor and
 - shorted across the load by the capacitor.
- At the resonance frequency $f = f_c$
 - the resistor adjusts the damping d .

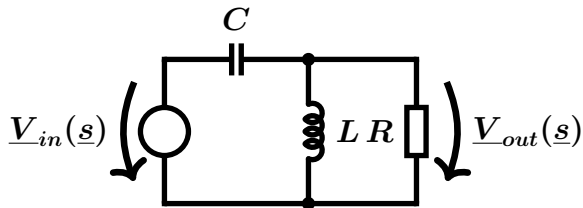
Passive Filters - Second Order

Lowpass - Bode Plot



Passive Filters - Second Order

Highpass - Circuit Diagram



Highpass - Transfer Function

$$\begin{aligned}
 \underline{H}_V(\underline{s}) &= \frac{\frac{\underline{s}RL}{R+\underline{s}L}}{\frac{1}{\underline{s}C} + \frac{\underline{s}RL}{R+\underline{s}L}} = \\
 &= \frac{\underline{s}^2 LC}{1 + \underline{s}\frac{L}{R} + \underline{s}^2 LC}
 \end{aligned} \tag{137}$$

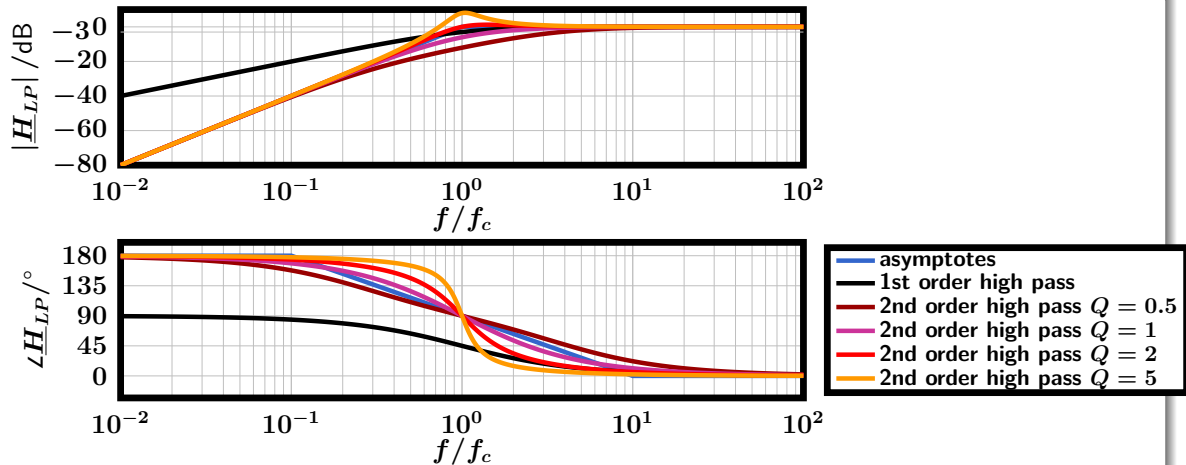
Passive Filters - Second Order

IMPORTANT Highpass - Meaning

- At low frequencies, an inductor is a short and a capacitor is an open.
- At high frequencies, a capacitor is a short and an inductor is an open.
- The resistor is frequency neutral.
- Therefore in the highpass, the low frequencies $f \ll f_c$
 - get blocked by series capacitor and
 - the inductor shorts them across the output.
- The high frequencies $f \gg f_c$
 - pass through the series capacitor and
 - the inductor is high impedance and has therefore no influence.
- At the resonance frequency $f = f_c$
 - the resistor adjusts the damping d .

Passive Filters - Second Order

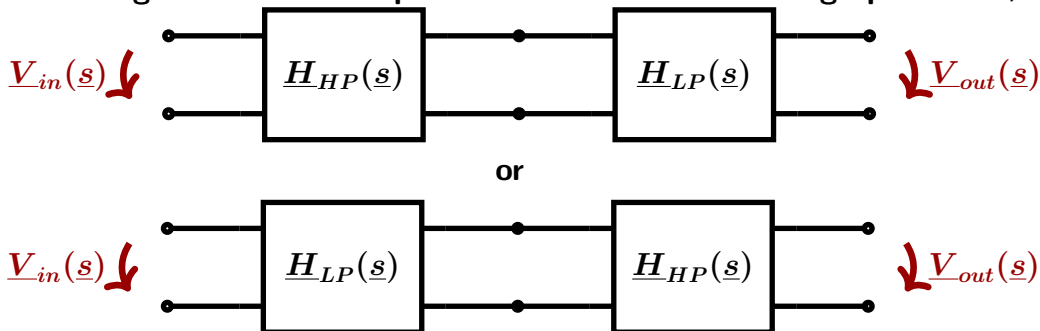
Highpass - Bode Plot



Passive Filters - Second Order

Bandpass - Cascade implementation - Block Diagrams

Cascading a first order lowpass filter and a first order high pass filter, like

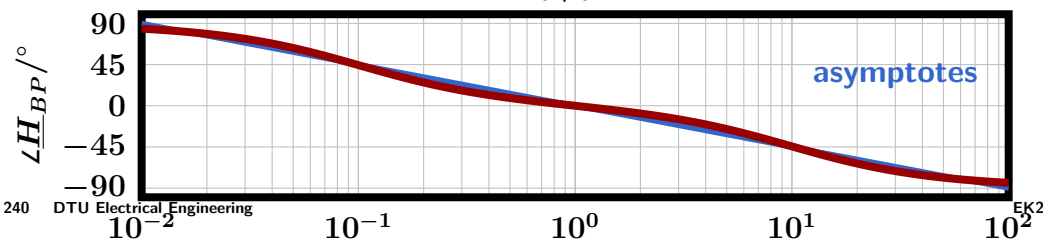
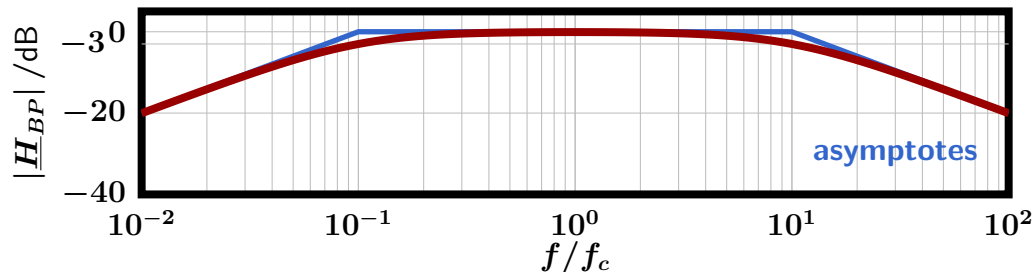


Bandpass - Cascade implementation - Transfer Function

$$H_{BP}(s) = H_{HP}(s)H_{LP}(s) = \frac{s\tau_{HP}}{(1 + s\tau_{HP})(1 + s\tau_{LP})} \quad \text{EK2} \quad (138)$$

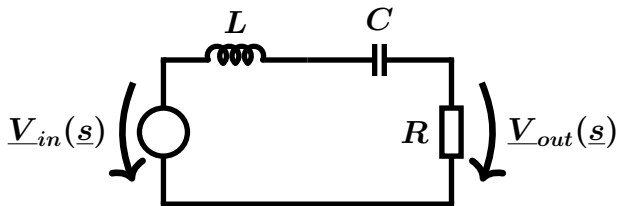
Passive Filters - Cascade implementation - Second Order

Bandpass - Bode Plot



Passive Filters - Second Order

Bandpass - LCR implementation - Circuit Diagram



Bandpass - LCR implementation - Transfer Function

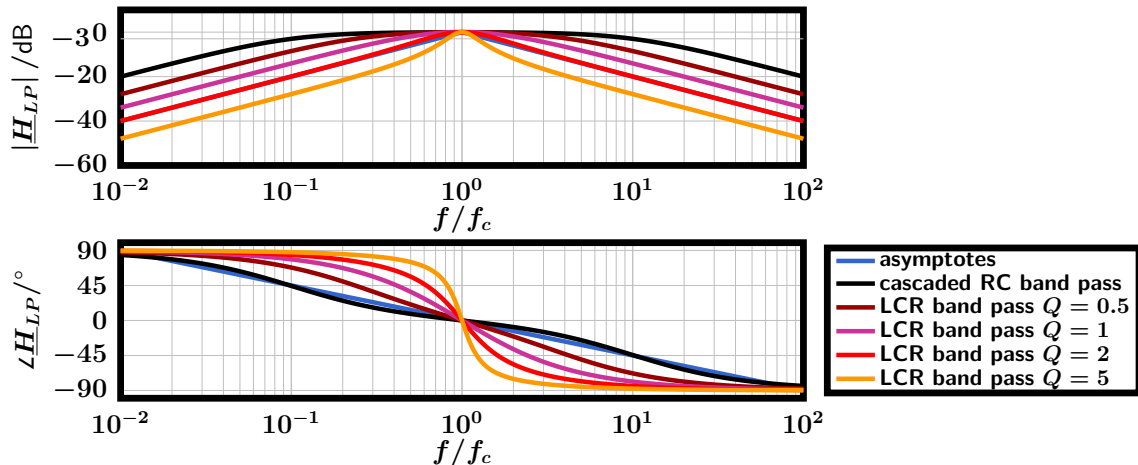
$$\begin{aligned}\underline{H}_V(\underline{s}) &= \frac{R}{R + \underline{s}L + \frac{1}{\underline{s}C}} = \\ &= \frac{\underline{s}RC}{1 + \underline{s}RC + \underline{s}^2LC}\end{aligned}\tag{139}$$

Passive Filters - Second Order

IMPORTANT Bandpass - Meaning

- At low frequencies, an inductor is a short and a capacitor is an open.
- At high frequencies, a capacitor is a short and an inductor is an open.
- The resistor is frequency neutral.
- Therefore in the bandpass, the low frequencies $f \ll f_c$
 - get blocked by series capacitor.
- The high frequencies $f \gg f_c$
 - get blocked by series inductor.
- Only at the resonance frequency $f = f_c$
 - the series impedance of the inductor and the capacitor are in the vicinity of the resistor and form a voltage divider while
 - the resistor adjusts the damping d .

Bandpass - Bode Plot



Passive Filters - Second Order

Bandstop - Stacking / Parallel implementation - Setup

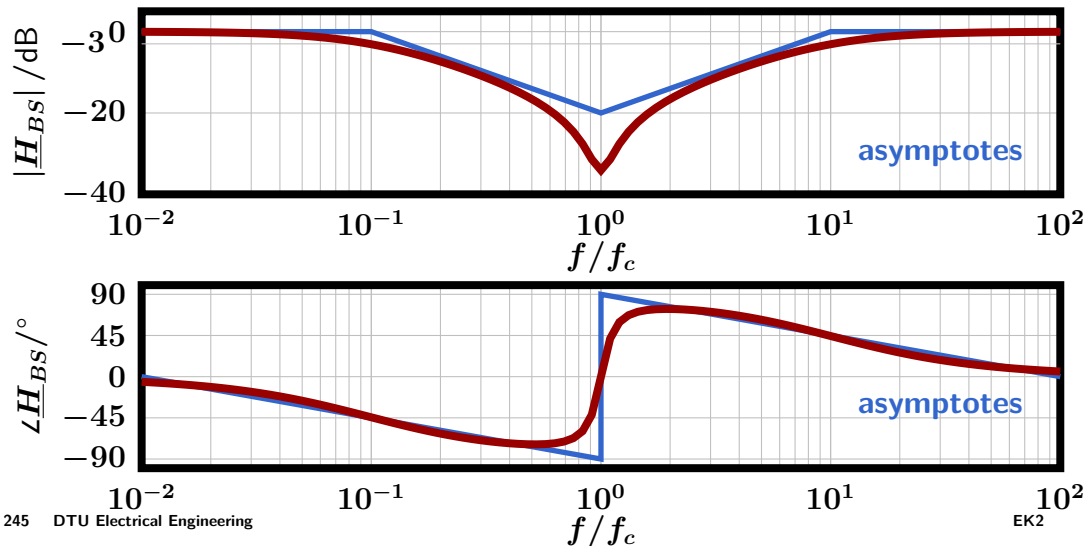
Adding a lowpass and a highpass filter. For voltages, adding means series connection (summing up in a loop), for currents adding means parallel connection (summing up in a node).

Bandstop - Stacking/ Parallel implementation - Transfer Function

$$\begin{aligned}\underline{H}_{BS}(\underline{s}) &= \underline{H}_{HP}(\underline{s}) + \underline{H}_{LP}(\underline{s}) \\ &= \frac{\underline{s}\tau_{HP}}{1 + \underline{s}\tau_{HP}} + \frac{1}{1 + \underline{s}\tau_{LP}} \\ &= \frac{\underline{s}\tau_{HP}(1 + \underline{s}\tau_{LP}) + (1 + \underline{s}\tau_{HP})}{(1 + \underline{s}\tau_{HP})(1 + \underline{s}\tau_{LP})}\end{aligned}\tag{140}$$

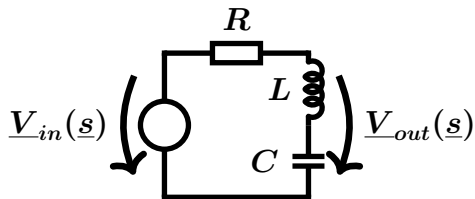
Passive Filters - Second Order

Bandstop - Stacking / Parallel implementation - Bode Plot



Passive Filters - Second Order

Bandstop - LCR implementation - Circuit Diagram



Bandstop - LCR implementation - Transfer Function

$$\begin{aligned}\underline{H}_V(\underline{s}) &= \frac{\underline{s}L + \frac{1}{\underline{s}C}}{R + \underline{s}L + \frac{1}{\underline{s}C}} = \\ &= \frac{1 + \underline{s}^2 LC}{1 + \underline{s}RC + \underline{s}^2 LC}\end{aligned}\tag{141}$$

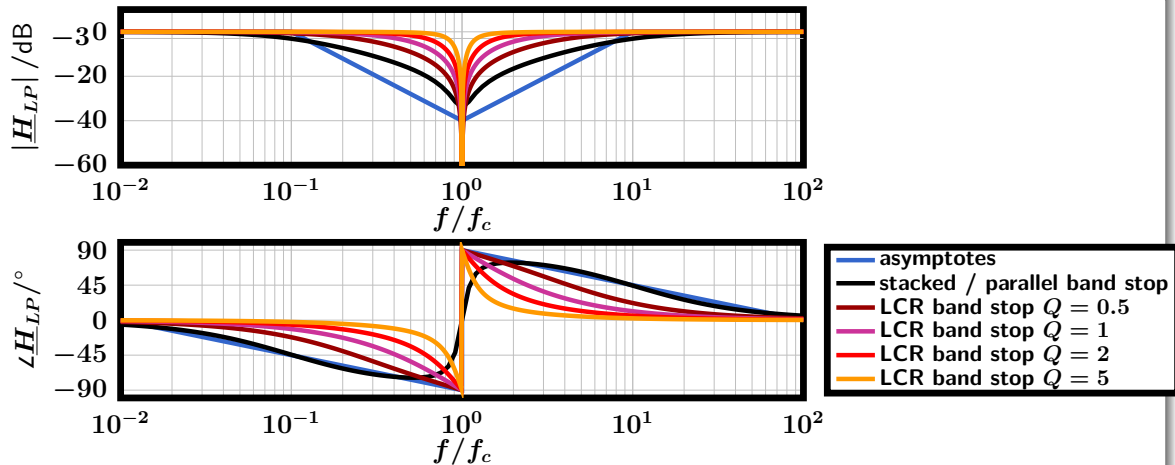
Passive Filters - Second Order

IMPORTANT Bandstop - Meaning

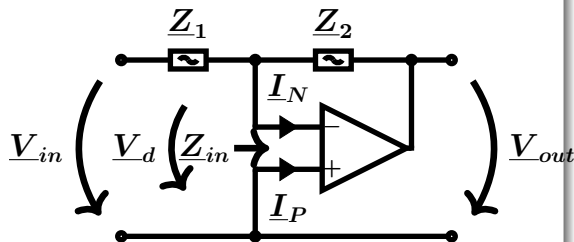
- At low frequencies, an inductor is a short and a capacitor is an open.
- At high frequencies, a capacitor is a short and an inductor is an open.
- The resistor is frequency neutral.
- Therefore in the bandstop, the low frequencies $f \ll f_c$
 - are kept at the output due to the high impedance of the capacitor.
- The high frequencies $f \gg f_c$
 - are kept at the output due to the high impedance of the inductor.
- Only at the resonance frequency $f = f_c$
 - the series impedance of the inductor and the capacitor are in the vicinity of the resistor and form a voltage divider with the resistor with a significant voltage drop across the resistor.
 - The resistor adjusts the damping d .

Passive Filters - Second Order

Bandstop - LCR implementation - Bode Plot



Inverting Amplifier - Circuit Diagram



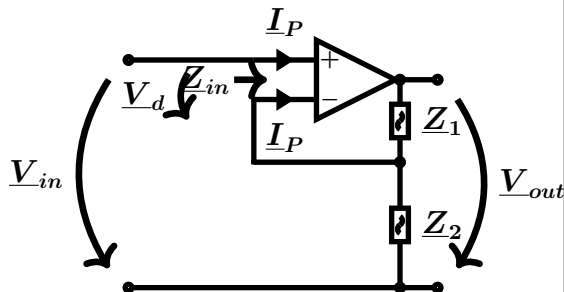
Inverting Amplifier - Conditions for Ideal Amplifier

$$\begin{aligned}
 \underline{Z}_{in} &= \infty \Omega & \underline{I}_N &= \underline{I}_P = 0 \text{ A} \\
 \underline{Z}_{out} &= 0 \Omega & \underline{V}_d &= \underline{V}_N - \underline{V}_P = 0 \text{ V} \\
 G &= \infty
 \end{aligned}
 \tag{142}$$

IMPORTANT Inverting Amplifier - Transfer Function

$$\begin{aligned}
 \underline{V}_{in} &= \underline{I} \underline{Z}_1 & \underline{V}_{out} &= -\underline{I} \underline{Z}_2 \\
 \Rightarrow \underline{H}_V &= \frac{\underline{V}_{out}}{\underline{V}_{in}} = -\frac{\underline{Z}_2}{\underline{Z}_1}
 \end{aligned}
 \tag{143}$$

Noninverting Amplifier - Circuit Diagram



Noninverting Amplifier - Conditions for Ideal Amplifier

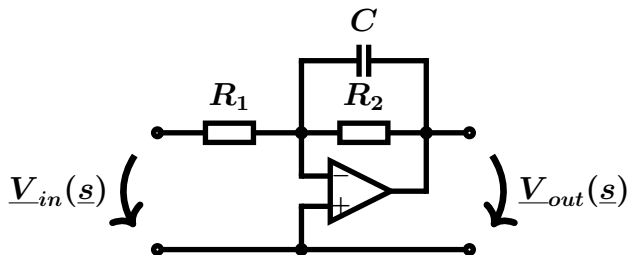
$$\begin{aligned}
 \underline{Z}_{in} &= \infty \Omega & \underline{I}_N &= \underline{I}_P = 0 \text{ A} \\
 \underline{Z}_{out} &= 0 \Omega & \underline{V}_d &= \underline{V}_N - \underline{V}_P = 0 \text{ V} \\
 G &= \infty
 \end{aligned}
 \tag{144}$$

IMPORTANT Noninverting Amplifier - Transfer Function

$$\begin{aligned}
 \underline{V}_{in} &= \underline{I} \underline{Z}_2 & \underline{V}_{out} &= \underline{I} (\underline{Z}_1 + \underline{Z}_2) \\
 \Rightarrow \underline{H}_V &= \frac{\underline{V}_{out}}{\underline{V}_{in}} = 1 + \frac{\underline{Z}_2}{\underline{Z}_1}
 \end{aligned}
 \tag{145}$$

Active Filters - First Order

Inverting Lowpass - Circuit Diagram



Active Filters - First Order

Inverting Lowpass - Transfer Function

$$\underline{H}_{LP} = -\frac{\frac{R_2}{1+sR_2C_2}}{R_1} = -\underbrace{\frac{R_2}{R_1}}_{DC\text{ gain}} \underbrace{\frac{1}{1+sR_2C_2}}_{\text{lowpass}} \quad (146)$$

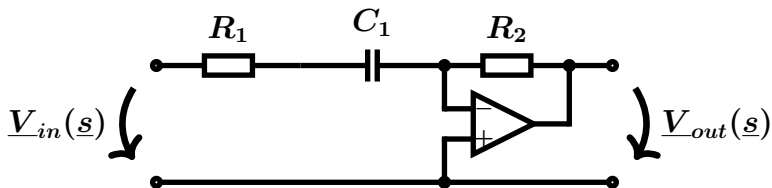
Further literature on active lowpass filters:

Noninverting lowpass:

https://www.electronics-tutorials.ws/filter/filter_5.html [7]

Active Filters - First Order

Inverting Highpass - Circuit Diagram



Active Filters - First Order

Inverting Highpass - Transfer Function

$$\underline{H}_{HP} = -\frac{R_2}{R_1 + \frac{1}{sC_1}} = -\frac{sR_2C_1}{1 + sR_1C_1} \quad (147)$$

Further literature on active highpass filters:

Noninverting highpass:

https://www.electronics-tutorials.ws/filter/filter_6.html [8]

Active Filters - Second Order

Further literature on second order filters:

Sallen-Key generic & specific implementations:

https://en.wikipedia.org/wiki/SallenKey_topology [9]

Further second order filters: <https://www.electronics-tutorials.ws/filter/second-order-filters.html> [10]

Further Sallen-Key literature:

<https://www.sciencedirect.com/topics/engineering/sallen-key> [11]

Applications of Circuits

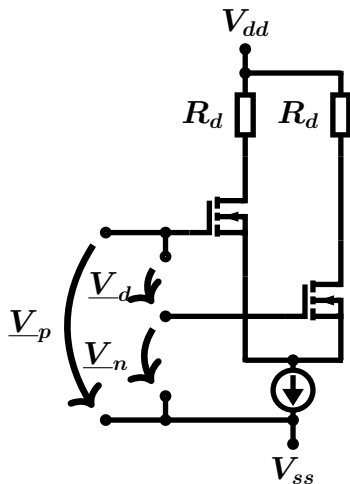
Motor Drives



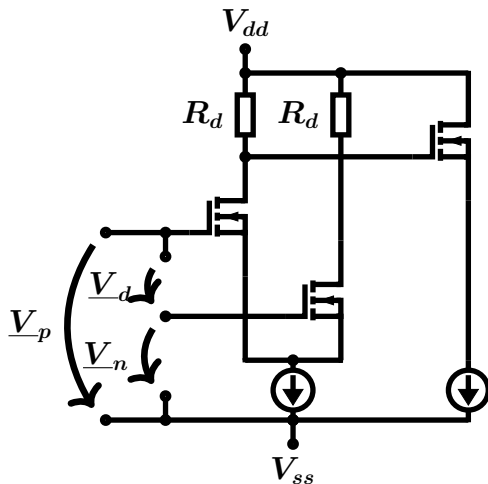
Guest Lecture

see slides from Nenad.

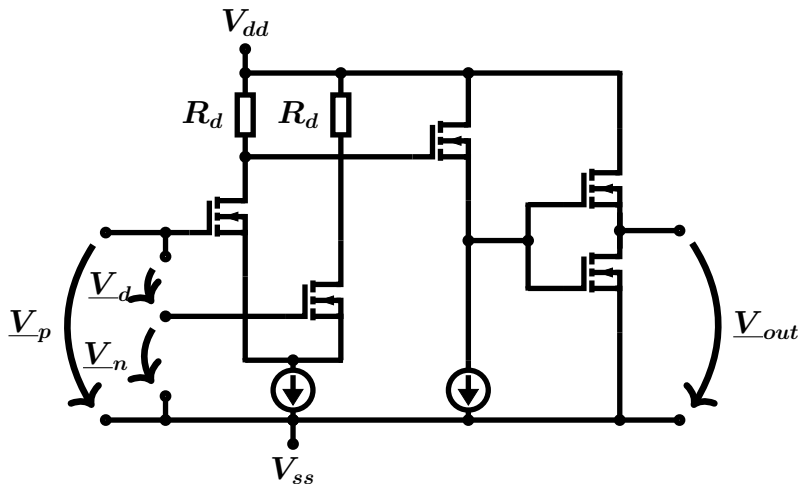
Three Stage Implementation: Differential Input Stage



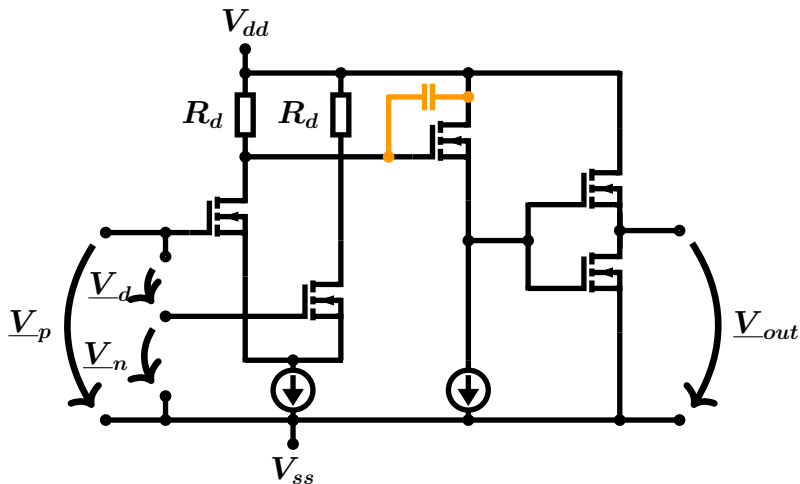
Three Stage Implementation: Voltage Amplification Stage



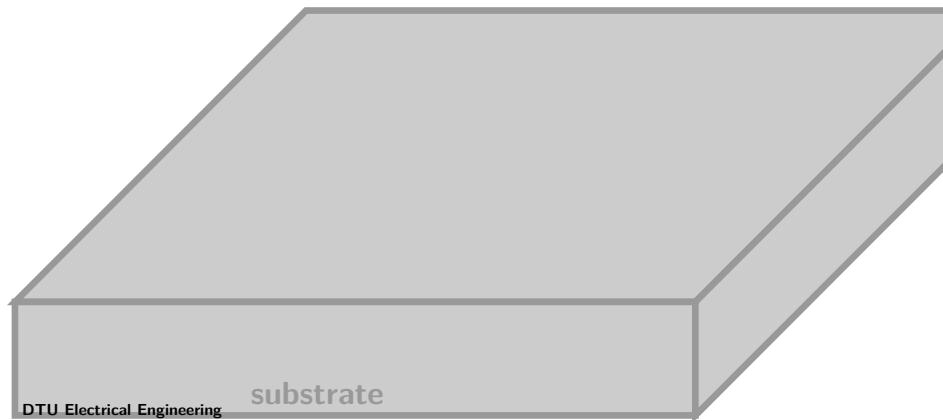
Three Stage Implementation: Current Amplification Stage



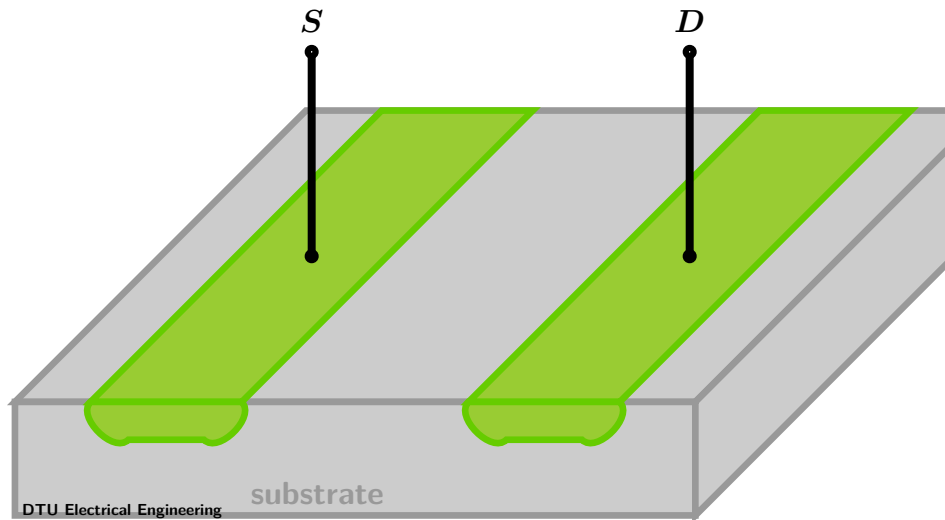
Three Stage Implementation: Dominating Pole



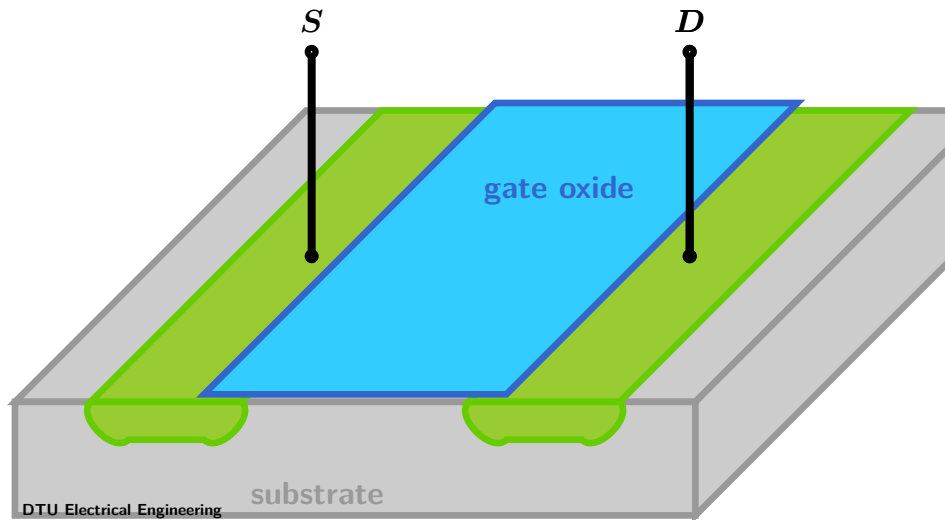
Crossection of a MOSFET



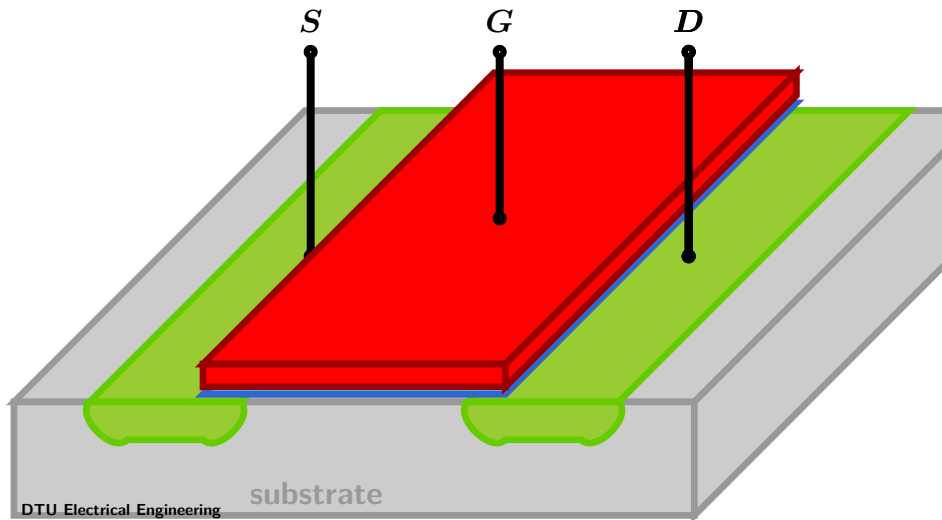
Crosssection of a MOSFET



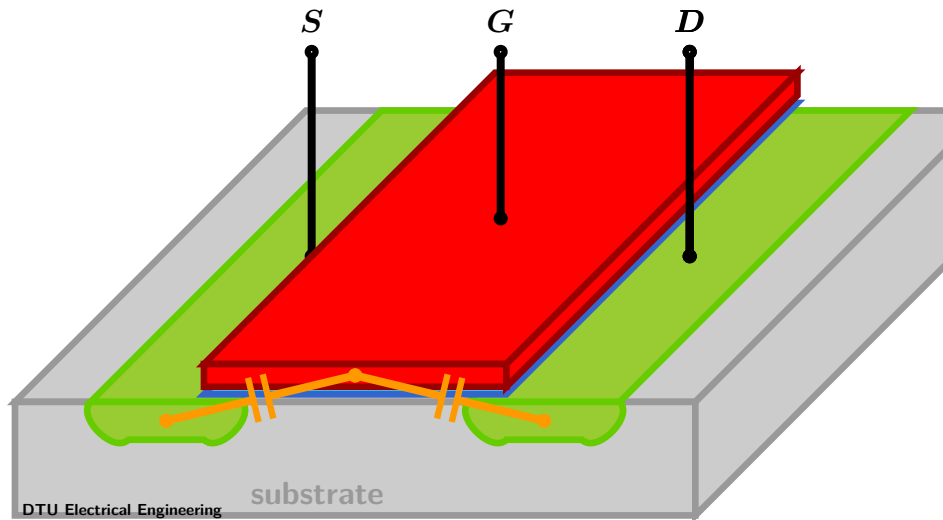
Crosssection of a MOSFET



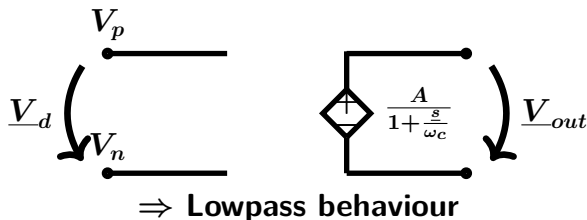
Crosssection of a MOSFET



Crosssection of a MOSFET with Dominating Pole



IMPORTANT AC Model of an Operational Amplifier

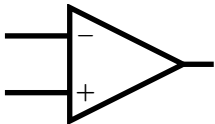


Nomenclature

- A DC gain of the operational amplifier
- ω_c angular corner frequency

Operational Amplifiers

Open Loop Circuit



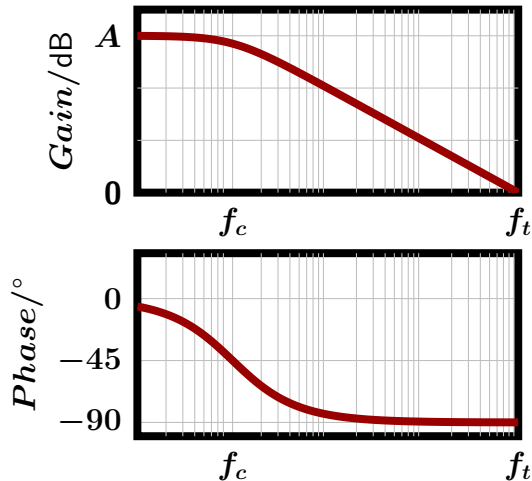
Gain Bandwidth Product

$$GBW = A\omega_c \quad (148)$$

with $\omega_c = 2\pi f_c$

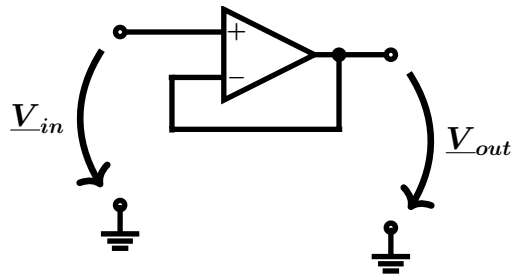
where f_c is the corner frequency and f_t is the transit frequency.

Bode Plot



Operational Amplifiers

Buffer



Derivation Buffer Transfer Function

$$\begin{aligned} \underline{V}_{out} &= \frac{A}{1 + \frac{s}{\omega_c}} (\underline{V}_{in} - \underline{V}_{out}) \\ \Rightarrow \underline{V}_{out} \left(1 + \frac{A}{1 + \frac{s}{\omega_c}} \right) &= \frac{A}{1 + \frac{s}{\omega_c}} \underline{V}_{in} \\ \Rightarrow \underline{V}_{out} \left(A + 1 + \frac{s}{\omega_c} \right) &= A \underline{V}_{in} \end{aligned}$$

(149)

AC Transfer Function of Buffer

$$\frac{\underline{V}_{out}}{\underline{V}_{in}} = \frac{A}{A + 1 + \frac{s}{\omega_c}} = \underbrace{\frac{A}{A + 1}}_{\text{DC-gain}} \frac{1}{1 + \frac{s}{(A+1)\omega_c}} \quad \text{with } (A + 1)\omega_c: \text{ bandwidth of the buffer}$$

Operational Amplifiers

IMPORTANT Buffer Gain
Bandwidth

$$GBW = \frac{A}{A+1} (A+1) \omega_c \quad (151)$$

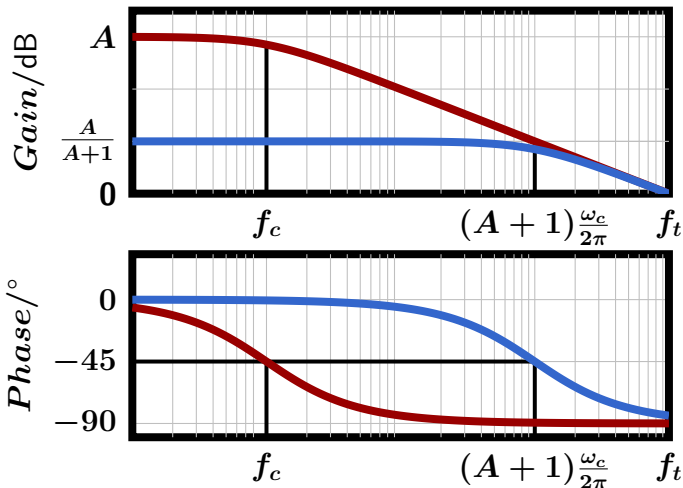
$$A \omega_c$$

Legend

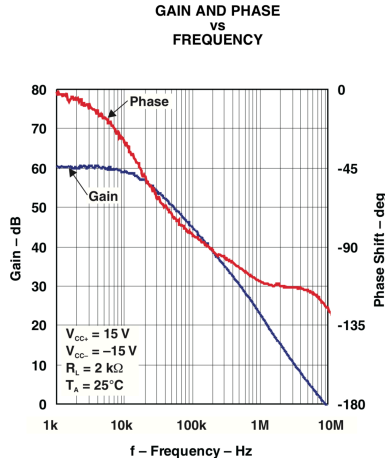
open loop opamp

buffer opamp

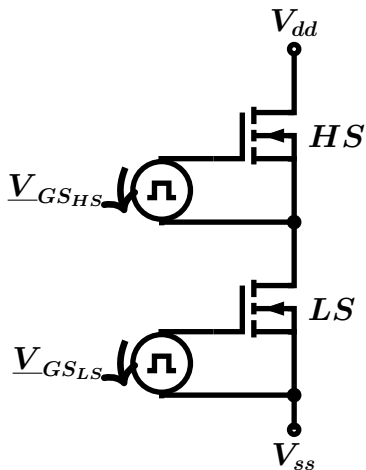
Buffer Bode Plot



Example of measured gain-phase (transfer function) of MC33078 operational amplifier

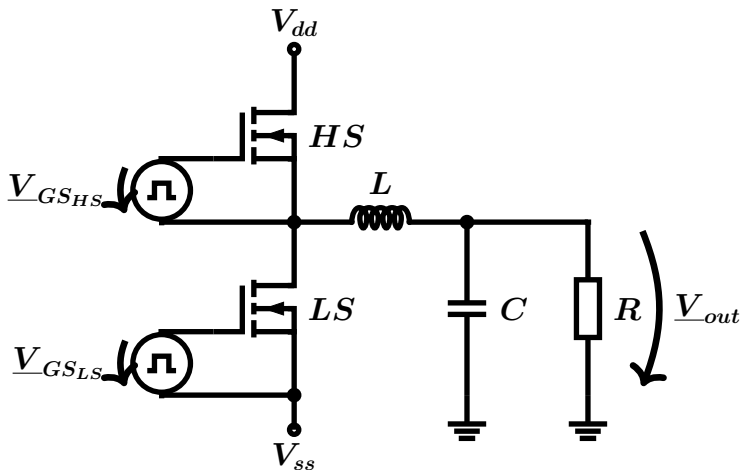


Circuit: Power Stage



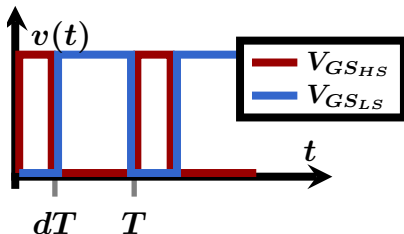
Switch-Mode Amplifiers

Circuit: Power Stage and 2nd Order Lowpass



Switch-Mode Amplifiers

Gate Signals



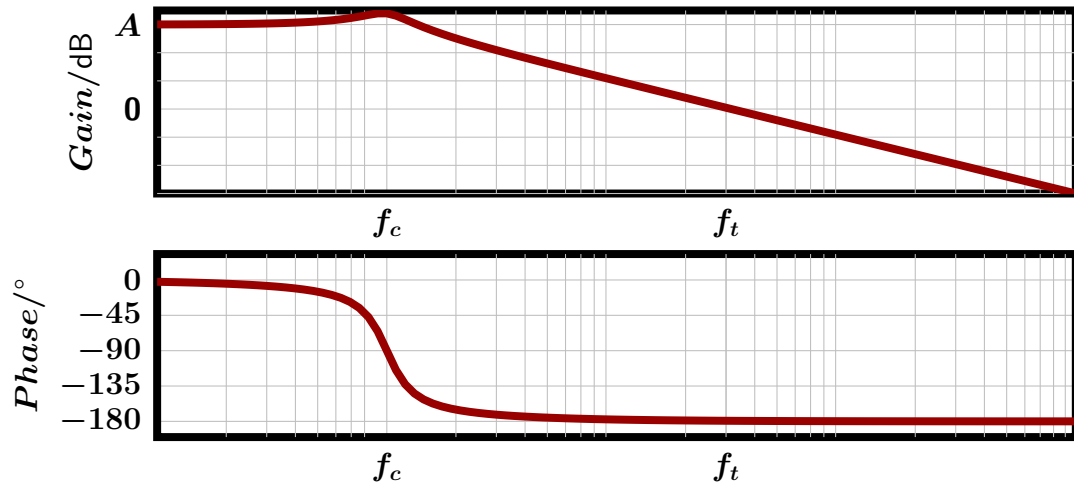
AC Transfer Function of Switch-Mode Amplifier

$$\frac{V_{out}}{V_{in}} = \frac{A}{1 + s\frac{L}{R} + s^2 LC} \quad (152)$$

Operation

- repetitive pulses V_{GSHS} and V_{GSLs} are driving the high-side (HS) and low-side (LS) transistors respectively
- for "high" drive signals, the corresponding transistor is low-impedance between its drain and source (on-state): nearly a short circuit
- for "low" drive signals, the corresponding transistor is high-impedance between its drain and source (off-state): nearly an open circuit
- with alternating gate signals the input to the 2nd order low pass filter gets alternating shorted to V_{dd} and V_{ss}
- a pulse width modulator (PWM) controls the duty cycle d , where $d \propto V_{in}$
- the average output V_{out} is proportional (amplified) to d

Bode Plot



Properties of Amplifiers

IMPORTANT Arbitrary Gain-Bandwidth Product

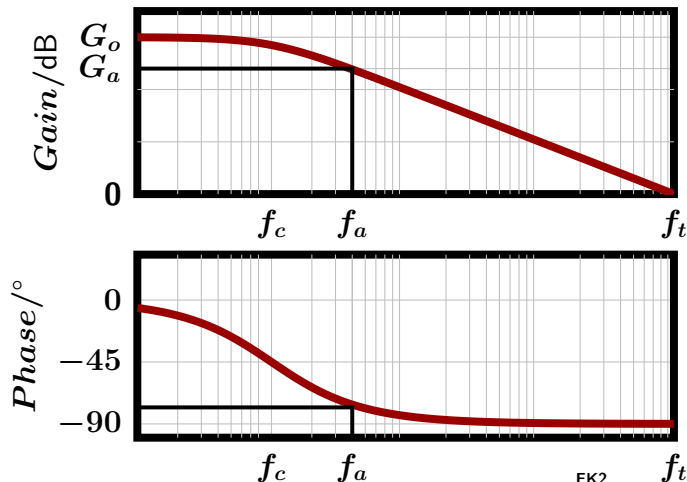
$$\begin{aligned} G_0 \omega_c &= G_a \omega_a \\ G_0 f_c &= G_a f_a \end{aligned} \quad (153)$$

Meaning

For an arbitrary frequency f_a , the maximum achievable gain G_a is bounded by the transfer function of the amplifier.

The open loop case is the maximising the gain, while the buffer configuration is maximising the bandwidth.

Bode Plot with an arbitrary Gain-Bandwidth Operating Point



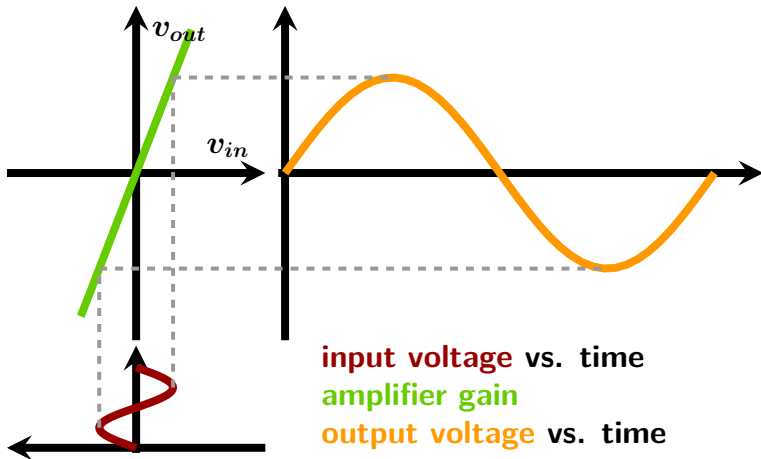
Properties of Amplifiers

An ideal Amplifier

An ideal amplifier is

- "a piece of wire"
- with "no resistance" and
- "a lot of gain".

Linearity



Properties of Amplifiers

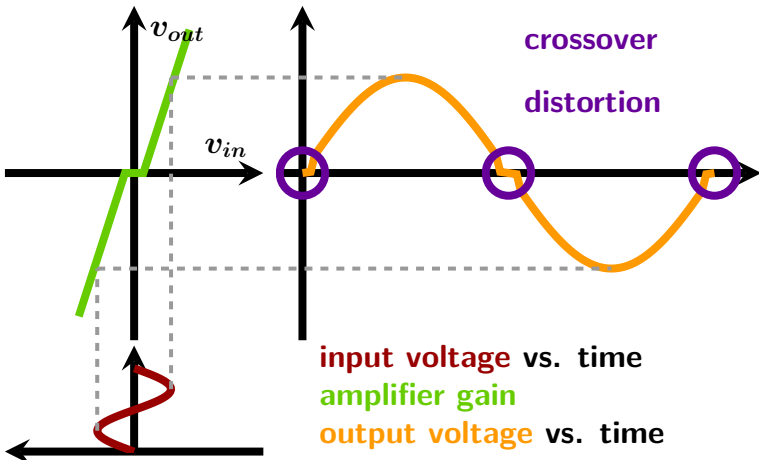
Cross-over

Especially linear amplifiers (like **Operational Amplifiers**), switch between components, when processing the positive and negative halfwaves.

⇒ Nonlinearity!

⇒ **Distortion**

Cross-over Nonlinearity



Properties of Amplifiers

Overdrive / Clipping

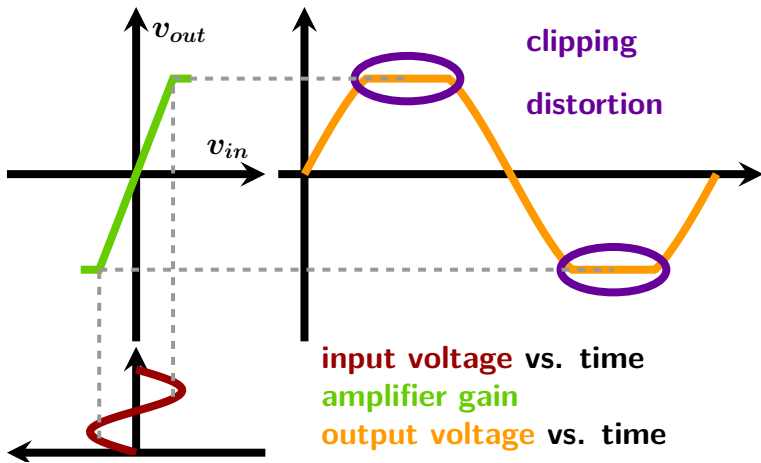
When the input of an amplifier is beyond the input stages input range, the amplifier is overdriven.

When the output stage is required to operate beyond its supply voltages, the amplifier is clipping.

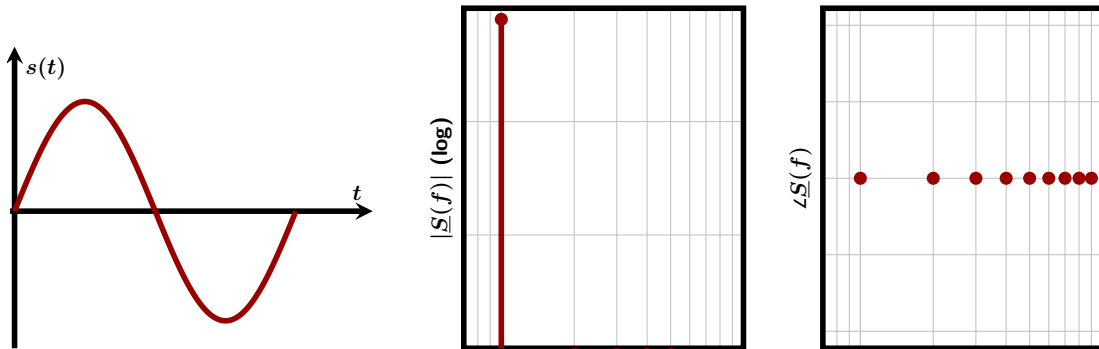
⇒ Nonlinearity!

⇒ Distortion

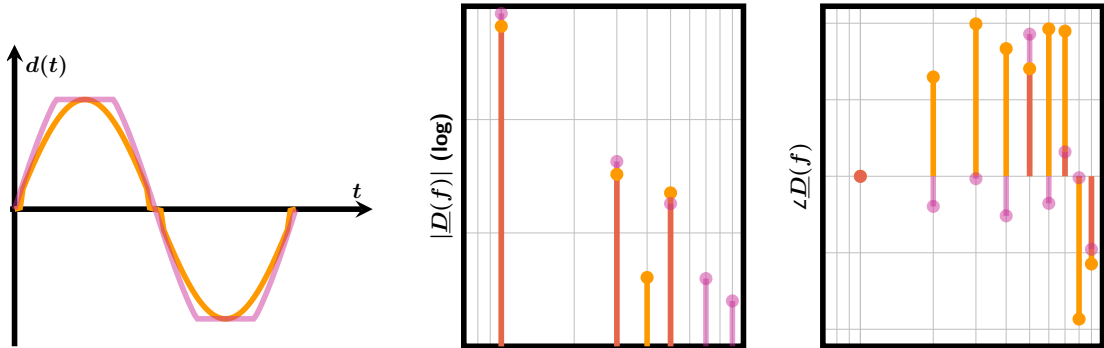
Overdrive / Clipping Nonlinearity



Sine wave in Time and Frequency Domain



Distorted Sine waves in Time and Frequency Domain



IMPORTANT Total Harmonic Distortion

$$THD = \frac{\sqrt{\sum_2^{\infty} V_n^2}}{V_1} \quad (154)$$

Nomenclature

- V_n is the amplitude of the n-th harmonic
- V_1 is the amplitude of the fundamental.

Examples of THD

For the waveforms above,

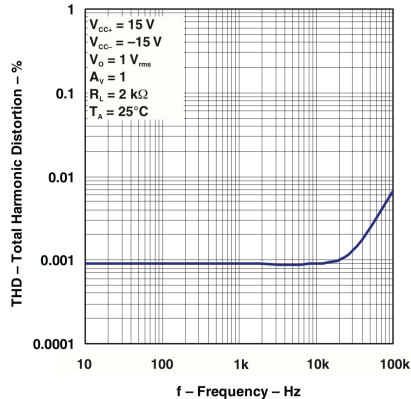
- the **crossover** distortion is 5.0 % and
- the **clipping** distortion is 5.4 %.

Dependencies of THD

The THD of the output signal of an amplifier is mainly dependent on

- the amplifiers bandwidth,
- the maximum power of the amplifier,
- the amplitude of the signal,
- the frequency of the signal, and
- the amplifiers control loop.

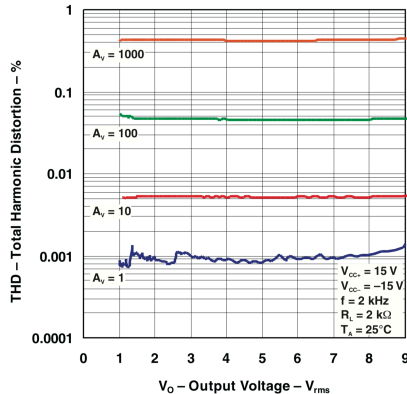
Example of measured THD as a function of frequency of MC33078 operational amplifier



<http://www.ti.com/lit/ds/symlink/mc33078.pdf> [13]

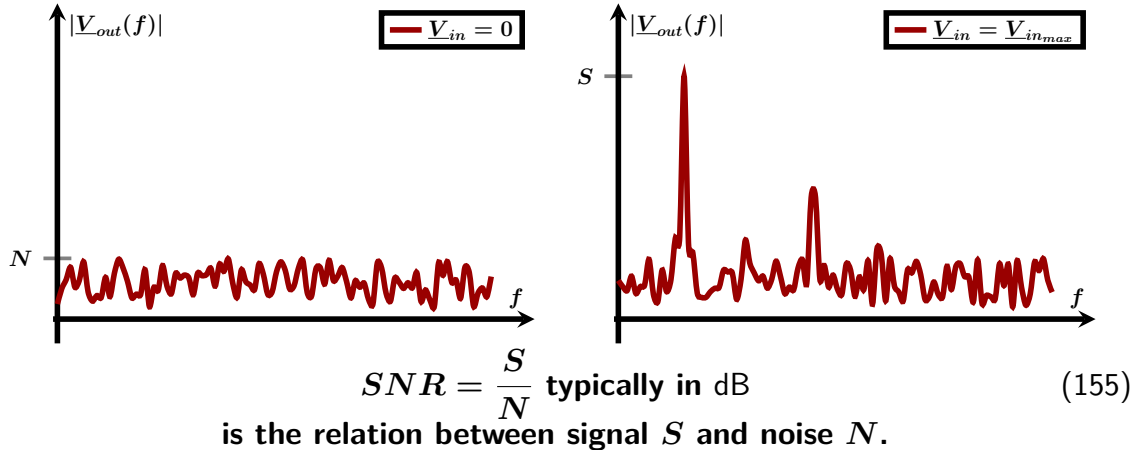
Properties of Amplifiers

Example of measured THD as a function of level of MC33078 operational amplifier



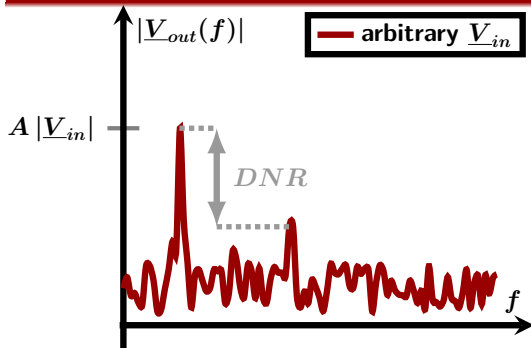
<http://www.ti.com/lit/ds/symlink/mc33078.pdf> [13]

Signal to Noise Ratio (SNR)



Properties of Amplifiers

Dynamical Range (DNR)

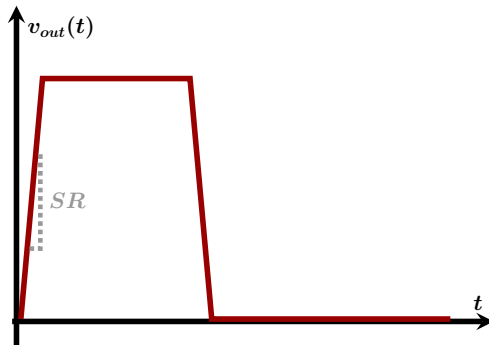
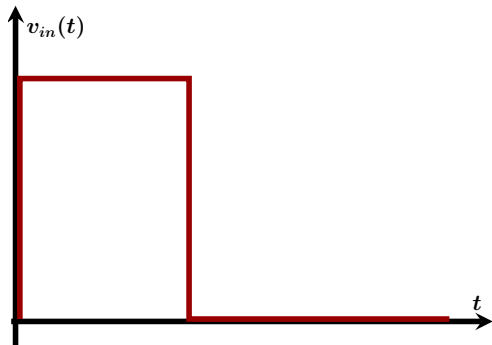


$$DNR = \frac{S_a}{N_a} \bigg|_{V_{in}} \quad \text{typically in dB} \quad (156)$$

is the relation between signal S_a at a given specific input level V_{in} and noise N at the same input level V_{in} .

- Anything, that was not present in the spectrum of the input V_{in} is limits the DNR , also distortion.
- DNR can not exceed SNR

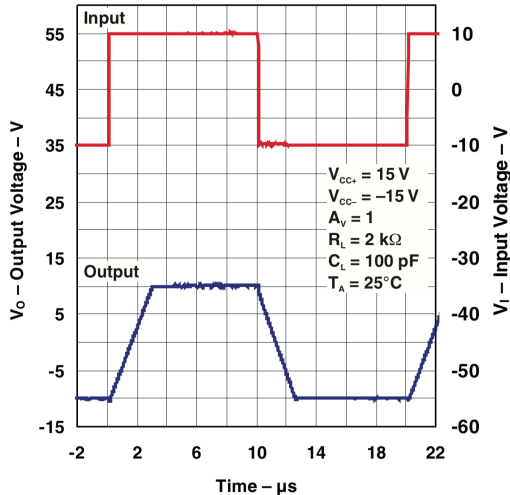
Slew Rate



$$SR = \frac{dv_{out}}{dt} \quad (157)$$

is the maximum output voltage v_{out} rise per time t at the output of an amplifier.

Example of measured slew rate of MC33078 operational amplifier



Properties of Amplifiers

Example of general parameters of MC33078 operational amplifier

FEATURES

- Dual-Supply Operation . . . $\pm 5\text{ V}$ to $\pm 18\text{ V}$
- Low Noise Voltage . . . $4.5\text{ nV}/\sqrt{\text{Hz}}$
- Low Input Offset Voltage . . . 0.15 mV
- Low Total Harmonic Distortion . . . 0.002%
- High Slew Rate . . . $7\text{ V}/\mu\text{s}$
- High-Gain Bandwidth Product . . . 16 MHz
- High Open-Loop AC Gain . . . 800 at 20 kHz
- Large Output-Voltage Swing . . . 14.1 V to -14.6 V
- Excellent Gain and Phase Margins

<http://www.ti.com/lit/ds/symlink/mc33078.pdf> [13]

Exercise 4.1 Gain-phase Simulation of Operational Amplifier

Conduct an AC-simulation of the LTSpice model "opamp" and plot the gain-phase diagram. Play with the parameters of the "opamp" and see what happens.

Exercise 4.2 Crossover Distortion of an Operational Amplifier

- Use the LTSpice model "opamp" to plot the output of an ideal buffer and use the spice directive to calculate the Fourier components and the total harmonic distortion. Observe how the simulation parameters (tstep, tstop, ..) influence the result.
- Add two anti-parallel diodes (with a forward voltage drop of 1 V) and a resistor in the adequate places at the input of the buffer to model cross-over distortion and repeat compare with the above plots and the Fourier calculation.

Exercise 4.3 Clipping distortion of Operational Amplifier

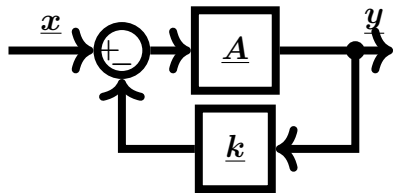
- a** Use the LTSpice model "UniversalOpamp2" to plot the output of an ideal buffer with symmetrical $\pm 10\text{ V}$ supply rails.
- b** Use the step function to step the input voltage from 5 V to 15 V and use the Fourier spice directive to calculate the spectrum and the total harmonic distortion.

Barkhausen Criterion

Further Literature on Oscillators

Chapters 12. to 12.5 in [14].

Block Diagram of an Oscillator



Transfer Function

$$\underline{y} = \underline{A} (\underline{x} - \underline{k}\underline{y}) \Rightarrow \underline{H}_{cl} = \frac{\underline{y}}{\underline{x}} = \frac{\underline{A}}{1 + \underline{k}\underline{A}} \quad (158)$$

with $\underline{H}_{cl}$: closed loop transfer function

IMPORTANT Barkhausen Oscillation Criterium

$$\frac{\underline{y}}{\underline{x}} \rightarrow \infty |_{\underline{s}_0} \quad (159)$$

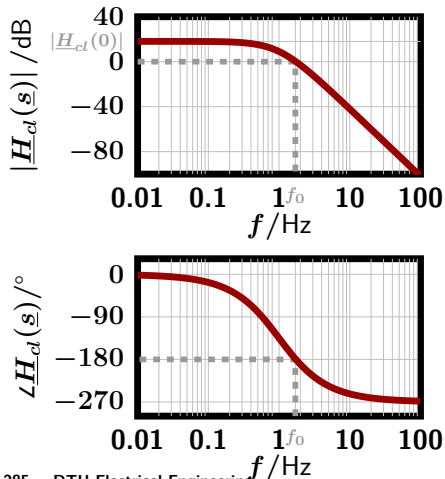
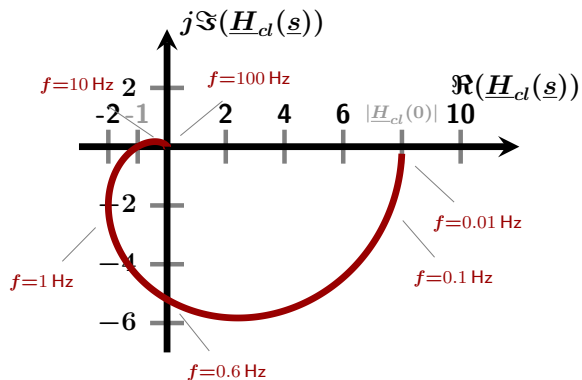
with $\underline{s}_0$: complex oscillation frequency

$$\Rightarrow \underline{H}_{ol}(\underline{s}_0) = \underline{k}(\underline{s}_0)\underline{A}(\underline{s}_0) \stackrel{!}{=} -1 = 1e^{-j180^\circ} \quad (160)$$

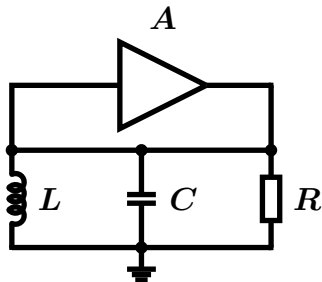
with $\underline{H}_{ol}$: open loop transfer function

Barkhausen Criterion

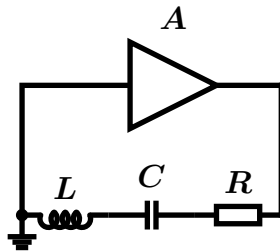
Visualization of Barkhausen's Criterion in a Bode Plot

Visualization of Barkhausen's Criterion in an s -plane

Parallel RLC Oscillator



Series RLC Oscillator



Operation

- L and C adjust the oscillation frequency $f_0 = \frac{1}{2\pi\sqrt{LC}}$,
- R adjusts the quality Q of the resonator (often parasitics only), and
- A provides the necessary gain to fulfill the Barkhausen criterion.

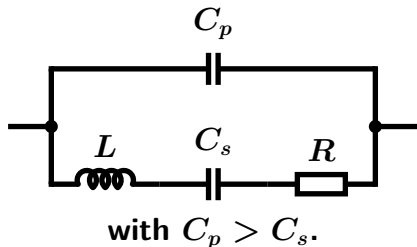
Applications of Circuits

Sinusoidal Oscillators

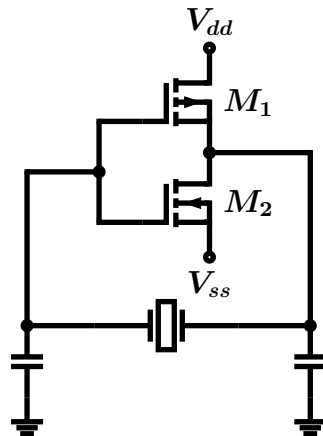
Crystal



Equivalent Circuit Diagram of a Crystal

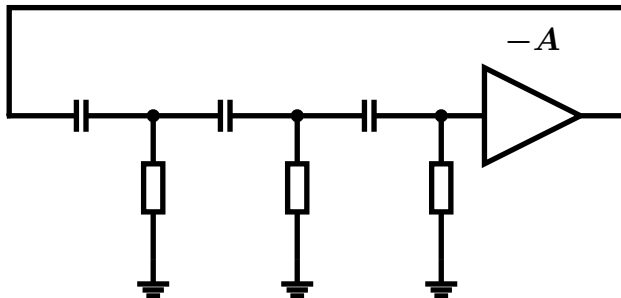


Colpitts Oscillator based on a Crystal



where M_1 and M_2 form an inverter.

Phase Shift Oscillator

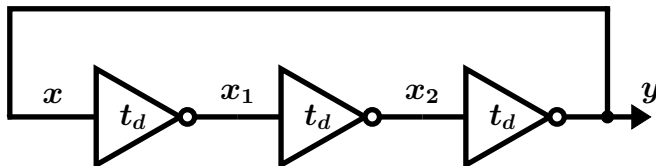


Operation

- The passive filter shifts the frequency by $3 \cdot (-90^\circ) = -270^\circ$ and
- the inverting amplifier adds another -180° .

Square Wave Oscillators

Delay Line Oscillator



Operation

- Each of the three inverters (NOT-gates) provides a logic "HIGH" or "LOW" on its output, for its respective input "LOW" or "HIGH".
- The output reaction to an input change is delayed by t_d in each inverter.

Square Wave Oscillators

Derivation Delay Line Oscillator Transfer Function

$$x_1(t) = -x(t - t_d)$$

$$x_2(t) = x(t - 2t_d)$$

$$y(t) = -x(t - 3t_d)$$



(161)

$$\underline{Y}(s) = e^{-t_d s} \underline{X}_2(s) = e^{-3t_d s} \underline{X}(s)$$

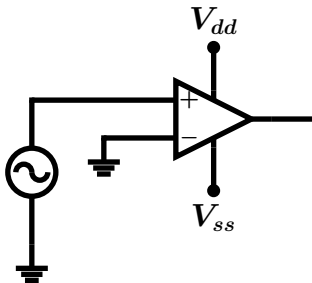
$$\underline{H}_{DL}(s) = \frac{\underline{Y}(s)}{\underline{X}(s)} = e^{-3t_d s}$$

Oscillation Frequency

- The phase of $\underline{H}_{DL}$ is decreasing linearly from 0° at $f = 0$ Hz towards $-\infty^\circ$ for $f \rightarrow \infty$ Hz.
- The delay line oscillates at the frequency, where $\angle \underline{H}_{DL} = -180^\circ$.

Square Wave Oscillators

Comparator

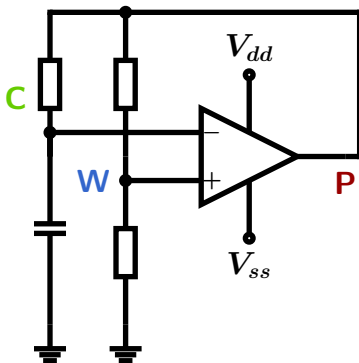


Operation

The comparator's output is the sign of the sine wave at its input.

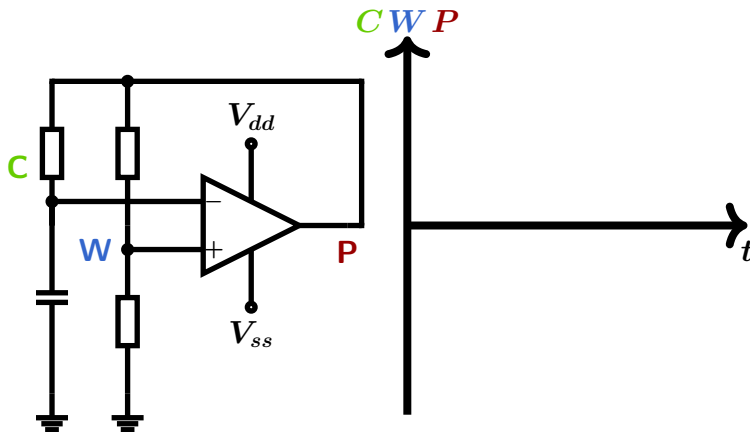
Pulse Width Modulator

Astable Integrating Modulator



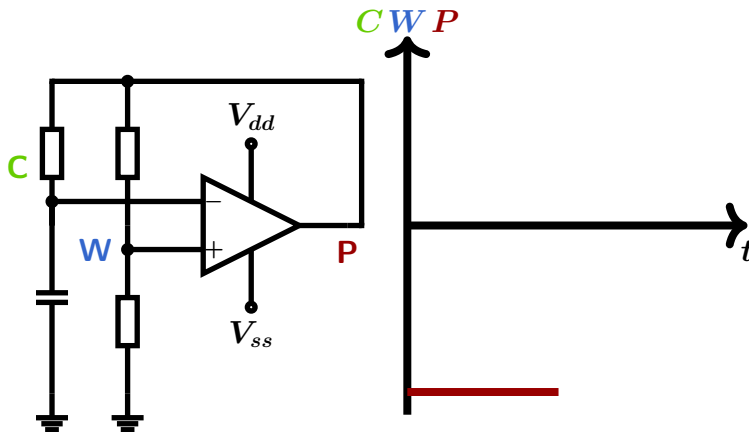
Pulse Width Modulator

Astable Integrating Modulator



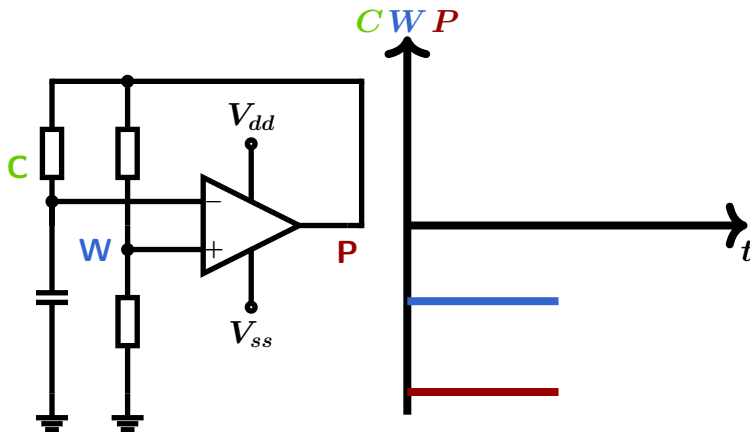
Pulse Width Modulator

Astable Integrating Modulator



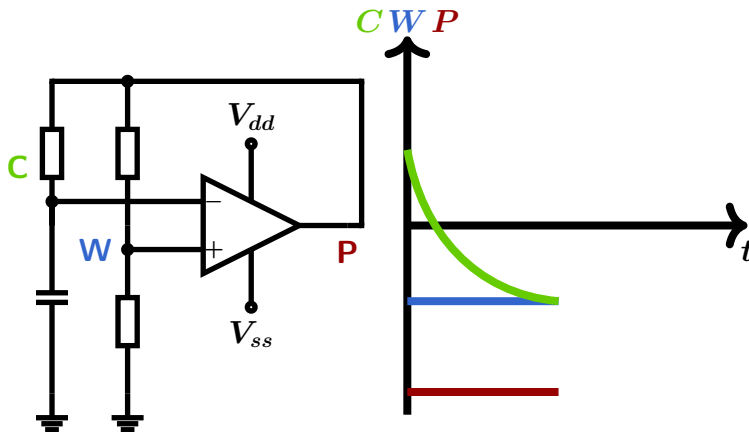
Pulse Width Modulator

Astable Integrating Modulator



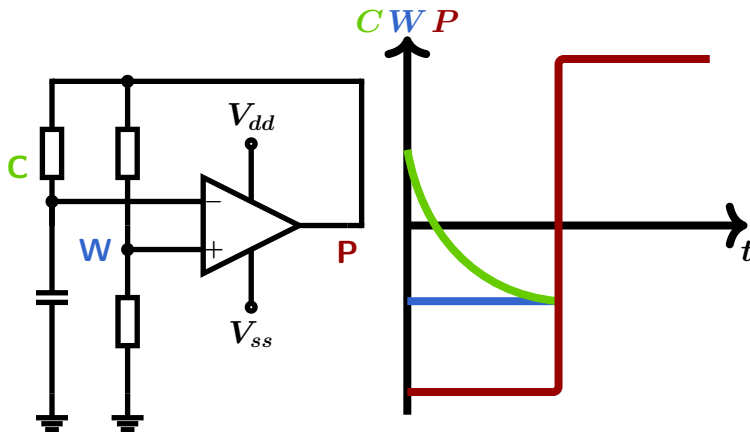
Pulse Width Modulator

Astable Integrating Modulator



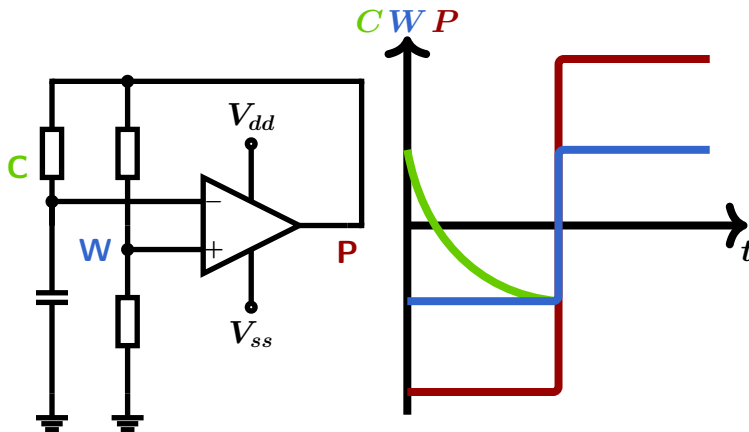
Pulse Width Modulator

Astable Integrating Modulator



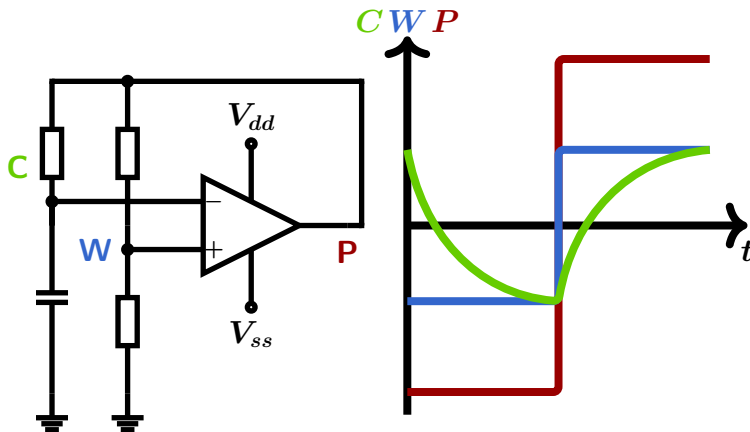
Pulse Width Modulator

Astable Integrating Modulator



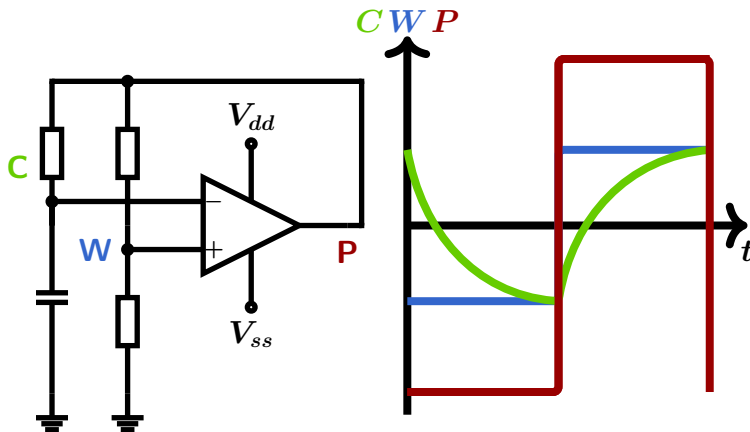
Pulse Width Modulator

Astable Integrating Modulator



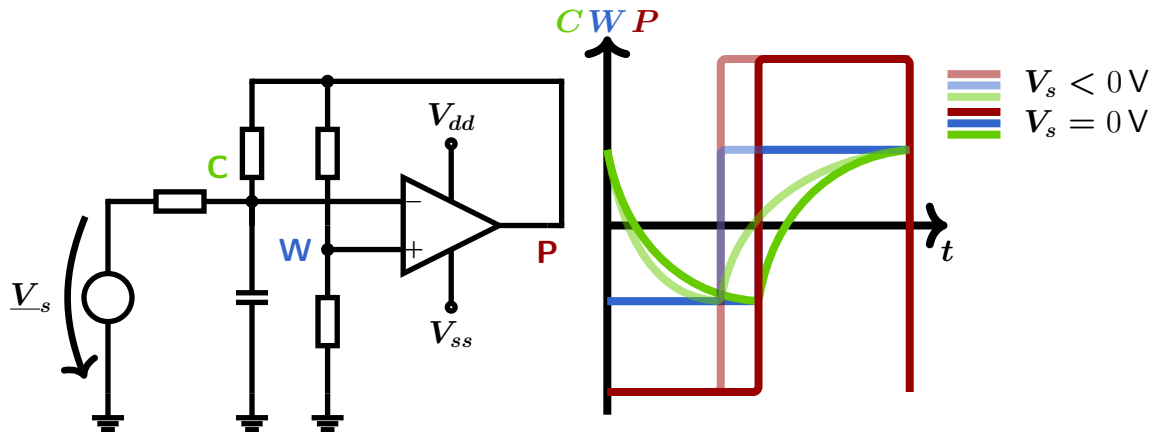
Pulse Width Modulator

Astable Integrating Modulator



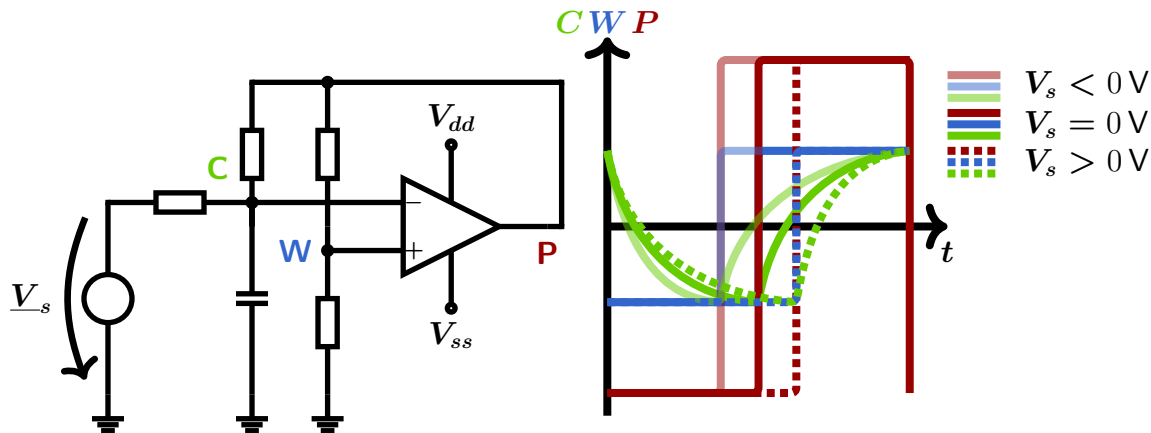
Pulse Width Modulator

Astable Integrating Modulator



Pulse Width Modulator

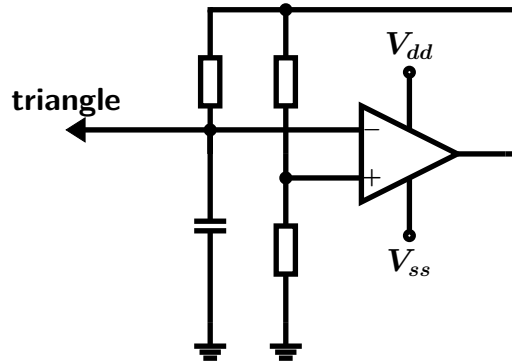
Astable Integrating Modulator



Applications of Circuits

Triangle Generators

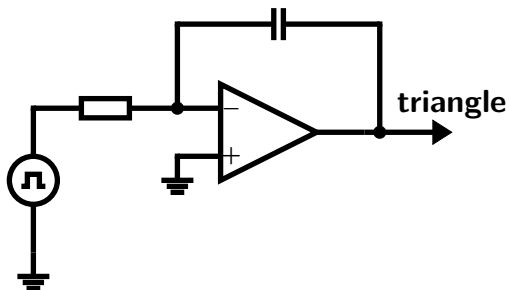
Bistable Multivibrator



Applications of Circuits

Triangle Generators

Integrator after Square Wave



Oscillators

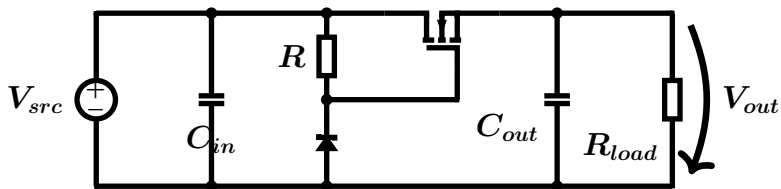
Generic Properties of Oscillators

- **Frequency Range:** the minimum, typical and maximum frequency an oscillator can provide. Crystal based oscillators are very precise and therefore used for clock generation in digital circuits.
- **Waveform:** sinusoidal, square, pulse, triangle or others.
- **Jitter:** the frequency stability - typically observed in eye-diagrams.
- **Temperature Drift:** the stability of other oscillator parameters versus temperature.
- **Voltage Range:** the operating voltage range of the output and the necessary supplies for the oscillator.

Exercise 4.4 Bistable Multivibrator with modulation - Astable Integrating Modulator (AIM)

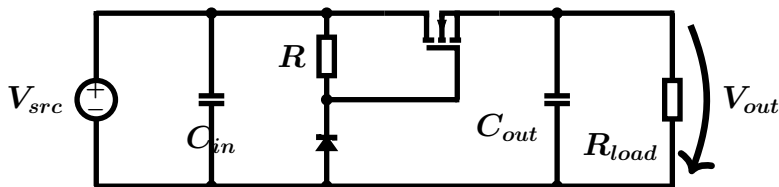
Consider section III in [15], which is modelling an Astable Integrating Modulator (AIM) (same as a bistable multivibrator with additional input). Simulate the circuit in LTSpice (or any other circuit simulation tool) and reproduce the diagrams in figure 18 and 19 from [15].

Circuit Diagram



Linear Power Supplies

Circuit Diagram



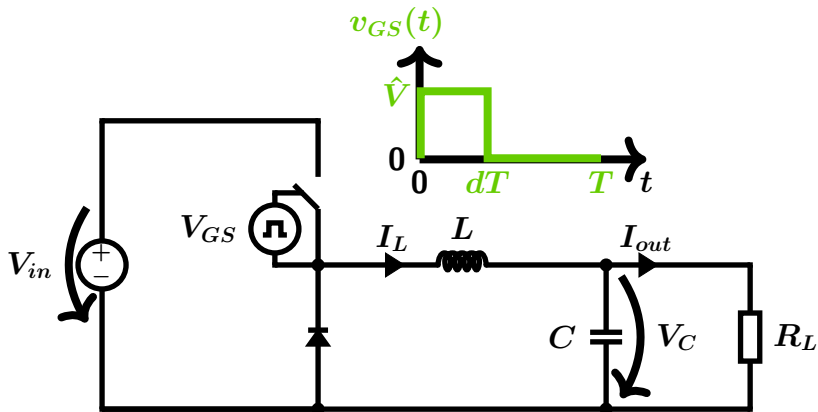
Load Regulation

$R_L \downarrow \Rightarrow C_{out} \text{ discharges} \Rightarrow V_{out} \downarrow$
 $V_{GS} = V_Z - V_{out} \Rightarrow V_{GS} \uparrow$
 $I_D = g_m V_{GS} \Rightarrow I_D \uparrow$
 $\Rightarrow C_{out} \text{ charges} \Rightarrow V_{out} \uparrow$

Line Regulation

$V_{in} \uparrow \Rightarrow I_D \uparrow \Rightarrow C_{out} \text{ charges} \Rightarrow V_{out} \uparrow$
 $V_{GS} = V_Z - V_{out} \Rightarrow V_{GS} \downarrow$
 $I_D = g_m V_{GS} \Rightarrow I_D \downarrow$
 $\Rightarrow C_{out} \text{ discharges} \Rightarrow V_{out} \downarrow$

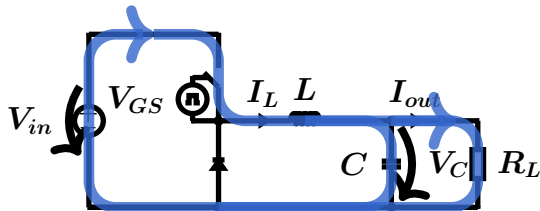
Circuit Diagram of a Buck Converter



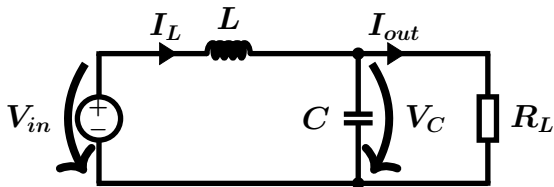
Applications of Circuits

Buck Converter

Operation of a Buck Converter during "on"-time of Transistor ($0 < t < dT$)



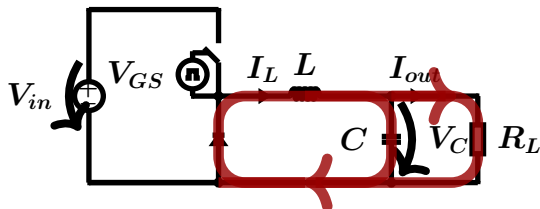
Equivalent circuit diagram during $0 < t < dT$



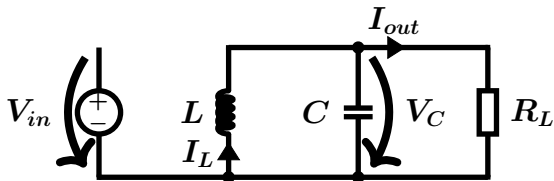
Applications of Circuits

Buck Converter

Operation of a Buck Converter during "off"-time of Transistor ($dT < t < T$)



Equivalent circuit diagram during $dT < t < T$



Buck Converter

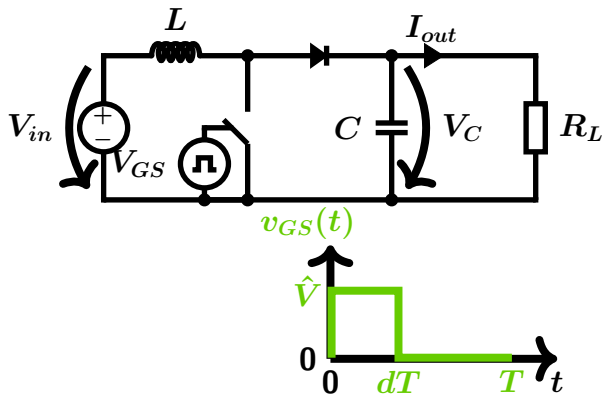
IMPORTANT DC-transfer function of Buck Converter

$$\frac{V_C}{V_{in}} = d \qquad I_{out} = I_L \qquad (162)$$

Applications of Circuits

Boost Converter

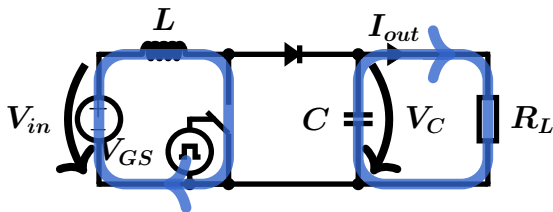
Circuit Diagram of a Boost Converter



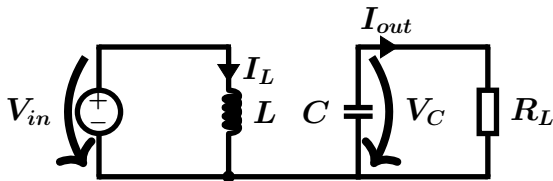
Applications of Circuits

Boost Converter

Operation of a Boost Converter during "on"-time of Transistor ($0 < t < dT$)



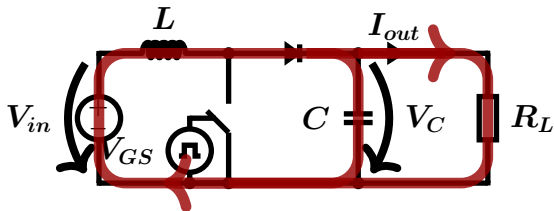
Equivalent circuit diagram during $0 < t < dT$



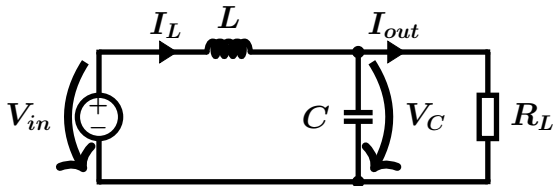
Applications of Circuits

Boost Converter

Operation of a Boost Converter during "off"-time of Transistor ($dT < t < T$)



Equivalent circuit diagram during $dT < t < T$



Applications of Circuits

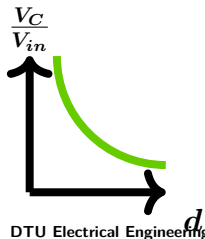
Boost Converter

IMPORTANT DC-transfer function of Boost Converter

$$\frac{V_C}{V_{in}} = \frac{1}{1 - d} \quad (163)$$

$$\frac{I_L}{I_{out}} = \frac{1}{1 - d} \quad (164)$$

$1/d$

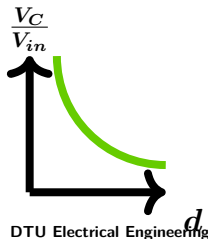


IMPORTANT DC-transfer function of Boost Converter

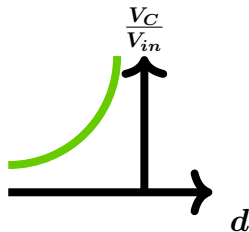
$$\frac{V_C}{V_{in}} = \frac{1}{1 - d} \quad (163)$$

$$\frac{I_L}{I_{out}} = \frac{1}{1 - d} \quad (164)$$

$1/d$



$1/(1 - d)$

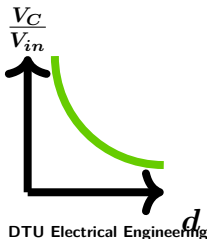


IMPORTANT DC-transfer function of Boost Converter

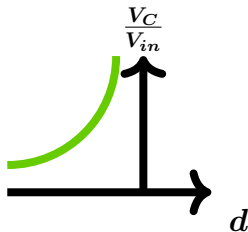
$$\frac{V_C}{V_{in}} = \frac{1}{1 - d} \quad (163)$$

$$\frac{I_L}{I_{out}} = \frac{1}{1 - d} \quad (164)$$

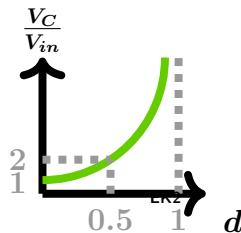
$1/d$



$1/(1 - d)$



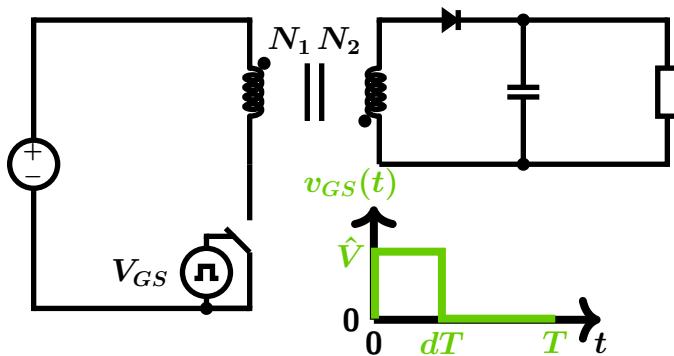
**Boost DC Transfer
Function $1/(1 - d)$**



Applications of Circuits

Flyback Converter

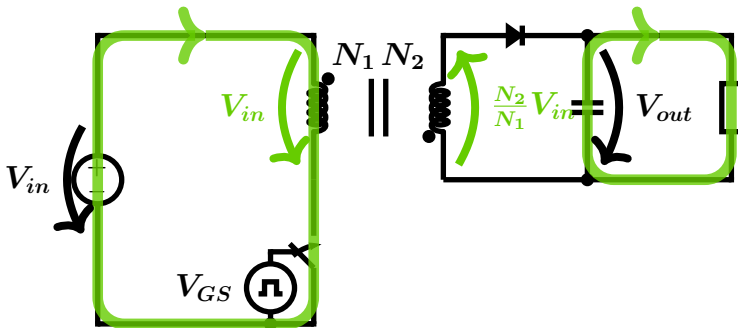
Circuit Diagram of Flyback Converter



Applications of Circuits

Flyback Converter

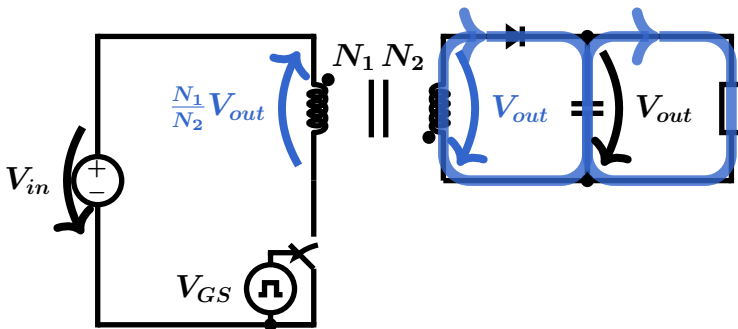
Operation of a Boost Converter during "on"-time of Transistor ($0 < t < dT$)



Applications of Circuits

Flyback Converter

Transistor off during $dT < t < T$

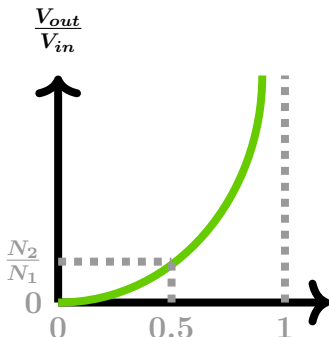


Power Supplies - Isolated: Flyback

IMPORTANT DC Transfer Function of Flyback Converter

$$\frac{V_{out}}{V_{in}} = \frac{N_2}{N_1} \frac{d}{1-d} \quad (165)$$

DC Transfer Function of Flyback Converter $d/(1-d)$ as a Function of Duty Cycle d



Applications of Circuits

Power Supplies

Generic Properties of Power Supplies

- Input and Output Voltage: DC and/or AC, amplitude, frequency range
- Output Power
- Efficiency $\eta = \frac{P_{out}}{P_{in}}$
- Isolation: Buck and boost are non-isolated, Flyback is isolated with Transformer

Exercise 4.5 Designing a Flyback Transformer

A flyback shall convert the peak-rectified European AC mains to 50 V with a duty cycle of 47.98 % at a switching frequency of 100 kHz. The AC-part of the input current shall not exceed 2 A.

- What is the inductance of the primary winding?
- b What is the inductance of the secondary winding?

Section 5

Course Literature

- **Course Literature**
- ## References

So long, and Thanks for All the Fish

Course Literature

References I

Main course literature: [1] [2] [4] [3]:



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